Search for axionlike particle as dark matter candidate with nuclear magnetic resonance

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Declaration of Authorship

I hereby declare that this thesis titled, *Search for axionlike particle as dark matter candidate with nuclear magnetic resonance* and the work presented in it are my own. I confirm that:

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Yuzhe Zhang

Abstract

Axions and axionlike particles (ALPs) are well-motivated candidates for the dark matter that makes up most of the matter content of the universe. In the nuclear-gradient interaction, an ALP dark-matter field acts as an effective oscillating pseudomagnetic field on nuclear spins, and nuclear magnetic resonance (NMR) provides a resonant method for detecting this weak drive.

This thesis develops the ALP-search program in the Cosmic Axion Spin Precession Experiment (CASPEr) from theory to data analysis. The introductory chapters motivate the dark-matter problem, review NMR concepts relevant to precision searches, and introduce the ALP framework used throughout the work. The theoretical chapters derive the ALP-induced spin response and the optimal strategy for frequency-scanning searches, and they distinguish two signal classes: a virialized Milky Way halo, which gives a stochastic broadband signal with finite coherence time, and gravitationally bound Earth-bound states, which can produce long-lived, narrow, and direction-dependent signals. For frequency scans, four regimes are identified according to the ordering of the NMR coherence time T_2^*, the intrinsic relaxation time T_2, and the ALP coherence time τ_a. In all long-dwell-time regimes, the coupling sensitivity improves only as $T_{\mathrm{tot}}^{-1/4}$ with total integration time, so better polarization, smaller field inhomogeneity, longer coherence times, and lower detector noise are more effective than simply measuring longer.

The experimental chapter describes the CASPEr-Gradient-Low-Field apparatus used in this thesis, including the superconducting magnet system, the methanol sample, the pickup coil, the SQUID detection chain, and the Python-based control and acquisition software. The measured sample is thermally polarized liquid methanol at about 180 K, with a very small proton polarization of approximately 1.8×10^{-7}, which sets the dominant sensitivity limitation. The same apparatus parameters are then used to project the gains expected from hyperpolarized methanol and liquid ^{129}Xe, as well as from improved field homogeneity and larger sample or pickup-coil performance.

The results chapters present the full data-analysis pipeline, including time-series sanity checks, Fourier transformation, low-pass correction, baseline removal, optional optimal filtering, thresholding, and candidate validation. A proof-of-principle Milky Way ALP search scanned a 240 Hz band around 1.348 570 MHz and found no statistically significant signal, yielding a 95 % confidence-level limit of $|g_{\mathrm{ap}}| \lesssim$

$3 \times 10^{-2}\,\mathrm{GeV}^{-1}$ over the scanned mass range. An Earth-bound ALP search was also carried out with long dwell times and a narrower scan window; its final candidate validation and limit setting are still in progress. The appendices provide the numerical Bloch-dynamics simulator, the statistical hypothesis-test framework, and the derivation of the optimal dwell-time strategy used in the scan optimization.

The concluding chapter summarizes these findings and outlines the next steps for CASPEr. The most promising upgrades are hyperpolarized samples, improved temperature control, lower noise, better field homogeneity, larger sample volume, and enhanced pickup performance. Together with future concepts such as dual-NMR-sample searches, these developments should extend CASPEr toward much higher sensitivity and broaden its reach to axions, dark photons, and other new-physics scenarios.

Zusammenfassung

Acknowledgments

Acknowledgments

Thank

I would like to express my sincere gratitude to Jiawei (Vivian) Gao and Julian Walter for their efforts on the Earth-bound ALP halo project. I would also like to thank Declan Smith, Derek Kimball, Alexander Sushkov for helpful discussion and great contribution on the data analysis.

Contents

1 Introduction

Astrophysical evidence for dark matter comes from galaxy rotation curves, gravitational lensing, large-scale structure, and the cosmic microwave background: visible matter alone cannot account for the observed dynamics. Axions and axionlike particles are theoretically attractive candidates because they are weakly coupled, naturally light, and can generate spin-precession signals accessible to NMR experiments.

This chapter introduces the key ingredients of the search. Section 1.1 reviews the NMR concepts most relevant to the experiment. Section 1.2 summarizes the dark-matter evidence. Section 1.3 describes the axion and ALP framework, including the pseudomagnetic-field analogy that connects ALP physics to NMR. Section 1.4 outlines the CASPEr project. The theory of signal estimation and scanning optimization is developed in Chapter 2; the apparatus is described in Chapter 3; measurements and results appear in Chapter 4.

1.1 Nuclear magnetic resonance

Nuclear magnetic resonance is a technique for studying nuclear spins in magnetic fields. In the presence of a bias magnetic field, spin states split by the Zeeman effect, resulting in nuclear polarization. A polarized sample acquires a non-zero net magnetization that precesses at the Larmor frequency. Because the Larmor frequency is proportional to the magnetic field, NMR can be a tool for magnetic field measurement.

1.1.1 Zeeman splitting and polarization

In a bias magnetic field B_0, a spin-1/2 nucleus splits into two energy levels separated by

$$\Delta E = \hbar\gamma B_0\,, \tag{1.1}$$

where γ is the nuclear gyromagnetic ratio. This energy difference sets the Larmor (resonance) frequency

$$\nu_L = \frac{\gamma B_0}{2\pi}\,. \tag{1.2}$$

For protons, $\gamma/(2\pi) = 42.577\,478\,461(18)$ MHz/T[33], so a bias field of 31.67 mT,as used in this work, gives $\nu_L \approx 1.348$ MHz.

At thermal equilibrium, the spin populations obey the Boltzmann distribution. In the high-temperature limit $\Delta E \ll k_B T$ (satisfied for all conditions in this thesis), the fractional spin polarization is

$$p \approx \frac{\hbar\gamma B_0}{2k_B T}\,. \tag{1.3}$$

At $B_0 = 31.67$ mT and $T = 180$ K, the proton thermal polarization is only $p \approx 2 \times 10^{-7}$. Hyperpolarization techniques—such as spin-exchange optical pumping (SEOP) or parahydrogen-induced polarization (PHIP)—can boost p by several orders of magnitud [19].

1.1.2 Spin-projection noise

Even in the absence of a bias magnetic field, an ensemble of N spins does not cancel perfectly to give a zero net magnetization. Statistical fluctuations leave a residual transverse moment of order $\sqrt{N}\,\mu$ (where $\mu = \hbar\gamma/2$ is the nuclear magneton), giving a magnetization noise floor

$$M_{\text{SPN}} \sim \frac{\sqrt{N}\,\mu}{V}\,. \tag{1.4}$$

This spin-projection noise (SPN) is the fundamental quantum limit for spin measurements [4, 9, 32].

1.1.3 Bloch equations

The evolution of the nuclear spin magnetization $\mathbf{M}$ in the magnetic fields is governed by the Bloch equations [1]:

$$\frac{dM_x}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_x - \frac{M_x}{T_2}\,, \tag{1.5}$$

$$\frac{dM_y}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_y - \frac{M_y}{T_2}\,, \tag{1.6}$$

$$\frac{dM_z}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_z - \frac{M_z - M_{\text{eqb}}}{T_1}\,, \tag{1.7}$$

where $\mathbf{B}_{\text{eff}}$ is the total effective field, M_{eqb} is the equilibrium magnetization, and T_1, T_2 are the longitudinal and transverse relaxation times. In most NMR experiments, the bias field $\mathbf{B}_0$ is much larger than any other applied magnetic field. Therefore, the equations are most efficiently solved in the rotating coordinate frame (RCF), which removes the fast Larmor precession at ν_L and exposes the much slower dynamics induced by the additional fields.

1.1.4 Relaxations

Relaxation describes the return of the spin ensemble toward equilibrium after it has been perturbed. Longitudinal relaxation T_1 controls how quickly the magnetization along the bias magnetic field rebuilds, while transverse relaxation determines how quickly coherent precession dephases. Two transverse relaxation times must be distinguished:

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_\delta}, \tag{1.8}$$

where T_2 characterizes irreversible spin-spin interactions and T_δ quantifies the additional dephasing caused by magnetic-field inhomogeneity across the sample volume. Because field inhomogeneity can be reduced by shimming (Chapter 3), T_2^* is experimentally controllable, but T_2 sets an intrinsic upper bound.

These time scales are central to the thesis because they set the window over which the ALP-induced signal can be observed coherently. Longer T_2^* improves sensitivity in all four signal regimes derived in Chapter 2.

T_1 relaxation

Longitudinal relaxation governs how the population difference between spin states rebuilds after perturbation. In practice, a short T_1 is helpful for rapid repetition, while a long T_1 can preserve polarization for longer measurements.

T_2^* relaxation

Transverse relaxation combines microscopic spin-spin interactions (T_2) and macroscopic field inhomogeneity (T_δ) as given in Equation (1.8). It determines the free-decay[1] signal and directly controls the linewidth in the frequency domain. For field searches, a narrower linewidth can improve sensitivity, but only if the coherence time remains long enough to integrate signal power efficiently.

1 Yuzhe: remove induction and mention there might not be any induction

Cross relaxation

Cross relaxation transfers energy between coupled spin species or spin reservoirs. In samples containing more than one spin system, it modifies the effective relaxation rates of each species.

1.1.5 Chemical shift

Chemical shielding modifies the local field at a nucleus and shifts its resonance frequency. The shift is small in absolute terms but distinguishes nuclei in different chemical environments.

In this thesis, chemical shift contributes to the spectral structure of the sample and offsets the resonance from the bare Larmor value.

1.1.6 Spin-spin coupling

Coupling between nuclear spins splits resonance lines into multiplets. The resulting J-coupling pattern provides a spectral fingerprint for molecular identification, which helps distinguish genuine NMR signals from noise.

For the calibration sample, J-coupling produces a resolved multiplet that serves as a reliable spectral reference.

1.1.7 Experimental schemes

Two pulse-based schemes are especially useful. Free induction decay measures the coherent response after a tipping pulse, while spin echo refocuses dephasing and gives access to the intrinsic transverse relaxation time. Both are diagnostic tools that help characterize the apparatus before an axion measurement.

Free decay

After a $\pi/2$ pulse, the magnetization is tipped into the transverse plane and then decays with the dephasing time T_2^*. The resulting free-induction signal is the simplest way to measure resonance frequency and linewidth.

Spin echo

Spin echo uses a π pulse to reverse dephasing from static field inhomogeneity. By recovering the signal at the echo time, it separates true spin-spin relaxation from broadening caused by imperfect field homogeneity, yielding a direct measurement of T_2.

CPMG sequence

The Carr–Purcell–Meiboom–Gill (CPMG) sequence extends the spin-echo scheme by applying a $\pi/2$ pulse followed by a train of π refocusing pulses. Each refocusing pulse produces an echo, and the decay of the echo amplitudes gives a robust measurement of T_2. In this thesis, CPMG is used as a pulsed-NMR diagnostic to characterize the sample relaxation before and after long ALP-search measurements.

1.2 Dark matter hypothesis

The prevailing cosmological framework is the Λ cold-dark-matter (ΛCDM) model. It combines a cosmological constant Λ, which describes the accelerated expansion of the universe, with a cold, nonrelativistic, and effectively collisionless dark-matter component. ΛCDM provides a successful common description of the cosmic microwave background, the growth of large-scale structure, galaxy clustering, and the expansion history. Its success establishes a powerful baseline, but it does not identify the physical nature of dark matter, and its assumptions remain less directly tested on galactic and subgalactic scales.

1.2.1 Astronomical evidence

The concept of dark matter arose from astronomical observations that could not be explained by visible matter and standard gravity alone. Galaxy rotation curves provide a classic example: orbital speeds remain approximately constant far beyond the bright stellar disk, rather than decreasing as expected if most of the mass followed the visible matter [36, 38]. This behavior indicates that galaxies are embedded in extended, non-luminous mass halos.

Evidence at larger scales comes from gravitational lensing and colliding galaxy clusters. In the Bullet Cluster, the dominant gravitational mass inferred from lensing is spatially separated from the hot X-ray-emitting gas that contains most of the baryonic mass [14]. This separation is naturally explained if most of the mass is effectively collisionless during the cluster collision. Independently, the cosmic microwave background, big-bang nucleosynthesis, and the growth of large-scale structure require substantially more gravitating matter than can be supplied by baryons. Cosmological fits assign roughly 5 % of the cosmic energy budget to baryonic matter and about 27 % to dark matter [2].

1.2.2 Recent developments

Despite the large-scale success of ΛCDM, its simplest collisionless cold-dark-matter realization faces open questions on smaller scales. Simulations and observations can differ in the predicted abundance of dwarf satellites and in the inner density profiles and diversity of low-mass halos. These tensions may reflect incompletely modeled baryonic processes, or they may indicate that the dark-matter sector is warmer, wave-like, or self-interacting rather than perfectly cold and collisionless [39].

Recent observations increasingly test these possibilities. Work involving Roberto Abraham and collaborators identified the ultra-diffuse galaxy NGC 1052–DF2 as having an unusually small dynamical dark-matter content [16]. Follow-up work proposed that this galaxy and related objects form a trail produced by a high-speed "bullet-dwarf" collision that separated stars and gas from their original dark-matter halos [17]. Such systems probe the normally tight connection between galaxies and their halos and may help clarify how baryonic and dark matter can become spatially separated. They challenge simple galaxy–halo relations, although collision-driven formation scenarios may accommodate them within the broader ΛCDM framework.

The central structures of halos provide a more direct test of the collisionless-CDM assumption. In self-interacting dark matter (SIDM) models, scattering between dark-matter particles redistributes energy and changes the central density profiles of halos. A recent strong-lensing analysis led by Simona Vegetti found an unusually compact perturber whose inferred structure, if dark-matter dominated, is difficult to reconcile with conventional cold or warm dark matter. One possible interpretation is an SIDM halo that underwent gravothermal core collapse and formed a central black hole [39]. This interpretation is not yet unique, but it illustrates how high-resolution lensing can distinguish between dark-matter microphysics.

1.2.3 Candidate landscape

The observations establish the gravitational role of dark matter but do not identify its microscopic nature. In this sense, ΛCDM specifies the required large-scale behavior of the dominant matter component rather than a unique particle. Proposed realizations cover an enormous range of mass and behavior, and several alternatives modify the small-scale predictions of minimal cold dark matter. At the low-mass end, fuzzy dark matter consists of bosons with masses around 10^{-22} eV; their kiloparsec-scale de Broglie wavelengths make wave effects relevant to galactic structure [27]. Other particle candidates include sterile neutrinos, warm dark matter, weakly interacting massive particles, and self-interacting particles. At the opposite, macroscopic end, primordial black holes formed in the early universe could contribute some or all of the dark matter within the remaining observationally allowed mass windows [12]. Laboratory searches consequently range from particle-collision and scattering experiments to gravitational surveys and precision measurements [28].

Axions and axionlike particles occupy a particularly interesting part of this landscape. Their low masses permit a coherent, wave-like dark-matter field, while their weak non-gravitational couplings make controlled laboratory searches possible. The following section introduces their origin, field properties, and coupling to nuclear spins.

1.3 Axions and axionlike particles

The QCD axion was originally proposed in 1977 by Peccei and Quinn as a solution to the strong *CP* problem [34, 35]. A global symmetry breaking at the scale f_a generates a pseudo-Goldstone boson whose potential is minimized at the *CP*-conserving point, naturally explaining why the strong interaction conserves *CP*. The axion acquires a mass $m_a \sim \Lambda_{\mathrm{QCD}}^2/f_a$ from QCD dynamics.

1.3.1 QCD axion

The QCD axion is a hypothetical pseudoscalar particle that appears when a global symmetry is introduced to explain why strong interactions conserve *CP* so well. Its properties are tightly linked to the QCD scale, which makes it a special case rather than the most general one. Cosmological production via the misalignment mechanism constrains $f_a \lesssim 10^{12}$ GeV (axion mass $\gtrsim 6\,\mu$eV) if the axion is to constitute all dark matter [25]. Yuzhe: check this paragraph by reading literatures.

1.3.2 Axionlike particles

Axion-like-particles (ALPs) emerge when a global symmetry is broken [15, 30, 37, 42, 43, 48]. ALPs share the oscillatory and weakly coupled character of the QCD axion, but their mass range can be much wider, since the mechanism of ALP production is different from QCD axion. There have been investigations in determining the ALP mass range from astronomical observations [31] and theoretical studies [48]. Yuzhe: check the literature However, in this thesis, the ALP mass is regarded as a free parameter, and the search is performed in a range which the experiment is sensitive to.

Wave nature

Yuzhe: check if speed of light c is defined. A nonrelativistic ALP with mass m_a and velocity $\mathbf{v}_a$ ($v_a \ll c$) has energy $E_a \simeq m_a c^2 + \frac{1}{2} m_a v_a^2$, ordinary frequency $\nu_a = E_a/h$, and wave vector $\mathbf{k} = m_a \mathbf{v}_a/\hbar$. Its de Broglie wavelength is

$$\lambda_{\mathrm{dB}} = \frac{h}{m_a v_a} \,. \tag{1.9}$$

For an ultralight ALP this wavelength can be macroscopic. Assuming axionlike dark matter is cold, its velocity is expected to be smaller than the escape velocity of the Milky Way halo $v_{\mathrm{esc}} = 550\,\mathrm{km/s}$ [20]. The minimum de Broglie wavelength is $\lambda_{\mathrm{dB}} \sim 7 \times 10^{-5}$ m for ALPs in the Milky Way ($m_a \leq 10\,\mathrm{eV/c^2}$) [2]. For the ALP mass range relevant to this thesis ($m_a \leq 2.5\,\mu\mathrm{eV}$), the de Broglie wavelength $\lambda_{\mathrm{dB}} \geq 270$ m, much larger than the size of the laboratory. Therefore, the amplitude of the ALP field can be approximated as uniform over the experimental volume.

The ALP field of a single free ALP mode can be written as a real field:

$$a(\mathbf{r}, t) = \sqrt{\frac{\hbar^3 c}{2m_a}} \left(\Psi(\mathbf{r}, t) + \Psi^*(\mathbf{r}, t)\right) \tag{1.10}$$

$$= \sqrt{\frac{\hbar^3 c}{m_a}} \, \Re\left[\Psi(\mathbf{r}, t)\right] , \tag{1.11}$$

where Ψ is wavefunction of the ALP. Consider an ensemble of N_a ALPs with local energy density ρ_a. Its number density is

$$n_a = \frac{\rho_a}{m_a c^2} \,, \tag{1.12}$$

and the mean occupation number within a de Broglie volume is approximately

$$\mathcal{N} \simeq n_a \lambda_{\mathrm{dB}}^3 = \frac{\rho_a h^3}{m_a^4 v_a^3 c^2} \,. \tag{1.13}$$

2 dB-wavelength.ipynb

Yuzhe: check if dm density has been mentioned before If the ALP is the dominant dark-matter component, ρ_a can be replaced by the local dark-matter density $\rho_{\mathrm{DM}} \sim 0.4\,\mathrm{GeV/cm^3}$ [20]. Considering ALP mass relevant to this thesis $m_a \leq 2.5\,\mu\mathrm{eV}$ and velocity $v_a \leq v_{\mathrm{esc}} = 550\,\mathrm{km/s}$, the occupation number is $\mathcal{N} \geq 3\times10^{27}$, much larger than unity [3]. Therefore, what is measured in the experiment is the collective effect of many ALPs instead of rare single-particle events.

For N_a ALPs described by a wavefunction normalized by $\int |\Psi|^2\, d^3r = 1$, the associated scalar field is the appropriately scaled real part of the wavefunction:

$$a(\mathbf{r},t) = \sqrt{\frac{2N_a\hbar^3 c}{m_a}}\,\Re[\Psi(\mathbf{r},t)]\,. \tag{1.14}$$

The local, time-averaged energy density is $\rho_a = m_a c^2 N_a \langle|\Psi(\mathbf{r},t)|^2\rangle_t$. Eliminating N_a gives

$$\begin{aligned} a(\mathbf{r},t) &= \sqrt{\frac{2\hbar^3\rho_a}{m_a^2 c}}\,\frac{\Re[\Psi(\mathbf{r},t)]}{\langle|\Psi(\mathbf{r},t)|^2\rangle_t^{1/2}} \\ &= a_0\,\frac{\Re[\Psi(\mathbf{r},t)]}{\langle|\Psi(\mathbf{r},t)|^2\rangle_t^{1/2}}\,. \end{aligned} \tag{1.15}$$

Thus, the local field amplitude is

$$a_0 = \sqrt{\frac{2\rho_a\hbar^3}{m_a^2 c}}\,. \tag{1.16}$$

If all N_a ALPs occupy the same mode, the corresponding local field takes the plane-wave form

$$a(\mathbf{r},t) = a_0\cos(2\pi\nu_a t - \mathbf{k}\cdot\mathbf{r} + \phi)\,, \tag{1.17}$$

over a region where the amplitude can be treated as uniform. A realistic ALP wavefunction generally contains many modes and may be written as

$$\Psi(\mathbf{r},t) = \sum_j C_j \exp\!\big[-i\big(2\pi\nu_j t - \mathbf{k}_j\cdot\mathbf{r} + \phi_j\big)\big]\,. \tag{1.18}$$

The coefficients C_j encode the populations of the modes and are fixed by the normalization of Ψ. The physical scalar field remains proportional to $\Re[\Psi]$ according

3 dB-wavelength.ipynb

to Equation (1.15). However, if ALPs occupy many modes, the resulting field is a superposition of those modes. The interference of many modes with different velocities makes the field stochastic. The stochasticity is characterized by the coherence time. Chapter 2 develops this behavior for thermalized Milky Way halo Earth-bound halo.

1.3.3 Non-gravitational interactions

An oscillating ALP field can produce three spin-sensitive non-gravitational interactions: its coupling to gluons induces oscillating *CP*-odd nuclear moments such as a nuclear electric dipole moment, while its derivative couplings to nucleons and electrons act as oscillating axial fields on nuclear and electron spins, respectively [22]. All three effects oscillate at a frequency set primarily by the ALP mass and can therefore be sought with resonant precision measurements.

This thesis focuses on the derivative coupling to nucleons, commonly called the gradient or "axion-wind" coupling. The nucleon Hamiltonian in the presence of an ALP field background can be written as

$$H_{\mathrm{aNN}} = c g_{\mathrm{aNN}}\, \boldsymbol{\nabla} a \cdot \mathbf{I}\,, \tag{1.19}$$

where g_{aNN} is the coupling strength in GeV^{-1} and $\mathbf{I}$ is the nuclear spin operator. In this form of Hamiltonian, $g_{\mathrm{aNN}} a_0$ becomes dimensionless. The amplitude a_0 is fixed by the local dark-matter density through Equation (1.16) and has dimensions of energy [8, 29, 38].

In some papers, nuclear spin operator $\mathbf{I}$ is substituted with $\hbar\boldsymbol{\sigma}_n$, where $\boldsymbol{\sigma}_n$ is dimensionless nuclear spin operator. Expanded:

$$H_{\mathrm{aNN}} = c g_{\mathrm{aNN}} a_0 \sin(2\pi\nu_a t - \mathbf{k}\cdot\mathbf{r} + \phi) m_a\, \mathbf{v}_a \cdot \mathbf{I}/\hbar\,, \tag{1.20}$$

Comparing this with the NMR Hamiltonian $H_{\mathrm{NMR}} = \gamma \mathbf{B}\cdot\mathbf{I}$, the ALP gradient acts as an effective oscillating pseudomagnetic field

$$\mathbf{B}_a(t) = \frac{c g_{\mathrm{aNN}}\, \boldsymbol{\nabla} a}{\gamma}\,. \tag{1.21}$$

This gradient coupling oscillates at the ALP Compton frequency and can drive resonant spin precession when the Larmor frequency matches ν_a. The search for this coupling is the subject of Chapter 4. Yuzhe: add some examples to show equivalent Ba field magnitude?

The r.m.s. Rabi frequency of the pseudomagnetic field can be found by the Hamiltonian in Equation (1.20) [13, 47]

$$\Omega_a = \frac{1}{2\hbar} c g_{\mathrm{aNN}} a_0 m_a v_\perp = \frac{1}{2} \mathrm{g}_{\mathrm{aNN}} \sqrt{2\hbar c \rho_{DM}} v_\perp \,, \tag{1.22}$$

where $v_\perp$ is the component of the ALP-detector relative velocity perpendicular to $\mathbf{B}_0$. Taking $v_\perp = 220\,\mathrm{km\,s^{-1}}$, we have $\Omega_a/2\pi = g_{\mathrm{aNN}} \times 0.2\,\mathrm{GeV \cdot Hz}$. For $g_{\mathrm{aNN}} \lesssim 10^{-10}\,\mathrm{GeV^{-1}}$, the Rabi period $2\pi/\Omega_a \gtrsim 1000\,\mathrm{yr}$, confirming that the ALP field is a weak drive on the spins. Yuzhe: What about that factor of one half? Check the values! Mention the weak-drive limit.

1.4 CASPEr project

Yuzhe: first mention a lot of magnetic resonance searches (haloscope). [28]

CASPEr (Cosmic Axion Spin Precession Experiment) is a family of experiments proposed in 2014 [11] that use nuclear spins to search for axion-induced effects. The idea is to translate the tiny axion coupling in Equation (1.21) into a measurable spin precession in a carefully controlled NMR setup.

The project has several branches, including electric and gradient-based searches, and both low-field and high-field implementations. The low-field regime is especially relevant when the goal is to scan a wide frequency band with high sensitivity to weak couplings.

1.4.1 Idea

The central idea is to tune the NMR resonance to a frequency that matches the axion Compton frequency. If the two frequencies align, the axion field acts as the pseudomagnetic field $\mathbf{B}_a(t)$ in Equation (1.21) and induces a coherent transverse magnetization in the sample.

In practice, the method reduces to a frequency scan: either the signal appears at a particular resonance or the absence of a signal constrains the coupling strength over the scanned band.

1.4.2 CASPEr-Electric

Yuzhe: shoten these subsections CASPEr-Electric (CASPEr-E) searches for an oscillating nuclear electric dipole moment induced by the ALP-gluon coupling. This moment generates an effective oscillating field along the nuclear spin axis in the presence of an electric field, enabling detection through NMR at the corresponding Compton frequency.

1.4.3 CASPEr comagnetometers

Comagnetometer configurations use two nuclear spin species in the same volume to suppress common-mode noise from magnetic-field fluctuations. The differential signal between the two species enhances sensitivity to a pseudomagnetic ALP field while rejecting electromagnetic backgrounds. Early results were obtained with a liquid-state comagnetometer [44] and an ultralow-field configuration [21].

1.4.4 CASPEr-Gradient

CASPEr-Gradient searches for the ALP-fermion gradient coupling described by Equation (1.19). Several setups at Mainz cover different ALP-mass ranges. The CASPEr-Gradient-Low-Field experiment uses bias fields up to about 0.1 T, corresponding to proton Larmor frequencies up to approximately 4.3 MHz. The CASPEr-Gradient-High-Field experiment will use a tunable 14.1 T magnet to reach frequencies up to 600 MHz. A conceptual diagram of the low-field apparatus is shown in Figure 1.1. The cryostat contains a liquid helium reservoir, excitation and pickup coils, and a SQUID magnetometer. A superconducting magnet produces the bias field $\mathbf{B}_0$, which sets the Larmor frequency and can be varied to scan ALP masses.

The resonance scheme is the main search strategy for CASPEr-Gradient-Low-Field. The bias field is stepped through a range of values, and the NMR response is recorded at each step. A resonant signal appears as a peak in the transverse magnetization when ν_L matches ν_a. The broadband scheme, which is more suitable for higher frequencies, continuously monitors the NMR response without stepping the bias field, looking for transient signals that could indicate an ALP interaction.

1.4.5 Candidate, exclusion, limit, and sensitivity

Exclusion and sensitivity in the mass-coupling (Yuzhe: what is mass coupling space? explain in advance) parameter space are often reported as one of the results of

1.4.6 Scientific targets beyond gradient coupling

Yuzhe: Mention what will be included. Will it appear in the coming chapters?

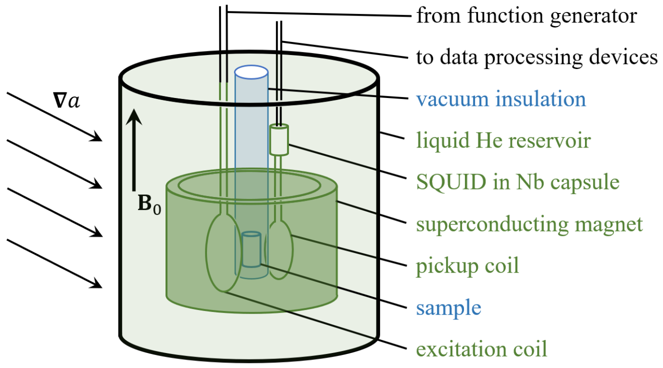

Figure 1.1: Conceptual diagram of the CASPEr-Gradient-Low-Field setup. Superconducting devices (dark green) are submerged in a liquid helium bath (light green). The excitation and pickup coils characterize the sample and detect the spin-precession signal. The sample temperature can be controlled independently of the helium bath, enabling measurements with both thermal and hyperpolarized samples. Reproduced from Ref. [47].

2 ALP models

Part of the content in this chapter related to frequency-scanning considerations is adapted from Earth bound halo paper Yuzhe: to be cited here. Yuzhe: modify?

In this chapter, two models of ALP dark matter are discussed, including the Milky Way halo model and the Earth-bound halo model. The Milky Way halo model is based on the standard halo model (SHM) of dark matter, which assumes that the dark matter in the Milky Way is virialized and follows a Maxwell-Boltzmann velocity distribution. The Earth-bound halo model, on the other hand, considers the possibility of a local overdensity of ALP dark matter bound to the Earth, which could lead to different experimental signatures.

2.1 Milky Way ALP DM halo

2.1.1 Virialized dark matter halo

In the standard halo model (SHM) [20], the Milky Way dark matter population is treated as a virialized [4] ensemble. In the Galactic rest frame, the dark-matter velocity distribution is approximated by a Maxwell–Boltzmann distribution truncated near the Galactic escape speed [20].[5] The average speed of local dark matter[6] is $v_{\mathrm{DM}} \approx 220\,\mathrm{km\,s^{-1}}$. The local dark matter density is estimated to be $\rho_{DM} \approx 0.4\,\mathrm{GeV/cm^3}$. Yuzhe: check if rms is written in a consistent way The r.m.s. Rabi frequency Ω_a can be computed from Equations (1.17), (1.16), and (1.19) [13]:

$$\Omega_a = \frac{1}{2\hbar} g_{\mathrm{aNN}} a_0 m_a c v_\perp \approx \frac{1}{2} g_{\mathrm{aNN}} \sqrt{2\hbar c \rho_{DM}}\, v_\perp\,, \tag{2.1}$$

Here it is assumed that the ALP-field gradient comes exclusively from the relative motion of the field and the detector. It is important to note that the ALP field is a stochastically varying quantity, so that the Ω_a in Equation (1.22) is the r.m.s value. For $v_\perp = 220\,\mathrm{km\,s^{-1}}$, $\Omega_a/2\pi = g_{\mathrm{aNN}} \times 0.2\,\mathrm{GeV}\cdot\mathrm{Hz}$. For an experiment searching for $g_{\mathrm{aNN}} \leq 10^{-10}\,\mathrm{GeV^{-1}}$, the Rabi period is $2\pi/\Omega_a \geq 1000\,\mathrm{yr}$; the ALP-field gradient is therefore a weak drive on the spins.

Though it is named “standard” halo model, it is not a perfect description of the Milky Way dark matter halo. It is more of a model built on phenomenological observations, rather than a model tested through direct measurements. Not every question about the Milky Way halo can be answered by the SHM. For example, the SHM does not account for the presence of substructure in galaxies. There have been alternative models or modifications to the SHM, including SHM^{++} [20], caustic rings [7], and the presence of substructure in the form of streams [8]. This thesis focuses on the SHM for simplicity. In the future, CASPEr can explore more complex models of the dark matter halo.

4 The term “virialized” means that the dark matter is gravitationally bound.
5 The velocity dispersion characterizes the width of the distribution; it is not another name for an individual velocity.
6 Here, “local” refers to the solar radius in the Milky Way.
7 $Sikiviecausticrings$
8 $Freese2004_Streams$, $Savage2006_Streams$, $Kuhlen2010_Streams$, $Lisanti2011_Streams$

2.1.2 Stochasticity due to inteference

If the ALP constitutes the dominant component of the Milky Way dark matter, the local ALP field is a superposition of many modes with different velocities and phases. Here the phases of the occupied modes are assumed to be uncorrelated. Each velocity class contributes one term to the multimode field in Equation (1.18), with

$$\nu(\mathbf{v}) = \frac{m_a c^2}{h} + \frac{m_a v^2}{2h}, \qquad \mathbf{k}(\mathbf{v}) = \frac{m_a \mathbf{v}}{\hbar} \,. \tag{2.2}$$

The relative spread of velocities is on the order of $v_a/c \sim 10^{-3}$. Due to the second-order Doppler effect, the relative spread of kinetic energies is on the order of $(v_a/c)^2 \sim 10^{-6}$. The inverse of this spread is defined as the quality factor of the ALP field, $Q_a = (c/v_a)^2 \sim 10^6$. Therefore, in the frequency domain, the ALP field is a narrowband signal with a fractional linewidth $\Delta\nu_a/\nu_a \sim 10^{-6}$. Considering the Maxwell–Boltzmann velocity distribution, the sum of the modes as shown in (1.18) is a random walk in the complex Ψ plane: the two quadratures approach Gaussian random variables, while the field (average) envelope follows a Rayleigh distribution [13, 24].

Consequently, in the time domain, the ALP field is a coherent oscillation only on times short compared with its coherence time; on longer time scales, the amplitude and phase drift stochastically, due to the spread of frequencies in the multimode superposition. The ALP coherence time τ_a can be estimated by the quality factor:

$$\tau_a = \frac{2\pi Q_a}{\nu_a} \,. \tag{2.3}$$

One example of stochasitc AlP field in time and frequency domains is shown in Figure 2.1.

Yuzhe: should I write a lot about this? a lot of the jobs are done at Gramolin's paper.

2.1.3 Periodic modulations

CASPEr-G is only sensitive to the component of $\boldsymbol{\nabla} a$ perpendicular to its sensitive axis. In the resonance scheme, the sensitive axis is aligned with the bias field $\mathbf{B}_0$. Since the bias field is always veritcal with repect to the ground, the setup is sensitive to gradient along $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\varphi}}$ directions, but not to the gradient along $\hat{\mathbf{r}}$ direction.

The ALP-field gradient driving spin precession depends on the direction of the relative velocity between the Earth and the local dark-matter halo. As the Earth

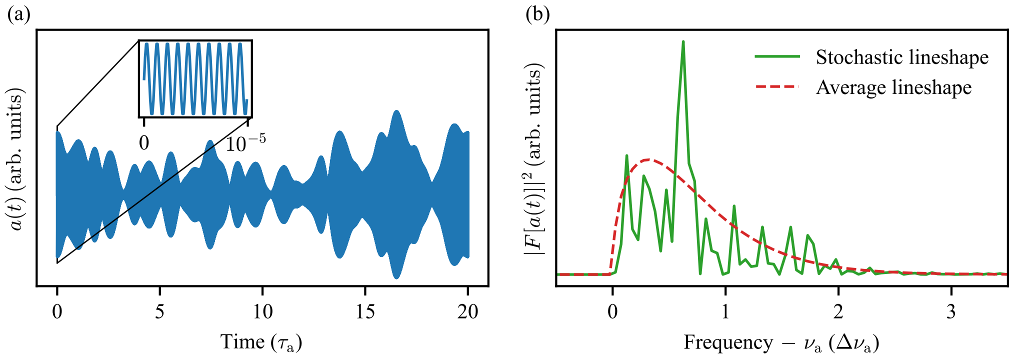

Figure 2.1: Simulated axion fields in the time and frequency domains. (a) Time-domain axion field $a(t)$, with a magnified view showing the oscillation at axion Compton frequency in the inset. The envelope of the $a(t)$ changes over time, indicating the coherence time τ_a is finite. Here τ_a is chosen as Q_a/ν_a. (b) The stochastic lineshape in the frequency domain is computed as $|F[a(t)]|^2$, where $F[a(t)]$ denotes the Fourier transform of the time-domain signal shown in (a). The average lineshape represents the expected amplitude spectrum after sufficient averaging. The linewidth $\Delta\nu_a$ is approximated as ν_a/Q_a. The figure is generated by the script scripts-and-data/MW-axion-time-frequency-domain.py. Reproduced from Ref. [5] with permission.

rotates, this direction sweeps through space with a period of one sidereal day. An ALP signal therefore exhibits a daily modulation of its amplitude and phase [24].

In addition to daily rotation, the Earth's orbital velocity around the Sun modulates the relative ALP-detector velocity with a period of one year. Both the daily and annual modulations have precisely predictable periods and phases, making them useful consistency checks for any claimed dark-matter-candidate signal. Examples of daily and annual modulations are shown in Figure 2.2, reproduced from Ref. [24].

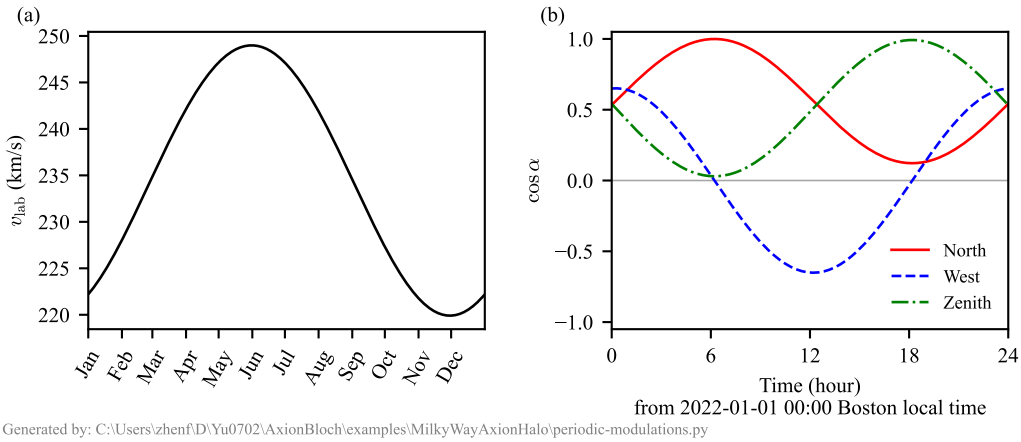

Figure 2.2: Periodic (annual and daily) modulations reproduced from Ref. [24]. (a) Annual modulation of laboratory speed due to the Earth's orbital motion around the Sun. (b) Daily modulations of angles between the ALP field and the three orthogonal orientations: towards the north, the west, and the zenith. The location is the Boston University and the date is January 1 (local time). Yuzhe: can we use figure width 6 or 6.5 for making font larger? TODO

2.2 Earth-bound ALPs

The galactic dark-matter halo provides the primary ALP signal target discussed in Chapter 4. However, a distinct and potentially enhanced source arises from ALPs that are gravitationally bound to the Earth. Such Earth-bound states are analogous to atomic bound states—the Earth's gravitational potential plays the role of the Coulomb well and the ALP plays the role of the electron—and can support a discrete spectrum of spatial eigenstates. This section derives the properties of these bound states, estimates the resulting spin-precession signal, and discusses the characteristic daily modulation that distinguishes this signal from the galactic background.

2.2.1 Trapping and accumulation of dark matter

For Earth-bound states to be populated, ALPs from the galactic halo must lose enough kinetic energy to fall below the local escape velocity ($v_{\mathrm{esc}} \approx 11.2\,\mathrm{km\,s^{-1}}$ for Earth). Several mechanisms have been proposed Yuzhe: cite: Banerjee et al. 2019 arXiv:1902.07720; OHare 2025 fine-grained; other relevant papers:

Gravitational three-body scattering A galactic ALP passing near the Earth can scatter off a third body (e.g. the Sun or Moon), losing kinetic energy and becoming trapped in a bound orbit. The rate depends on the local density and the ALP-body cross section.

Self-interaction or wave-packet spreading Yuzhe: placeholder: if ALPs have self-interactions or if wave-packet decoherence provides an effective dissipation, this may also populate bound states.

Resonant capture Near specific orbital resonances the exchange of energy between the ALP and the Earth-Moon or Earth-Sun system may be efficient.

The total mass M_{bound} of ALPs trapped in Earth-bound states is strongly constrained by astrophysical and geophysical observations Yuzhe: leave the plot here. cite and expand: relevant constraints on exotic mass inside Earth from geophysics / seismology / gravitational measurements. An upper bound comes from the requirement that the Earth's measured gravitational multipole moments not be distorted beyond current geodetic precision. Yuzhe: Add quantitative upper limit on M_{bound} once reference is identified. Yuzhe: We note that.... skip the mechanism of capturing... Yuzhe: check if max dm mass is mentioned

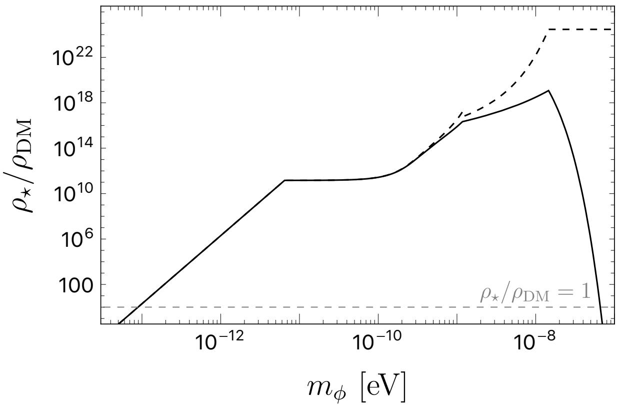

Figure 2.3: Density Distribution of ground state Earth-bound ALP halo. The solid line indicates the density at the Earth surface, whereas the dashed line represents the density at the center of the halo. The density value $\rho_\star$ is derived from the maximum Earth ALP halo mass permitted by gravitational constraints, as assessed in [6, 7].Yuzhe: discard this pic?

2.2.2 Gravitational potential and the quantum picture

To simplify the calculation, Earth is considered as a spherical object. The density of Earth varys over radii. From geophysical measurements on earthquake waves, Preliminary reference Earth model (PREM)[18] was constructed, giving the Earth density profile as shown in Figure 2.4.

The gravitational potential of Earth V_g/m per unit mass can be found by

$$V_g(r)/m = -\frac{GM_{\rm enclosed}}{r} , \tag{2.4}$$

Yuzhe: check if we are wrong by a constant or so. where r is the distance measured from the Earth's center, G is Newton's constant, and $M_{\rm enclosed}$ is the Earth's mass enclosed by r. Here we take the infinity as the reference point and have potential $V_g(\infty)/m = 0\,\mathrm{J\,kg^{-1}}$. The potential over radii can be calculated numerically based on PREM (Figure 2.4). The Earth escape velocity can therefore be found by

$$v_{\rm escape} = \sqrt{\frac{2GM_\oplus}{R_\oplus}} , \tag{2.5}$$

where $M_\oplus = 6 \times 10^{24}\,\mathrm{kg}$ and $R_\oplus = 6378.1\,\mathrm{km}$ are the mass and average radius of Earth, respectively. By plugging the values, the Earth escape velocity is found

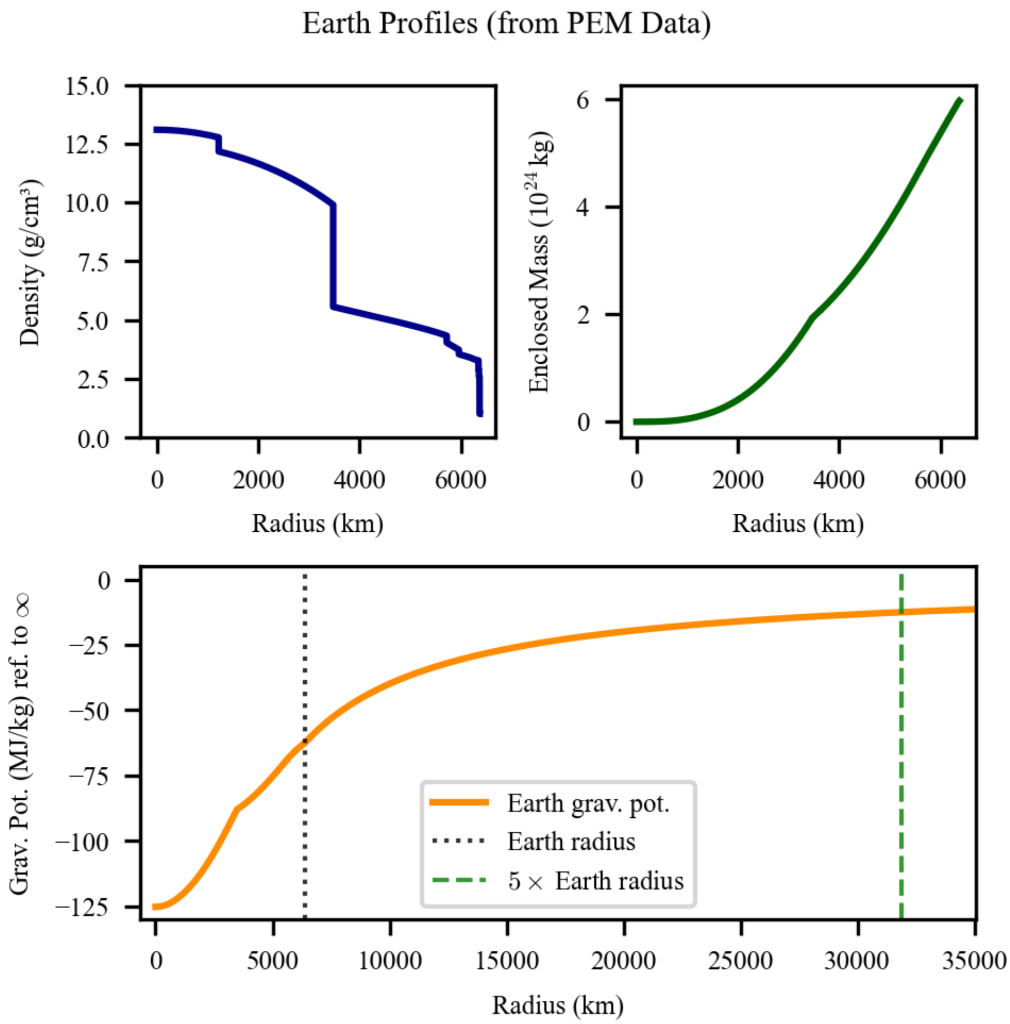

Figure 2.4: to be modified later.

to be 11.2 km/s. Since the escape speed is much smaller than the speed of light, non-relativistic limit is assumed for the discussion.

For ultralight dark matter particles bound by Earth, they can no longer be treated as classic objects due to their large de Broglie wavelengths. A rough estimation of the wavelength is made as follows. The ALP masses that CASPEr-G can probe range from ~ 4 peV to 2.5 μeV. Consider ALPs trapped by the earth potential with velocities roughly smaller than the escape veclocity. The de Broglie wavelength as a function of particle mass $\lambda_{\mathrm{dB}}(m_a, v_a) = \frac{h}{m_a v_a}$ gives

- $\lambda_{\mathrm{dB}}(4\,\mathrm{peV}, 11.2\,\mathrm{km/s}) \approx 8.3\times10^{9}\,\mathrm{m} \approx 1300R_{\oplus}$ and
- $\lambda_{\mathrm{dB}}(2.5\,\mu\mathrm{eV}, 11.2\,\mathrm{km/s}) \approx 13\,\mathrm{km} \approx 0.0021R_{\oplus}$.

Therefore, across the scan range of CASPEr, the de Broglie wavelength spans values from much smaller to much larger than the Earth radius. In later chapters, a measurement probing MHz ALPs is presented. This corresponds to a de Broglie wavelength ~ 1 $\mathrm{R}_{\oplus}$. To prepare for the analysis, we focus on this scenario and take the wave nature into account. Yuzhe: mention Na here.

2.2.3 Bound-state wavefunctions

The general relation between an ALP wavefunction, its local energy density, and the real scalar field was established in Equations (1.14) and (1.15). For the Earth-bound model, the wavefunction is expanded in the discrete gravitational eigenstates:

$$\Psi(\mathbf{r}, t) = e^{-im_a c^2 t/\hbar} \sum_{nlm} c_{nlm} e^{-iE_n t/\hbar} \psi_{nlm}(\mathbf{r}) \,. \tag{2.6}$$

Here $\Psi(\mathbf{r}, t)$ is the wavefunction, ψ_{nlm} and E_n are the eigenfunction and energy of the state (n, l, m), and c_{nlm} are expansion coefficients determined by the state populations. The total particle number N_a enters through the density below. The fast oscillation at ALP Compton frequency is indicated by $e^{-im_a c^2 t/\hbar}$, where the small variation of oscillation frequency is indicated by $e^{-iE_n t/\hbar}$. The corresponding local density, $\rho^E_{DM}(\mathbf{r}) = m_a c^2 N_a \langle|\Psi(\mathbf{r}, t)|^2\rangle_t$, fixes the field amplitude through Equation (1.15). The remaining task is therefore to determine the eigenfunctions, eigenenergies, and state populations in the Earth's potential.

In the non-relativistic limit, the eigenstates can be found by the time-independent Schrödinger equation (TISE)

$$\left[-\frac{\hbar^2}{2m_a}\nabla^2 + V_g(r)\right]\psi_{nlm}(\mathbf{r}) = E_n\, \psi_{nlm}(\mathbf{r}) \,. \tag{2.7}$$

In analogy to the hydrogen atom, the ALP-Earth system is a gravitational atom Yuzhe: cite: relevant paper on gravitational atom, e.g. Arvanitaki et al. and more.... Yuzhe: what about quantum number m? zero?

Considering the spherical symmetry of the Earth gravitational potential, spherical coordinate system (r, θ, φ) is adopted, where r is the radial distance from the origin, θ is the polar angle, and φ is the azimuthal angle. In spherical coordinates,

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\,, \tag{2.8}$$

so that TISE can be rewritten as

$$\left[-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}\right) + V_g(r)\right]\psi_{nlm}(r,\ \theta,\varphi) = E_n\psi_{nlm}(r,\ \theta,\varphi)\,, \tag{2.9}$$

where $\hat{L}$ is the angular momentum operator. The eigen-function can be separated as

$$\psi_{nlm}(\mathbf{r}) = R_{nl}(r)\, Y_l^m(\hat{\mathbf{r}})\,, \tag{2.10}$$

where R_{nl} is the radial wavefunction and Y_l^m is the spherical harmonic. The spherical harmonics satisfies

$$\hat{L}^2 Y(\theta,\ \varphi) = l(l+1)\hbar^2 Y(\theta,\varphi)\,. \tag{2.11}$$

Therefore, to derive the radial wavefunction of eigen-state with angular quantum number l, TISE is transformed to

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial R_{nl}(r)}{\partial r}\right] + \left[\frac{l(l+1)\hbar^2}{2mr^2} + V_g\right]R_{nl}(r) = E_n R_{nl}(r)\,. \tag{2.12}$$

Since the earth density is not uniform, it is difficult to derive the radial wavefunction analytically. Hence, numerical method is adopted. In a discretized space, a matrix for the Hamiltonian in Equation (2.12) is constructed. The eigen-values (eigen-energies) and eigen-vectors (eigen-states) can be derived using Python package `SciPy`. This method is implemented in the Python class `EarthBoundAxionHalo` of `axionbloch`.

Note that normalization is necessary for deriving the right magnitude of the eigen-functions. The wavefunctions are normalized so that the total probability

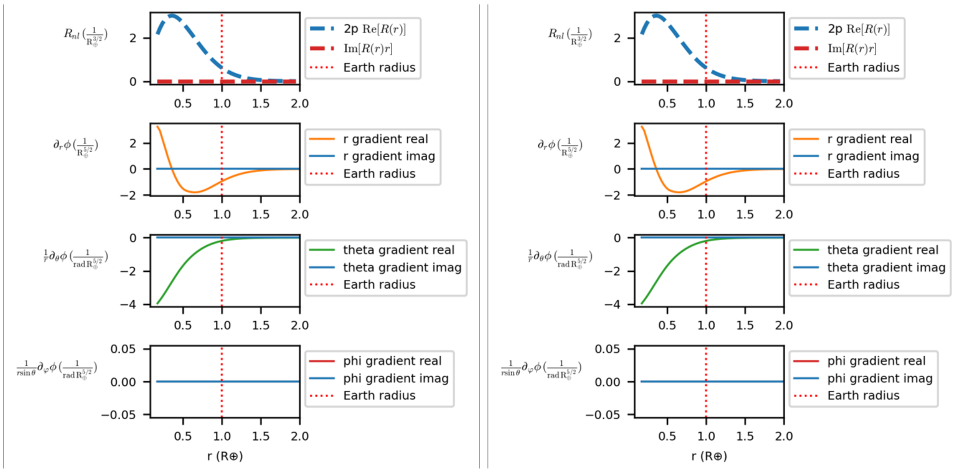

Figure 2.5: radial wavefunctions of 1s 2p... states. include higher states and mention that we are only interested in low energy states.

integrates to unity,

$$\int |\psi_{nlm}(\mathbf{r})|^2 \, d^3r = \int_0^\infty |R_{nl}(r)|^2 \, r^2 \, dr \int |Y_l^m(\hat{\mathbf{r}})|^2 \, d\Omega = 1 \,. \tag{2.13}$$

Here Ω is solid angle coordinate. Examples of normalized wavefunctions are illustrated in Figure 2.5. Yuzhe: eigen states plotted here:

2.2.4 ALP field gradient

The ALP field gradient at the detector can be derived from the spatial gradient of the wavefunction ψ_{nlm} evaluated at the detector position. The gradient operator in spherical coordinates is

$$\boldsymbol{\nabla} = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\boldsymbol{\varphi}} \frac{1}{r \sin\theta} \frac{\partial}{\partial \varphi} \,. \tag{2.14}$$

The eigenstates are defined in the Earth-rest frame, whereas the gradient coupling is measured in the laboratory frame L. Since the ALP field is a Lorentz scalar, $a'(x') = a(x)$, while its derivatives transform according to the inverse Lorentz transformation. Applying the chain rule, $\partial'_\mu = (\Lambda^{-1})^\nu{}_\mu \partial_\nu$, mixes the temporal derivative with the component of the spatial gradient parallel to the laboratory velocity. Here $\Lambda^\mu{}_\nu$ is the 4×4 Lorentz-boost matrix relating the Earth-rest-frame coordinates $x^\mu = (ct, \mathbf{r})$ to the laboratory-frame coordinates through $x'^\mu = \Lambda^\mu{}_\nu x^\nu$.

Its inverse appears because the derivative basis transforms oppositely to the coordinates; the Greek indices run over the temporal and three spatial components. To obtain the spatial transformation explicitly, the unboosted gradient is decomposed into components parallel and perpendicular to $\mathbf{v}$:

$$\nabla_{\parallel} a = \frac{\mathbf{v}}{v^2}(\mathbf{v} \cdot \nabla a), \qquad \nabla_{\perp} a = \nabla a - \nabla_{\parallel} a\,. \tag{2.15}$$

The inverse Lorentz boost leaves the perpendicular component unchanged and transforms the parallel component as

$$(\nabla_L a)_{\perp} = \nabla_{\perp} a, \qquad (\nabla_L a)_{\parallel} = \gamma_L \left(\nabla_{\parallel} a + \frac{\mathbf{v}}{c^2} \partial_t a \right). \tag{2.16}$$

Adding these two components gives

$$\nabla_L a = \nabla a + \frac{\gamma_L - 1}{v^2} \mathbf{v}(\mathbf{v} \cdot \nabla a) + \gamma_L \frac{\mathbf{v}}{c^2} \partial_t a\,, \tag{2.17}$$

where $\mathbf{v}$ is the velocity of the laboratory relative to the Earth-rest frame and $\gamma_L = (1 - v^2/c^2)^{-1/2}$. In the limit $v/c \longrightarrow 0$, the laboratory and Earth-rest frames coincide. Although the second term in Equation (2.17) contains v^2 in the denominator, $\gamma_L - 1 \propto v^2/c^2$, and the complete term has the regular limit

$$\lim_{v \to 0} \frac{\gamma_L - 1}{v^2} \mathbf{v}(\mathbf{v} \cdot \nabla a) = 0\,. \tag{2.18}$$

The temporal-gradient term also vanishes, so Equation (2.17) reduces to

$$\nabla_L a = \nabla a \qquad (v = 0)\,, \tag{2.19}$$

as expected.

For $0 < v/c \ll 1$, the Lorentz factor can be expanded as

$$\gamma_L = 1 + \frac{1}{2}\frac{v^2}{c^2} + O\!\left(\frac{v^4}{c^4}\right), \qquad \frac{\gamma_L - 1}{v^2} = \frac{1}{2c^2} + O\!\left(\frac{v^2}{c^4}\right). \tag{2.20}$$

Consequently,

$$\nabla_L a = \nabla a + \frac{\mathbf{v}}{c^2} \partial_t a + \frac{\mathbf{v}}{2c^2}(\mathbf{v} \cdot \nabla a) + \cdots\,. \tag{2.21}$$

To first order in v/c, this becomes

$$\nabla_L a \simeq \nabla a + \frac{\mathbf{v}}{c^2} \partial_t a\,. \tag{2.22}$$

The condition $v/c \ll 1$ does not necessarily make the temporal-gradient contribution negligible because $\partial_t a$ contains the rapid ALP Compton oscillation. It is the laboratory-frame gradient $\boldsymbol{\nabla}_L a$ that enters the interaction Hamiltonian in Equation (1.19). The unboosted gradient can be inferred from Equations (1.14) and (2.6) as

$$\boldsymbol{\nabla} a(\mathbf{r}, t) = \sqrt{\frac{N_a \hbar^3 c}{2m_a}}\,\Re\left[\boldsymbol{\nabla}\Psi(\mathbf{r}, t)\right] \tag{2.23}$$

$$= \sqrt{\frac{N_a \hbar^3 c}{2m_a}}\,\Re\left[e^{-im_a c^2 t/\hbar} \sum_{nlm} c_{nlm} e^{-iE_n t/\hbar} \boldsymbol{\nabla}\psi_{nlm}(\mathbf{r})\right], \tag{2.24}$$

where

$$\boldsymbol{\nabla}\psi_{nlm}(\mathbf{r}) = \hat{\mathbf{r}}\,\frac{\partial R_{nl}}{\partial r} Y_l^m + \hat{\boldsymbol{\theta}}\,\frac{R_{nl}}{r}\frac{\partial Y_l^m}{\partial \theta} + \hat{\boldsymbol{\varphi}}\,\frac{R_{nl}}{r\sin\theta}\frac{\partial Y_l^m}{\partial \varphi}. \tag{2.25}$$

Using Equation (2.6), the two Lorentz-boost terms in Equation (2.17) become

$$\frac{\gamma_L - 1}{v^2}\mathbf{v}(\mathbf{v}\cdot\boldsymbol{\nabla} a) = \sqrt{\frac{N_a \hbar^3 c}{2m_a}}\,\Re\left[e^{-im_a c^2 t/\hbar} \sum_{nlm} c_{nlm} e^{-iE_n t/\hbar} \frac{\gamma_L - 1}{v^2}\mathbf{v}(\mathbf{v}\cdot\boldsymbol{\nabla}\psi_{nlm})\right], \tag{2.26}$$

$$\gamma_L \frac{\mathbf{v}}{c^2}\partial_t a = \sqrt{\frac{N_a \hbar^3 c}{2m_a}}\,\Re\left[e^{-im_a c^2 t/\hbar} \sum_{nlm} c_{nlm} e^{-iE_n t/\hbar}\left(-i\gamma_L \frac{m_a c^2 + E_n}{\hbar c^2}\mathbf{v}\,\psi_{nlm}\right)\right]. \tag{2.27}$$

The size of these corrections can be estimated from the rotation of the Earth. Using $R_\oplus \simeq 6.37\times 10^6$ m and the sidereal rotation period $T_\oplus \simeq 8.62\times 10^4$ s gives the maximum surface speed

$$v_{\mathrm{rot}}^{\mathrm{max}} = \frac{2\pi R_\oplus}{T_\oplus} \simeq 4.65\times 10^2\,\mathrm{m\,s^{-1}}, \qquad \frac{v_{\mathrm{rot}}^{\mathrm{max}}}{c} \simeq 1.55\times 10^{-6}. \tag{2.28}$$

At latitude λ, this speed is reduced by a factor $\cos\lambda$. The corresponding Lorentz factor satisfies

$$\gamma_L - 1 \simeq \frac{1}{2}\left(\frac{v_{\mathrm{rot}}^{\mathrm{max}}}{c}\right)^2 \simeq 1.2\times 10^{-12}, \tag{2.29}$$

so the first correction in Equation (2.27) is negligible compared with the unboosted gradient. For an ALP field oscillating at ordinary frequency ν_a, the time-derivative term instead produces an effective gradient

$$\frac{|\delta\boldsymbol{\nabla} a|_t}{|a|} \sim \frac{2\pi\nu_a v_{\rm rot}^{\rm max}}{c^2} \simeq \frac{0.21}{R_\oplus}\left(\frac{\nu_a}{1\,\mathrm{MHz}}\right). \tag{2.30}$$

Relative to an intrinsic bound-state gradient characterized by a speed $v_{\rm bound}$, the temporal boost contribution is approximately $v_{\rm rot}/v_{\rm bound}$. For $v_{\rm bound} \sim v_{\rm esc} = 11.2\,\mathrm{km\,s^{-1}}$, this ratio is about 4% at the equator. The correction proportional to $\gamma_L - 1$ is therefore of order 10^{-12}, whereas the temporal boost correction is of order 4% for this velocity scale and can be larger when the intrinsic transverse gradient is suppressed.

The laboratory velocity relative to the Earth-bound halo is written generally as

$$\mathbf{v} = v_r\hat{\mathbf{r}} + v_\theta\hat{\boldsymbol{\theta}} + v_\varphi\hat{\boldsymbol{\varphi}}, \qquad v^2 = v_r^2 + v_\theta^2 + v_\varphi^2, \tag{2.31}$$

where the radial, polar, and azimuthal components are all retained. The boost correction to the complex gradient amplitude of one mode is then

$$\begin{aligned}\delta\boldsymbol{\mathcal{G}}_{nlm} = {}& \frac{\gamma_L - 1}{v^2}\mathbf{v}\left(v_r\frac{\partial R_{nl}}{\partial r}Y_l^m + v_\theta\frac{R_{nl}}{r}\frac{\partial Y_l^m}{\partial\theta} + v_\varphi\frac{R_{nl}}{r\sin\theta}\frac{\partial Y_l^m}{\partial\varphi}\right) \\ & - i\gamma_L\frac{\omega_n}{c^2}\mathbf{v}R_{nl}Y_l^m, \qquad \omega_n = \frac{m_a c^2 + E_n}{\hbar}.\end{aligned} \tag{2.32}$$

For the spherically symmetric state with $l = m = 0$, the angular derivatives vanish, but the first correction retains a radial contribution proportional to $v_r\,\partial R_{n0}/\partial r$. It vanishes only when $v_r = 0$, while the contribution from the temporal derivative remains. If multiple modes are populated, the total gradient is a superposition of the gradients from each mode, weights of which correspond to the population distribution, as can be seen in Equation (2.24).

contributed by the wavefunction ψ_{nlm}

The gradient magnitude depends on the eigen-state (n, l, m) and the detector location. The characteristics of the gradient are addressed as follows:

- For the eigen-states with zero angular momentum (s state), the wavefunction is spherically symmetric, so the gradient is purely radial and its magnitude is proportional to the radial wavefunction gradientYuzhe: is this paragraph right??.

- When angular momentum is non-zero ($l >= 1$), the gradient has non-radial components may vary over different locations, depending on the angular structure of the wavefunction.

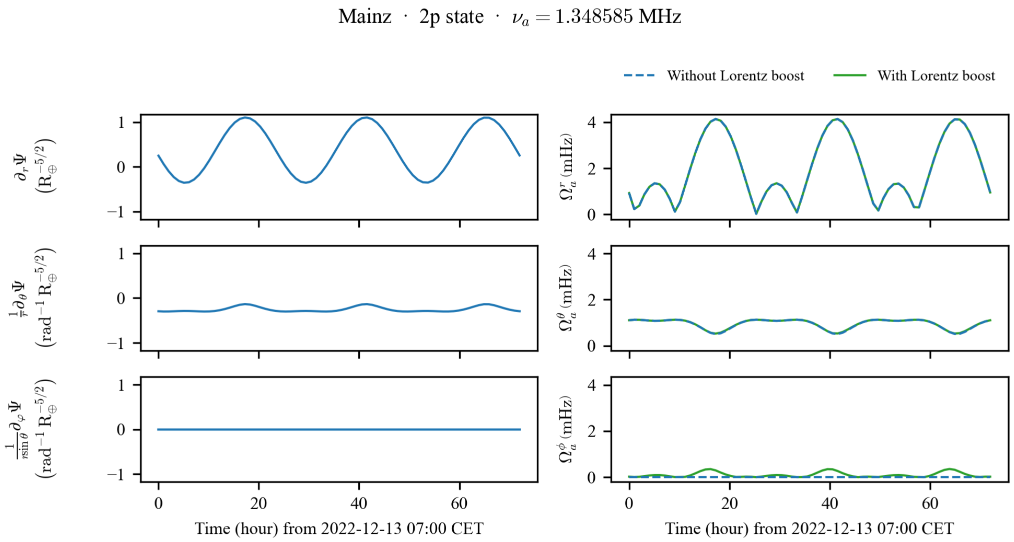

Figure 2.6: gradients of 1s 2p... states. Three rows would be enogh.

- The choice of the quantization axis is critical for considering $l > 0$ states. Since the solar gravity breaks the rotational symmetry, the z-axis is chosen along the Earth–Sun line. The x-axis is chosen along the instantaneous velocity of the Earth in the Solar System, and the y-axis is defined by $\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$. This coordinate system generally does not align with the Earth's rotation axis, so the laboratory velocity is retained in all three spherical components, v_r, v_θ, and v_φ. The gradient at the detector location therefore exhibits a daily modulation as the Earth rotates, as discussed in Section 2.2.6.

- **Azimuthal dependence** For modes with $m \neq 0$, the gradient has an azimuthal dependence. However, for the brevity of this discussion, only $m = 0$ is considered in this work.

The values of the gradients are computed numerically from the eigen-states and are used to estimate the expected signal strength in Chapter 4. Yuzhe: mention the time for computing the gradients. Examples of the gradient magnitude for selected modes are illustrated in Figure 2.5.

Yuzhe: Add a numerical estimate for $\Omega_a^{\mathrm{Earth}}$ as a function of m_a and M_{bound}. Add a comparison with the galactic-halo Rabi frequency Ω_a (Equation (1.22)). A rough estimation of the Rabi frequency $\Omega_a^\oplus$ of the pseudomagnetic field can be made using Equation (1.22) and escape speed v_{esc} as the typical velocity of the Earth-bound ALP:

$$\Omega_a^{\oplus}/(2\pi) \approx \frac{g_{\mathrm{aNN}}}{10^{-9}\,\mathrm{GeV}^{-1}} \times 6\,\mathrm{mHz}\,. \tag{2.33}$$

A more accurate estimation can be made using the numerical values of the wavefunction gradients at the detector location. With Equations (1.21, 2.24, 2.25), it can be estimated as

$$\Omega_a^{\oplus} = c\, g_{\mathrm{aNN}} \sqrt{\frac{N_a \hbar^3 c}{2 m_a}} \left|(\nabla \psi_{nlm})_\perp\right|, \tag{2.34}$$

where $|(\nabla\psi_{nlm})_\perp|$ is the magnitude of the wavefunction gradient component perpendicular to the bias field $\mathbf{B}_0$ (i.e., the $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\varphi}}$ terms from Equation (2.25)). Using the numerical values of the wavefunction gradients, the value can be obtained as

$$\Omega_a^{\oplus,\theta}/(2\pi) \approx \frac{g_{\mathrm{aNN}}}{10^{-9}\,\mathrm{GeV}^{-1}} \times 3\,\mathrm{mHz}\,. \tag{2.35}$$

Yuzhe: What about the radial gradient? TODO: add it. The two estimations in Equation (2.33) and (2.35) yield consistent results within an order of magnitude. The calculations can be found at Yuzhe: add reference to the script used for the calculation.

Such Rabi frequency is ~7 orders of magnitude larger than the expected Rabi frequency from the galactic halo ALP, which is $\Omega_a/(2\pi) = \frac{g_{\mathrm{aNN}}}{10^{-9}\,\mathrm{GeV}^{-1}} \times 2 \times 10^{-10}$ Hz, as estimated after Equation (1.22)Yuzhe: TODO: I may have to choose a different equation after finishing theory section. If indeed such amount of axionlike dark matter can be accumulated in the Earth gravitational potential, it would produce a much stronger signal than the galactic halo ALP, making it an target that CASPEr-G is sensitive to.

2.2.5 Population, interference and coherence

Unlike the galactic halo signal, which has a continuous Maxwell-Boltzmann spectrum, Earth-bound ALPs occupy discrete energy levels. Only the lowest few eigenstates are of interest because the higher eigen-states have small wavefunction amplitudes at the detector location. The population of the eigen-states is determined by the dark matter capture / accumulation process and potential ALP self-interaction or interactions with other particles, which is not well understood. The population distribution is therefore treated as a free parameter in this thesis. A few simple scenarios are considered: (1) only one of the lowest eigenstates are populated; (2) the first few eigenstates are populated uniformly.

Each level n contributes an ALP oscillation at a slightly different Compton frequency

$$\nu_n = \nu_a + \frac{|E_n|}{h}\,, \tag{2.36}$$

where $\nu_a = m_a c^2/h$ is the rest-mass Compton frequency. If multiple modes are populated simultaneously, the phases of the individual oscillations drift relative to each other over time. This leads to finite coherence time for the multi-mode ALP field. A rough estimate of the coherence time can be made from the spacing ΔE between eigen-states:

$$\tau_{\mathrm{coh}} \approx \frac{1}{\Delta\nu} = \frac{h}{\Delta E}\,. \tag{2.37}$$

The maximum spacing can be estimated from the ALP mass and the Earth gravitational potential which is $|\Delta V_g| = 125\,\mathrm{MJ\,kg^{-1}}$:

$$\Delta E \approx m_a\,\Delta V_g \approx m_a \frac{GM_\oplus}{R_\oplus} \tag{2.38}$$

$$= 6\times 10^{-18}\,\mathrm{eV}\frac{\nu_a}{1\,\mathrm{MHz}}\,. \tag{2.39}$$

Yuzhe: check the value of Vg later. The corresponding frequency spacing and coherence time are

$$\Delta\nu \approx \frac{\Delta E_{\mathrm{max}}}{h} \approx 10^{-9}\nu_a \tag{2.40}$$

and

$$\tau_{\mathrm{coh}} \approx \frac{1}{\Delta\nu} = \frac{10^9}{\nu_a}\,. \tag{2.41}$$

[9] It can be seen that the coherence time is $\sim 10^3$ times longer than the galactic-halo coherence time $\tau_a = Q_a/\nu_a \sim 10^6/\nu_a$ due to the much smaller velocity spread of the Earth-bound ALP. A more precise estimate of the coherence time can be made by computing the eigen-energies numerically from the Earth gravitational potential, as shown in Figure 2.7. The relative frequency spacing of the lowest 6 eigen-states is $\sim 10^{-9}$ or 1 ppb, which is consistent with the rough estimate in Equation (2.40).

To determine the exact coherence time, numerical solutions of Equation (2.7) and simulations of eigen-states interference are needed. However, when the ALP mass is small, the level spacing is small and the coherence time can be very long, which may exceed the relaxation times of the NMR sample and duration of a typical CASPEr-G measurement. In such scnarios, the ALP field will result in a one-bin signal in the frequency spectrum, and the amplitude of the signal will be limited

9 Calculations can be found at scripts-and-data/Theory-EarthHaloALP.ipynb.

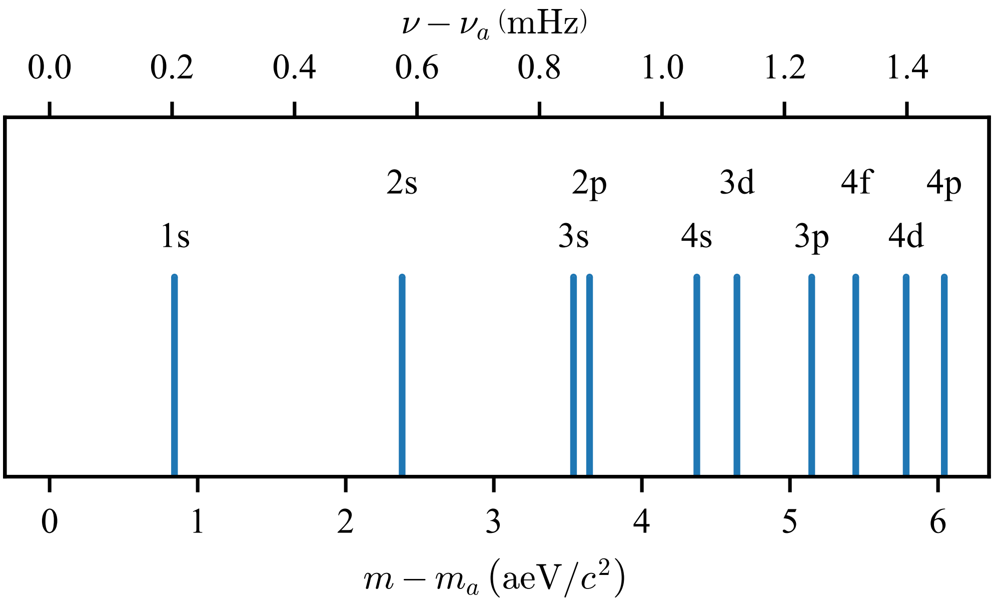

Figure 2.7: ALP eigen-energy for the lowest few eigen-states. The ALP mass is chosen to be $m_a c^2 = h \times 1.348\,585\,\mathrm{MHz}$, which is close to the Larmor frequency of the CASPEr-G measurements. The eigen-energies are computed numerically from the Earth gravitational potential.

by the relaxation times of the NMR sample. Therefore, the exact value of the coherence time is not always needed for the analysis of the CASPEr-G data.

2.2.6 Periodic modulations

The Earth-bound ALP wavefunction ψ_{nlm} is defined in the Earth-centered frame introduced above, with the z-axis along the Earth–Sun line, the x-axis along the instantaneous velocity of the Earth in the Solar System, and $\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$. This frame is generally not aligned with the Earth's rotation axis. As the Earth rotates, the laboratory co-rotates, so the orientation of the component of $\boldsymbol{\nabla} a$ at any given location therefore exhibits a daily modulation. Due to Earth's orbital motion, the orientation of the Earth–Sun line also changes over the course of a year, producing an annual modulation of the gradient component at a given location. Examples of daily and annual modulations are illustrated in Figure 2.8 and Figure 2.9, which show the daily and annual modulations of the gradient component for the $2p$ state $(n, l, m) = (2, 1, 0)$ at the location of CASPEr-G. In summary, the Rabi frequency exhibits changes within one order of magnitude due to periodic modulations. Therefore, the date and time of measurements would not have a significant effect for the sensitivity of CASPEr-G.

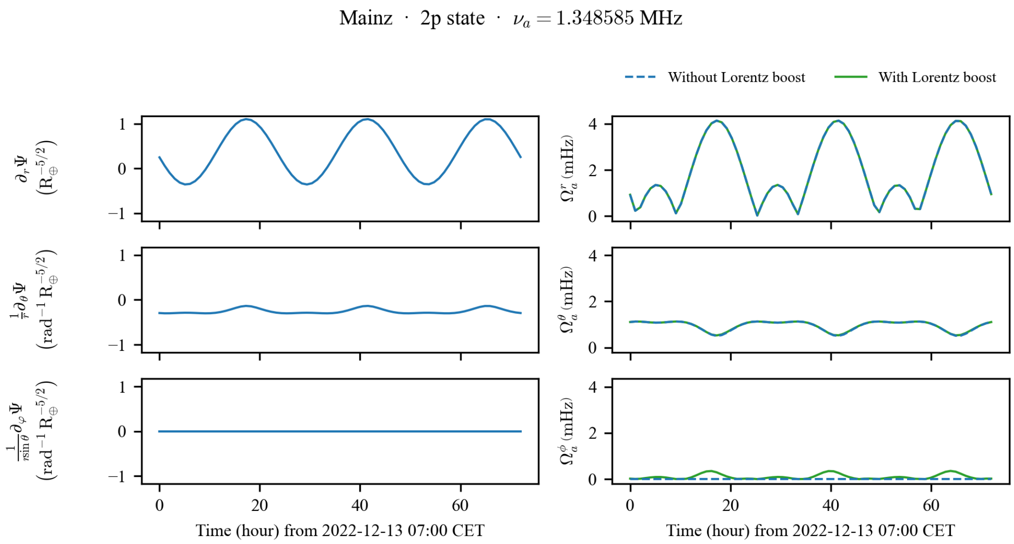

Figure 2.8: Daily modulation of the gradient component for the $2p$ state $(n, l, m) = (2, 1, 0)$ at the location of CASPEr-G (Helmholtz Institute Mainz). The modulation is due to the rotation of the Earth. Calculations with and without Lorentz boost are shown for comparison. In general, Lorentz boost modifies the Rabi frequency by less than 1 mHz at 1 MHz ALP Compton frequency. Note that for Ω_a^ϕ, which is zero without Lorentz boost (since magnetic quantum number m is zero), is non-zero with Lorentz boost due to Earth rotation.

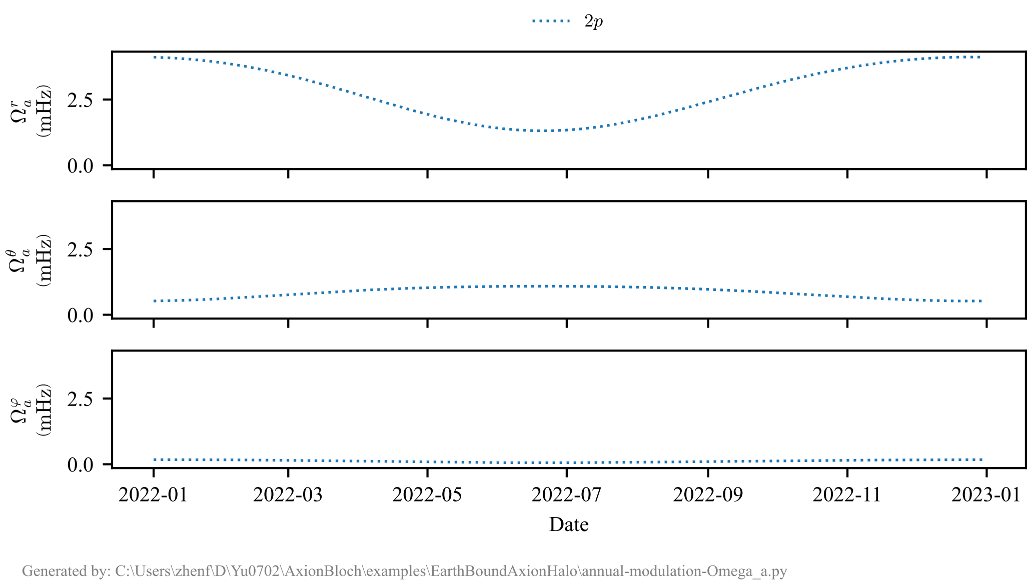

Figure 2.9: Annual modulation of the gradient component for the $2p$ state $(n, l, m) = (2, 1, 0)$ at the location of CASPEr-G (Helmholtz Institute Mainz). The modulation is due to the Earth's orbit around the Sun. Lorentz boost is considered in the calculation.

3 Experiment

The apparatus description is adapted from Ref. [41]. Sections 3.7–3.9 are adapted from Ref. [47].

The CASPEr-Gradient-Low-Field apparatus converts a nuclear-spin signal into a detectable electrical output through three functional blocks: the magnet system sets the operating bias field and thereby the Larmor frequency of the sample; the sample and pickup coil generate and capture the magnetic response; and the SQUID and lock-in amplifier chain convert, amplify, and record the signal. This chapter describes each block, summarizes the key parameters, and then develops the strategy used to scan an ALP-mass range efficiently.

3.1 Magnet system

The apparatus is built around a flow-through liquid-helium cryostat housing a superconducting solenoid that can generate a bias field B_0 up to 0.1 T. The DM mass range accessible to the experiment depends on the gyromagnetic ratio of the sample and on B_0; for proton spins, the covered Larmor frequency range is approximately 1 kHz to 4.3 MHz.

Eight superconducting shim coils surrounding the solenoid are used to reduce the field inhomogeneity to less than 2 ppm over a spherical volume with radius 4 mm. In the measurements reported here, an inhomogeneity of approximately 10 ppm was achieved after tuning the shim fields using a Bayesian-optimization algorithm [40]. The larger-than-design inhomogeneity results from using a longer sample (24 mm height) and from the insertion of an additional superconducting pickup coil.

Helmholtz coils wound outside the cryostat provide a controllable offset on B_0, enabling fine-tuning of the Larmor frequency without ramping the main solenoid. This arrangement is used to step the Larmor frequency in 6 Hz increments during the scan.

3.1.1 Cryostat and superconducting magnet

The cryostat housing the superconducting magnet is placed inside a Faraday room, shielded from external electric fields. It consists of an outer shell, reservoirs for liquid nitrogen (LN_2) and liquid helium (LHe), and vacuum insulation layers in between. Liquid nitrogen cools the cryostat and reduces the evaporation of liquid helium, while the superconducting devices and niobium shield are submerged in the LHe bath at 4.2 K. When filled, the cryostat can maintain a superconducting environment for approximately one week. The niobium shield repels external magnetic fields, isolating internal devices from external magnetic noise.

The magnet system consists of the main solenoid coil and eight shim coils (named Z1, Z2, X, Y, XY, XZ, YZ, and X^2-Y^2), each capable of generating magnetic field components for field homogenization. The shim coil channel names indicate the geometry of the magnetic field they generate, enabling systematic correction of field inhomogeneities.

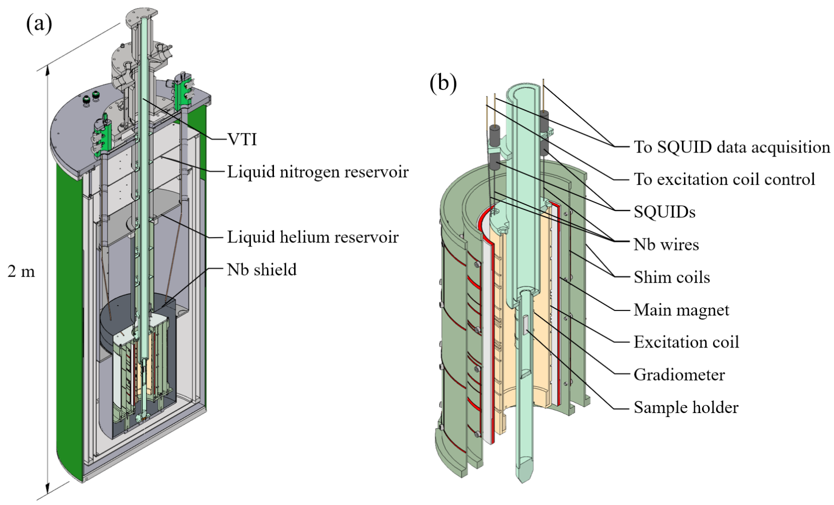

Figure 3.1: (a) Cryostat for maintaining superconducting temperature. LHe bath provides 4.2 K environment for superconducting apparatus. (b) Superconducting apparatus including main coil, shim and excitation coils, SQUIDs with niobium capsule, niobium wires for connections, and gradiometer made of niobium wires.

3.1.2 Excitation generation

To perform pulsed-NMR schemes, an excitation coil made of niobium wires is positioned near the sample to avoid Johnson–Nyquist noise from thermal agitation in normal-conducting materials. The excitation coil provides B_x, B_y, B_z, and dB_z/dz channels, with the first three capable of producing homogeneous fields along the respective axes and the last generating a linear gradient along the z-axis.

Radiofrequency pulses are generated by a function generator and amplified by an rf amplifier. A tuning circuit composed of capacitors and inductors is designed for impedance matching between the function generator and the excitation coil across different frequencies.

3.2 Sample

Liquid methanol (CH_3OH) was chosen as the sample because of its high proton density (99 mmol/cm^3) and low freezing point (175.6 K), which allows it to remain liquid at the operating temperature of approximately 180 K. With four protons per molecule, the 1.2 cm^3 cylindrical sample (radius 4 mm, height 24 mm) contains 7.2×10^{22} protons.

A variable-temperature insert (VTI) positions the sample in the homogeneous region of the bias field and thermally isolates it from the liquid-helium bath. The sample temperature is regulated by flowing warm nitrogen gas through a spiral surrounding the sample holder. Temperature stability is monitored by repeated pulsed-NMR measurements: a constant NMR linewidth indicates a stable temperature.

At thermal equilibrium in the bias field B_0, the proton spins acquire a net longitudinal magnetization

$$M_0 = \frac{n\,\gamma^2 \hbar^2\, I(I+1)\, B_0}{3\, k_B\, T}, \tag{3.1}$$

where n is the proton number density, $\gamma = 2\pi \times 42.577$ MHz/T is the proton gyromagnetic ratio, $I = 1/2$, k_B is the Boltzmann constant, and T is the sample temperature. At $B_0 = 31.67$ mT and $T \approx 180$ K, the thermal polarization is $\approx 1.8 \times 10^{-7}$ and $M_0 \approx 1.86 \times 10^{-10}\ \mathrm{T}/\mu_0$.

3.3 Pickup coil and shielding

A niobium (Nb) pickup coil mounted on the outside of the VTI in the liquid-helium bath detects the transverse magnetization of the sample. The coil consists of three saddle-pair coils in a gradiometer configuration: the area of the central coil is twice that of the two outer coils, and the coils are wound in opposing senses so that far-field flux contributions cancel. The filling factor—the fraction of the coil volume occupied by the sample—is approximately 8 %.

A superconducting Nb shield built into the cryostat attenuates low-frequency magnetic noise. Additional shielding is provided by a μ-metal shield outside the cryogenic reservoir and by a Faraday cage enclosing the entire apparatus, which suppresses electromagnetic interference (EMI). The pickup coil geometry and pulsed-NMR scheme are illustrated in Figure 3.3.

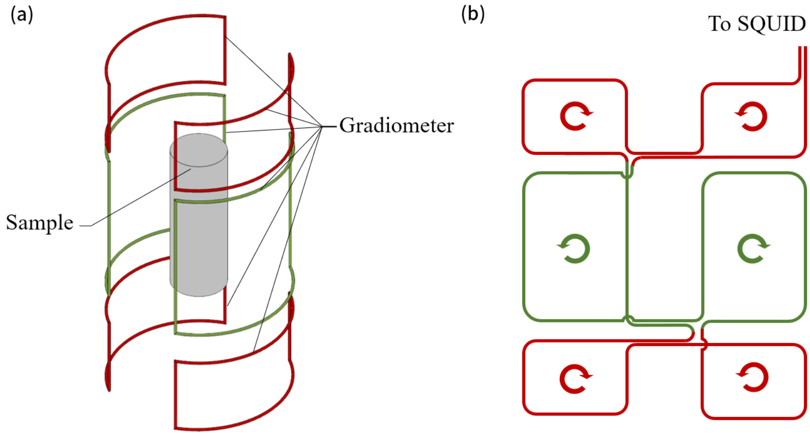

Figure 3.2: (a) Simplified schematic of gradiometer and sample positioned at the center. (b) Details of gradiometer loop connection showing the three-coil saddle-pair configuration with opposing windings for noise rejection.

3.4 Detection chain

3.4.1 SQUID detector and flux-locked-loop operation

Magnetic flux in the pickup coil is inductively coupled into a direct-current superconducting quantum interference device (dc-SQUID) operated in flux-locked-loop (FLL) mode. The SQUID converts magnetic flux into electrical signals through the combination of flux quantization and Josephson tunneling effects, offering exceptional sensitivity for weak magnetic signals.

The relationship between the magnetic flux coupled to the SQUID and the feedback voltage is given by equation (3.2), where the conversion factor depends on the coupling strengths and feedback parameters summarized in Table 3.1. The SQUID circuit is enclosed in a niobium capsule for shielding and protection from external disturbances.

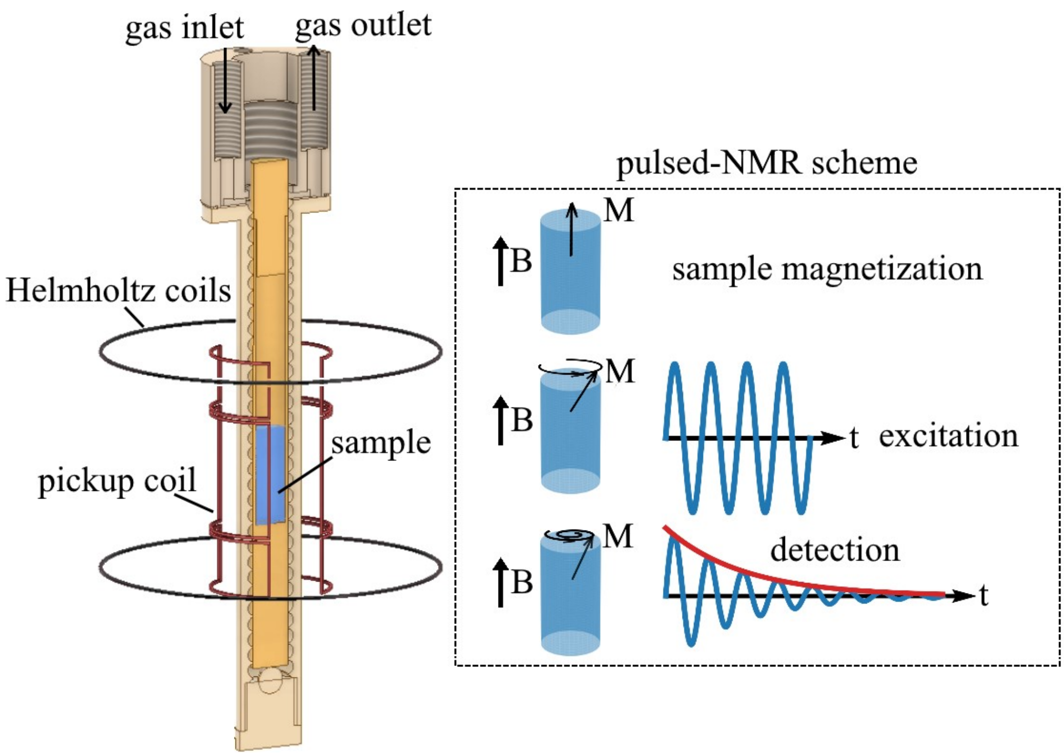

Figure 3.3: Left: the methanol sample (blue) sits at the center of the three-coil Nb saddle-pair pickup (red). Helmholtz coils (black) fine-tune the bias field. Right: pulsed-NMR scheme used for sample characterization. The net magnetization **M** is tipped into the transverse plane by a $\pi/2$ pulse and precesses, generating a signal in the pickup and SQUID. Reproduced from Ref. [41].

3.4.2 Signal chain and processing

Magnetic flux in the pickup coil is inductively coupled into a direct-current superconducting quantum interference device (dc-SQUID) operated in flux-locked-loop (FLL) mode. The properties of the SQUID sensor are listed in Table 3.1. The feedback voltage V_f is related to the magnetic flux in the pickup coil Φ_{pick} by

$$V_f = \frac{R_f}{M_f} \frac{M_{\text{in}}}{L_{\text{pick}} + L_{\text{in}}} \Phi_{\text{pick}} = \varepsilon \, \Phi_{\text{pick}} \,, \tag{3.2}$$

where R_f is the feedback resistance, M_f is the feedback-coil mutual inductance, M_{in} is the mutual inductance between pickup and SQUID input coils, and L_{pick}, L_{in} are

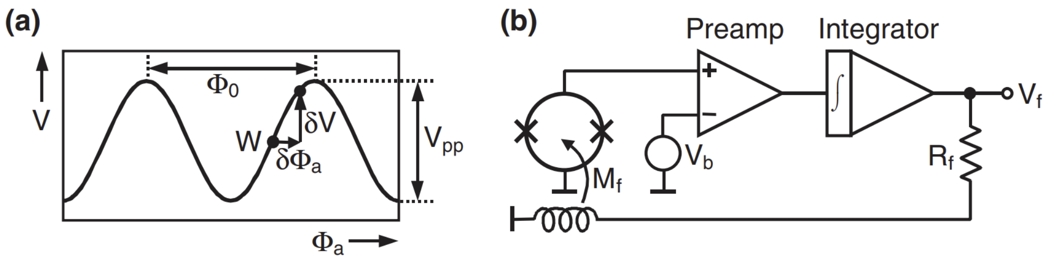

Figure 3.4: (a) V–Φ characteristic of dc SQUID showing the operating point W where the feedback voltage has approximately linear dependence on magnetic flux. (b) Direct-coupled flux-locked-loop (FLL) circuit with built-in pre-amplifier and integrator for maintaining the SQUID at the linear operating point.

the respective coil inductances. The conversion factor $\varepsilon \approx 0.549\,\mathrm{mV}/\Phi_0$, where Φ_0 is the magnetic flux quantum.

Table 3.1: Properties of the dc-SQUID sensor used in the measurements.

Pickup coil inductance (L_{pick})	553 nH
SQUID input coil inductance (L_{in})	400 nH
Mutual inductance (M_{in})	3.98 nH
Feedback-coil inductance (M_{f})	22.83 Φ_0/mA
Feedback resistance (R_{f})	3 kΩ

The detection chain is characterized by the transfer coefficient β, defined as the ratio of the SQUID feedback-voltage amplitude $V_\perp$ to the transverse sample magnetization $\mu_0 M_\perp$:

$$\beta = \frac{V_\perp}{\mu_0\, M_\perp}\,. \tag{3.3}$$

The transfer coefficient was determined at each scan step from the feedback voltage following a $\pi/2$ pulse, for which $M_\perp = M_0$. The average value is $\beta \approx 4 \times 10^6\,\mathrm{V/T}$.

During ALP-search measurements the SQUID and FLL electronics are battery-powered to avoid electromagnetic interference from the mains. All other electronics (lock-in amplifier, field-ramp supply) are outside the Faraday cage; only shielded BNC cables penetrating the cage are used to carry signals.

The FLL output is fed into a Zurich Instruments MFLI Lock-in Amplifier (LIA), which demodulates the signal into in-phase (I) and quadrature (Q) channels. The complex output $I + iQ$ is recorded at a sampling frequency of 13.4 kHz and is used to generate power spectral densities (PSDs) via fast Fourier transformation (FFT).

3.5 Experimental control and data acquisition

Comprehensive centralized control of the CASPEr-Gradient-LF apparatus is implemented in Python to coordinate the complex timing requirements of pulsed-NMR experiments and manage communication between diverse hardware components. This control system enables synchronized operation of shim coils, rf pulse generation, and SQUID data acquisition.

The main functions of the control system include:

1. **Magnet and shim control:**
   a) Remote switching of shim power supply on and off
   b) Connection and disconnection of communication between power supply and shim coils
   c) Real-time setting of shim currents to desired values, with optional heater control for current stabilization
2. **Pulse generation and timing:**
   a) Command interface for injecting radiofrequency pulses
   b) Recording rf pulse timing via oscilloscope into HDF5 format
   c) Analysis of function generator data and pulse characterization
3. **Data acquisition and processing:**
   a) Recording SQUID data and automated creation of HDF5 files for storage
   b) Real-time analysis and monitoring of acquired signals
   c) Generation and analysis of power spectral densities

The control system orchestrates multiple measurement schemes including pulsed-NMR diagnostics for sample characterization, automated shimming procedures with feedback optimization, axion wind measurements, and magnetic field monitoring. Additionally, the system includes capabilities for NMR kinetic simulations, stochastic axion wind simulations, apparatus properties computation, and synchronized file management across multiple terminals.

3.6 Parameters summary

Table 3.2 summarizes the key parameters of the CASPEr-Gradient-LF apparatus for the methanol measurement reported in this thesis and for two projected future configurations using hyperpolarized samples. The thermal polarization of methanol is the dominant limiting factor in the current measurement. Hyperpolarized methanol or liquid ^{129}Xe could boost the polarization by several orders of magnitude, substantially improving the sensitivity.

3.7 Single-frequency sensitivity

The transverse magnetization M_1 induced by the ALP field couples to the pickup coil and SQUID via the pickup transfer coefficient α:

$$\alpha = \frac{\Phi}{\mu_0 M_1}, \tag{3.4}$$

where Φ is the magnetic flux in the SQUID and μ_0 is the vacuum permeability. Using Equations ??, the average ALP-induced flux power is

$$\langle \Phi_a^2 \rangle = \frac{1}{2}(\alpha\, u_n\, \mu_0 M_0 \Omega_a)^2 \begin{cases} \tau_a T_2^*, & \tau_a \ll T_2^* \\ \frac{\tau_a^2}{\pi}, & T_2^* \ll \tau_a \ll T_2 \\ T_2^2, & T_2 \ll \tau_a \end{cases}. \tag{3.5}$$

The signal amplitude in the power spectral density (PSD) is

$$S = \frac{\langle \Phi_a^2 \rangle}{\Gamma/2\pi}, \tag{3.6}$$

where Γ is the expected signal linewidth. Depending on which linewidth is narrower,

$$\Gamma = \begin{cases} \Gamma_a, & \Gamma_a \ll \Gamma_n \\ \Gamma_n, & \Gamma_a \gg \Gamma_n \end{cases}. \tag{3.7}$$

Within a spectral bin of width Γ, the number of PSD values available for averaging from a measurement of dwell time T is

$$N_\Gamma = \frac{\Gamma T}{2\pi} \gg 1\,, \tag{3.8}$$

and the noise standard deviation after averaging is

$$\sigma = \sqrt{\frac{2\pi}{\Gamma T}}\,\sigma_0\,, \tag{3.9}$$

where σ_0 is the single-bin noise level of the SQUID system.

Using a detection threshold of $S \geq 3.355\,\sigma$—corresponding to a true signal $5\,\sigma$ above the noise detected at 95% confidence [3]—the ALP coupling strengths excluded when no signal is found satisfy

$$|g_{\mathrm{aNN}}| \geqslant \begin{cases} \eta\, T^{-1/4}\,(T_2^*)^{-3/4}\,\tau_a^{-1/2}, & \tau_a \ll T_2^* \\ \eta\,\pi^{-1/4}\,T^{-1/4}\,(T_2^*)^{-1}\,\tau_a^{-1/4}, & T_2^* \ll \tau_a \ll T_2 \\ \eta\,\pi^{-3/4}\,T^{-1/4}\,T_2^{-1}\,(T_2^*)^{-1}\,\tau_a^{3/4}, & T_2^* \ll T_2 \ll \tau_a \\ \eta\,\pi^{-3/4}\,T^{-1/4}\,T_2^{-2}\,\tau_a^{3/4}, & T_2^* = T_2 \ll \tau_a \end{cases}\,, \tag{3.10}$$

where the prefactor η collects the apparatus-dependent constants:

$$\eta = \frac{2\sqrt{3.355\,\sigma_0}}{\pi^{1/4}\,\alpha\,\mu_0\,M_0\,v_\perp\,\sqrt{\hbar c\,\rho_{DM}}}\,. \tag{3.11}$$

Larger magnetization M_0 and lower noise σ_0 both improve η and hence the sensitivity.

3.8 Scanning procedure

The CASPEr-Gradient-Low-Field experiment scans the ALP-mass parameter space by sweeping the magnitude of $\mathbf{B}_0$, which tunes the Larmor frequency of the sample to a series of values $\nu_i \in [\nu_{\mathrm{start}}, \nu_{\mathrm{end}}]$. One step of the scan consists of:

A. Pulsed-NMR measurement to determine the resonance frequency and the effective linewidth (T_2^*).

B. Spin-echo measurement to determine T_2.

C. Free-precession data acquisition with the SQUID magnetometer—no transverse pulse applied—to record the ALP-signal channel over the dwell time T.

D. Fourier transform of the magnetometer data to produce a power spectrum.

E. Binning of PSD values within frequency bins of width Γ, and averaging to reduce noise.

F. Statistical analysis of binned PSD values (e.g., histogram comparison with the expected noise distribution) to identify DM-candidate frequencies.

G. If a candidate is found: repeat the measurement at the same field value and perform additional checks.

H. If no candidate is found: record the exclusion contour for this frequency step and ramp $\mathbf{B}_0$ to the next value.

The procedure is illustrated in Figure 3.5. A significant excess surviving repeated measurement can be tested further using the daily and annual modulation signatures of ALP dark matter [24].

The frequency-step size Γ_{scan} is set by the sensitive bandwidth of a single step. An experiment is sensitive to signals within a bandwidth equal to the maximum of Γ_n and Γ_a around a given Larmor frequency. Choosing a step size larger than Γ_{scan} leaves gaps; choosing a smaller step size multiplies the number of steps without improving coverage.

3.9 Scanning optimization

With fixed total measurement time T_{tot}, the dwell time can in principle depend on frequency: $T \propto \nu^n$. Maximizing the sensitive area in the $\log \nu$–$\log |g_{\mathrm{aNN}}|$ parameter space over the full scan determines the optimal exponent n. The derivation is given in Appendix A.7; the result is $n = 0$ (uniform dwell time) for regimes (1)–(3) and $n = -1$ (dwell time decreasing as $1/\nu$) for regime (4).

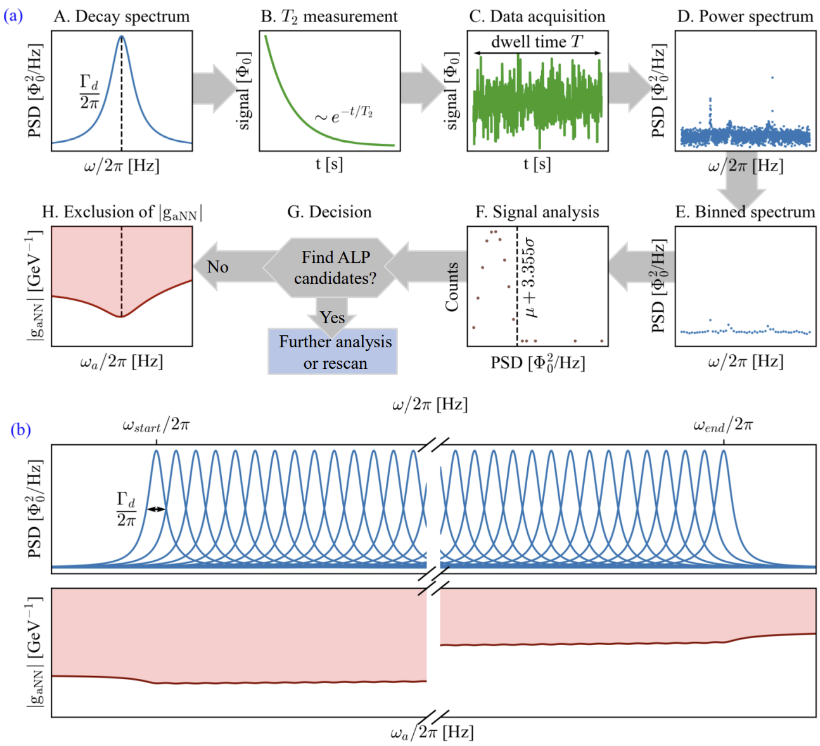

Figure 3.5: Schematic of the experimental procedure. (a) One step of the scan: pulsed-NMR diagnostics, free-precession data acquisition, PSD computation and binning, and statistical analysis using a histogram of PSD values. (b) Example decay spectra (blue curves) and the resulting sensitivity region in the ALP mass-coupling space (reddish area) after scanning over a frequency range. Reproduced from Ref. [47].

With optimal dwell-time allocation, the scanning sensitivity is

$$|g_{aNN}| \geqslant \begin{cases} \eta\left(\dfrac{T_{\rm tot}}{Q_a \ln(\nu_{\rm end}/\nu_{\rm start})}\right)^{-1/4} (T_2^*)^{-3/4}\,\tau_a^{-1/2}, & \begin{array}{c}\tau_a \ll T_2^* \text{ or}\\ T_2^* \ll \tau_a \ll T_2\end{array} \\ \eta\left(\dfrac{\pi^2 T_{\rm tot}}{Q_a \ln(\nu_{\rm end}/\nu_{\rm start})}\right)^{-1/4} (T_2^*)^{-3/4}\,T_2^{-1}\,\tau_a^{1/2}, & \begin{array}{c}T_2^* \ll T_2 \ll \tau_a \text{ or}\\ T_2^* = T_2 \ll \tau_a\end{array} \end{cases}. \tag{3.12}$$

Several conclusions follow directly:

- **Time scaling.** The sensitivity scales as $T_{\rm tot}^{-1/4}$ in all regimes, a consequence of the long-dwell-time limit $T \gg \tau_a$, T_2^*, T_2 [4, 11]. An order-of-magnitude increase in measurement time improves the exclusion bound by only a factor $10^{1/4} \approx 1.8$.

- **Effect of T_2^*.** Longer T_2^* is beneficial in all regimes. When $\tau_a \ll T_2^*$, it increases the r.m.s. tipping angle and narrows the signal linewidth. When $T_2^* \ll \tau_a$, it increases the fraction u_n of spins on resonance.
- **Effect of T_2.** Longer T_2 improves sensitivity in the regime $T_2 \ll \tau_a$, where T_2 directly limits the r.m.s. tipping angle.

The sensitivity plots computed with uniform ($n = 0$) and frequency-adaptive ($n = -1$) dwell-time allocation for an example scan range $\nu_{\text{end}}/\nu_{\text{start}} = 10$ are shown in Figure 3.6. The area covered in the log ν–log $|g_{\text{aNN}}|$ plane is larger with optimal n.

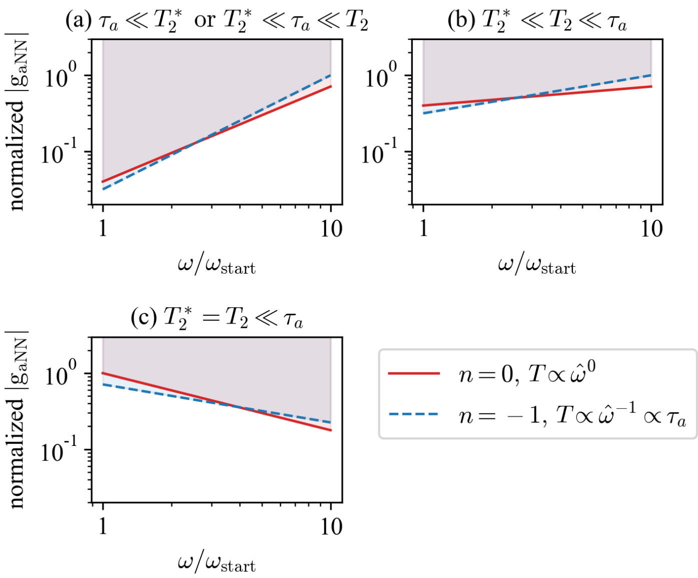

Figure 3.6: Sensitivity plots on a log-log scale for two dwell-time allocation schemes: uniform ($n = 0$) and frequency-adaptive ($n = -1$). The shaded regions indicate the parameter space accessible to the experiment. The coupling strength is normalized in each regime so that the highest point equals 1. The scan range is $\nu_{\text{end}}/\nu_{\text{start}} = 10$. Reproduced from Ref. [47].

In summary, the optimal strategy for a frequency-scanning ALP search is to work at the maximum achievable T_2 and T_2^*, use a step size equal to the sensitive bandwidth, and distribute the total measurement time uniformly across frequency steps (or as $T \propto \nu^{-1}$ in regime (4)).

Table 3.2: Parameters of the CASPEr-Gradient-LF apparatus for the current measurement (thermally polarized methanol) and two projected future configurations. Polarization is given as molar polarization (polarization×concentration).

	This work	Examples of future CASPEr-Gradient-LF	
Sample	Methanol	Methanol[a]	Liquid ^{129}Xe[b]
Spin density (cm^{-3})	5.9×10^{22}	5.9×10^{22}	$\sim 10^{22}$
Sample volume (cm^3)	1.2	1.2	~ 1
Polarization technique	Thermal	BF[c]/ PHIP[d]/ DNP[e]	SEOP[f]
Polarization	1.8×10^{-7}	$> 1.1 \times 10^{-5}$	~ 1
T_2 (s)	0.72	1	1000
B_0 inhomogeneity (ppm)	10	≤ 2	≤ 2
Electronic noise ($\mu\Phi_0/\mathrm{Hz}^{1/2}$)	7.1	1.0	1.0
Flux of SPN[g]($\mu\Phi_0$)	0.39	0.39	0.16
Scan range	1.348 450 MHz–1.348 690 MHz	~1 kHz–4.3 MHz	~1 kHz–1.2 MHz
$\lvert g_{\mathrm{aNN}}\rvert$ sensitivity (GeV^{-1})	3×10^{-2}	1.0×10^{-6}–1.2×10^{-5}	10^{-12}–10^{-11}

[a] Methanol at 180 K prepolarized at 2 T; measurement time 100 τ_a per scan step.
[b] ^{129}Xe-enriched liquid sample.
[c] Brute-force prepolarization in a strong field at low temperature.
[d] Parahydrogen-induced polarization (PHIP).
[e] Dynamic nuclear polarization (DNP).
[f] Spin-exchange optical pumping (SEOP).
[g] Spin-projection noise (SPN).

4 Results

Data analysis procedures and searches for Milky Way ALP are adapted from Ref. [41], while the Earth-bound ALP search is adapted from Yuzhe: add Earth-bound ALP search reference here.

This chapter first presents the data analysis pipeline including data processing and outlier validation check. The data analysis pipeline is later applied to two proof-of-principle ALP searches using the CASPEr-Gradient-LF apparatus with a thermally polarized methanol sample. Since the search is sensitive to the proton NMR signal in methanol, the results are interpreted in terms of the ALP-proton gradient coupling constant g_{ap}, i.e. g_{aNN} evaluated for proton spins. The full MW and Earth-halo candidate lists are given in Appendix A.8.

4.1 Data analysis

The data analysis of CASPEr-Gradient aims at identifying potential ALP signals in the frequency spectra obtained from the time-domain measurements and setting exclusion / sensitivity limits on the ALP-nuclear spin gradient coupling in the absence of a detection.

The development of analysis framework took the data analysis pipelines of HAYSTAC, SHAFT, and CASPEr-Electric [3, 10, 23] as reference.

4.1.1 Methodology

In general, the raw time-series data are first converted to the frequency domain using a Fourier transform, and then processed with filters to suppress noise. The outliers in the filtered spectra, judged by a threshold determined from Monte-Carlo (MC) simulations, are later evaluated for their consistency across repeated scans. Outliers that survive the consistency checks are considered as ALP candidates. If no candidate is found at a frequency band, the analysis sets an exclusion limit on the ALP-nuclear spin gradient coupling in this band.

A more detailed description of the data-analysis procedure is given below and summarized in Figure 4.1. There are four main blocks of the analysis:

- **Frequency spectra generation and filtering.** Recorded time-series are evaluated by its sanity. Sane data will generate frequency spectra by Fourier transform. Digital filtering, such as Savitzky–Golay (SG) filter and optimal filters are applied to the spectra to remove the baseline and optimize SNR. The filtered spectra are later used for finding outliers.

- **Monte-Carlo simulation for determining outlier threshold.** To determine the threshold in picking up outliers in the frequency domain, MC simulation are ran. Artificial ALP signals are generated in simulations in determining the threshold.

- **NMR charaterization.** The NMR sample, bias magnetic field, and detection chain are characterized by pulsed-NMR measurements. These NMR configurations are later used for examining the outliers and determining experimental sensitivity to ALP.

- **Outlier examination.** Based on the information from previous steps, outliers in the frequency spectra are picked up and examined. The valided outliers become ALP signal candidates for further investigation. Exclusion or sensitivity plots can also be generated after the validation check.

In addition to these procedures, the analysis also includes procedures such as data sanity checks and calibration by artificial signal injection. The technical details of these procedures are discussed as follows.

4.1.2 Procedure

Time-series sanity check

Data with Gaussian noise are expected from the SQUID and LIA in the absence of ALP signal. In the presence of ALP signal:

- If the ALP signal is weak compared to the noise, the data is approximately Gaussian.

- If the ALP signal is strong, the data is no longer Gaussian, and stochastic drift with a characteristic period of ALP coherence time is expected, as illustrated in Figure 2.1.

Moreover, when recording the data, instability and fluctuations in hardware and software could result in insane data. Therefore, the first step in data processing is to check the sanity of time-series. The time-series are checked for

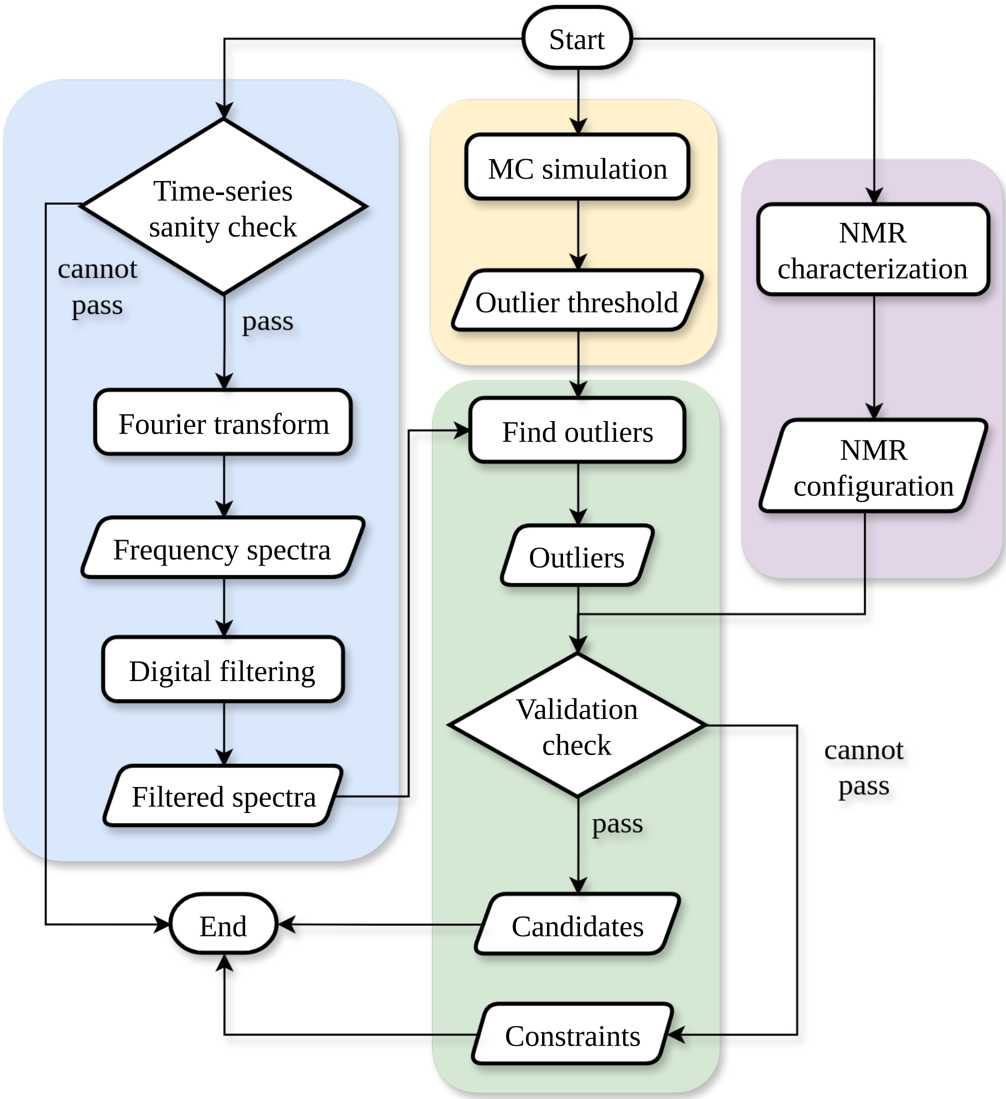

Figure 4.1: Data-analysis pipeline for the CASPEr-Gradient search. Details of this pipeline are discussed in Section 4.1.2.Yuzhe: to be updated

1. **finite values.** The data are first checked for `NaN` (not-a-number) or infinite values. Since the time-series are stored as `numpy` arrays, NaN and infinity are examined by `numpy.isfinite` and `numpy.isnan`.

2. **long-term stability ("drift").** Slow (compared with the ALP coherence time) changes of the mean or variance due to instable SQUID flux-lock-loop mode are tested within individual scan steps.

3. **short-term stability ("jumps").** Local discontinuities or clusters of outlying samples in chunks much shorter than the ALP coherence time are searched for after the drift check. Such jumps may originate from magnetic noise due to mechanical or accoustic vibration, and can produce spectral leakage and artificial outliers in the search spectrum.

4. **Gaussianity.** Drift-clear data are tested for consistency with the Gaussian-noise assumption used in the threshold estimation.

Depending on the problems found, the following approaches may be applied:

1. **Window or mask on the time-series.** If values in the data are found to be `NaN`, infinite values, jumps exceeding a certain threshold, or drifts, a window or a mask is applied on the time-series to set the values to zeros, so their effects are excluded. The window is designed in a way so that its envelope is smooth. The effects of the window on the average power of the data is later considered in the Fourier transform.

2. **Baseline correction.** The baseline of the time-series is corrected by setting the mean values of chunks of data to consistent if the drift is diagnosed to arise from instable SQUID. Otherwise, a window with zero values is applied to exclude the data. Otherwise low-frequency spectral structures would be created.

3. **Discarding.** If the amount of insane data exceeds a certain ratio of the total data, the corresponding measurement should be discarded. This ratio is arbitrarily chosen to be 1 %.

Note that features that do not match ALP models investigated in this thesis, such as fluctuations in period much shorter than ALP coherence time, may not be noise or faulty hardware/software behavior, but signals of new physics. It is wise to keep track of these features for future considerations.

Fourier transform and filtering

Frequency spectra are generated by the Fourier transform from time-series data that can pass the sanity check. There are several forms of the frequency spectra, including power spectrum (PS), power spectral density (PSD), and amplitude / linear spectra density (ASD or LSD). The definition of these spectra are

$$\mathrm{PS} \propto |\mathrm{FT}[\mathrm{window(t)} \cdot s(t)]|^2, \tag{4.1}$$

$$\mathrm{PSD} = \frac{\mathrm{PS}}{\mathrm{ENBW}}, \tag{4.2}$$

and

$$\mathrm{ASD} = \mathrm{LSD} = \sqrt{\mathrm{PSD}}, \tag{4.3}$$

where FT is Fourier transform, window(t) is the window function, $s(t)$ is the time-series, and ENBW is effective noise bandwidth. The formula of ENBW can be found at Equation (22) of Ref. [26]. In the data analysis of CASPEr, ENBW is approximately the resolution bandwidth (RBW) [10] since the applied window function is roughly rectangle. The power spectrum is normalized in a way that the sum of the spectra is the average "power" of the signal:[11]

$$\text{Average power} = \frac{\sum_{k=0}^{N-1} |\mathrm{window(t_k)} \cdot s(t_k)|^2}{\sum_{k=0}^{N-1} |\mathrm{window(t_k)}|^2}, \tag{4.4}$$

where N is the number of samples in the time-series, and t_k is the time of the k-th sample. When window function is rectangular (window(t) $= 1$ for all t), the average power is simply the sum of the power spectrum divided by the number of samples. When window function is not rectangular, the amplification or attenuation due to the window function is corrected by the sum of the window function squared in the denominator.

After the frequency spectra are obtained, digital filterings are applied for correction and SNR optimization. Specifically, filterings here consist of low-pass filter attenuation correction, baseline removal, and optimal filtering.

The demodulated lock-in time-series experience digital low-pass filtering by the Zurich Instrument LIA [49]. The output time-series can be calculated using the

10 as known as spectral resolution or binwidth

11 Usually, power is defined as the amount of energy transferred or converted per unit time. For a voltage signal in unit of V, the power is in the units of V^2/Ω. However, the "power" in the spectral analysis is defined as the square of the signal, therefore the "power" of the signal has units of V^2. This special but common definition is adopted in this thesis unless otherwise noted.

following recursive formula

$$X_{\mathrm{out}}[k] = e^{-T_s/T_C} X_{\mathrm{out}}[k-1] + (1 - e^{-T_s/T_C}) X_{\mathrm{in}}[k]\,, \tag{4.5}$$

where X_{in} and X_{out} are the input and output time-series arrays, k is the sample index, T_s is the sampling time, and T_C is the time constant configured by the user. For the first stored sample $X_{\mathrm{out}}[-1]$ is chosen to be $X_{\mathrm{in}}[0]$ by convention, This exponential moving average is also described as digital discrete-Time RC (DTRC) Filter as it resembles the low-pass response of analog resistor-capacitor (RC) filters. The time constant T_C determines the 3 dB bandwidth in the frequency domain. Equation (4.5) describe a first order DTRC filter. Higher-order filters are implemented by cascading several filters in practice for suppressing signals of frequencies much larger than the Larmor frequency ($|\mathrm{frequency} - \mathrm{Larmorfrequency}| \gg \mathrm{NMRlinewidth}$). The response of the n-th order filter in the frequency domain is approximated by the formula

$$H_n(\nu) = \frac{1}{(1 + i2\pi\nu T_C)^n}\,, \tag{4.6}$$

where $H_n(\nu)$ is the n-th order attenuation at frequency ν. This attenuation is corrected by dividing the PSD by the square of the absolute value of the attenuation. If the time-series is Gaussian noise, the frequency spectra should be white noise (flat noise in the frequency domain) after the low-pass filter correction, and the distribution of the PSD should follow a χ^2 distribution. Since there are two channels in the lock-in amplifier, the degrees of freedom of the distribution should be two.

Since the frequency-dependent attenuation in (4.6) is only an approximation, the baseline of the PSD is not absolutely flat. The curve of the baseline is estimated by a SG filter and removed from the PSD. In practice, the spectrum is divided into chunks of a bandwidth much larger than the expected ALP linewidth, and the SG filter of the first order is applied to each chunk to estimate the baseline. Later, the baseline is removed by subtracting and dividing the spectrum by the estimated baseline, the product of which is also called the normalized power excess (NPE):

$$\mathrm{NPE}(\nu) = \frac{\mathrm{PSD}(\nu)}{\mathrm{Baseline}(\nu)} - 1\,.$$

The next step is to decide whether optimal filtering[12] should be applied or not. The optimal filter is a weighted average of neighboring frequency bins, with weights

12 as known as matched filtering

determined by the expected signal lineshape; this choice maximizes the signal-to-noise ratio among linear filters for random noise with known covariance, as shown in Appendix A.6. If the ALP linewidth is broader than the RBW of the spectrum, the signal is resolved, which means the signal is spread over multiple frequency bins. In this case, optimal filtering is applied to combine the signal power into a single bin. Alternatively, if the ALP linewidth is narrower than the RBW, the signal is unresolved and optimal filtering is not applied.

One of the procedures that was adopted in the analysis of Ref. [41] is the removal of narrow spikes in the frequency spectra. The philosophy behind this procedure is that spikes with linewidths much narrower than the expected ALP linewidth are likely to be radio-frequency interference (RFI), and are unlikely to be ALP signals, thus can be removed to avoid false positives and reduce noise. However, this procedure is not applied in the analysis of the data presented in this thesis for three reasons. First of all, since the stochastic nature of ALP signals allows for spiky signals as simulated in Figure 2.1. Moreover, RFI lines are expected to be stable in frequency and amplitude, and can be identified by repeated scans. Lastly, the spikes are suppressed by the optimal filtering.

After the filterings mentioned above, the filtered spectra are ready for outlier search.

Outlier threshold

If the probability of a signal being a false positive is less than 2.87×10^{-7}, which is the probability of a Gaussian random variable exceeding five times of its standard deviation σ, the signal is considered as a $5\,\sigma$ signal. An illustration of the Gaussian distribution and the probability of a $5\,\sigma$ signal is shown in Figure 4.2 (a). Such signals are considered as indications of new physics.

In addition to understanding what a $5\,\sigma$ signal is, it is also important to understand what a $5\,\sigma$ signal is not, and what is not a $5\,\sigma$ signal:

- Due to random noise, the amplitude of a $5\,\sigma$ signal fluctuate, instead of being $5\,\sigma$ constantly [3, 10]. If the noise is Gaussian, and if the noise is added to the signal linearly, the measured signal amplitude would be a Gaussian random variable with a mean of $5\,\sigma$ and a standard deviation of σ, as illustrated in Figure 4.3.

- A $5\,\sigma$ signal is not always a signal that exceeds five times the standard deviation of the noise, unless the noise is Gaussian. For instance, if the noise follows a χ^2 distribution with two degrees of freedom, the threshold for a so-called $5\,\sigma$ signal would be $15.06\,\sigma$, as shown in Figure 4.2 (b).

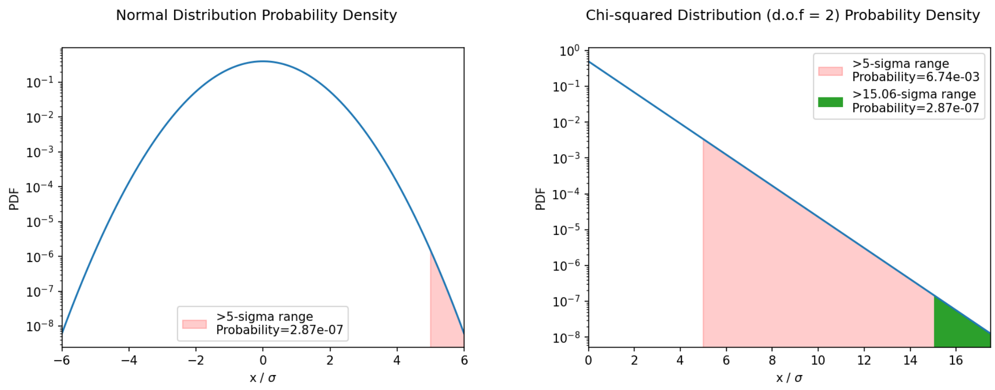

Figure 4.2: (a) The probability of a Gaussian random variable exceeding n times of its standard deviation. The probability of a $5\,\sigma$ signal is less than 2.87×10^{-7}. (b) The χ^2 distribution with two degrees of freedom. Yuzhe: to be updated

- $5\,\sigma$ signals are not rare, in terms of the amount of events they represent. On one hand, $5\,\sigma$ signals due to non-Gaussian noises are common in real experiments. On the other hand, experiments can accumulate huge amounts of data ($\gg 1/2.87 \times 10^{-7}$ data points), and the probability of one $5\,\sigma$ signal appearing due to random fluctuation is not negligible. This is also known as the look-elsewhere effect.

The exotic $5\,\sigma$ signals measured by CASPEr would appear as peaks in the NPE spectra. To capture the $5\,\sigma$ signals, a threshold is set, above which frequency bins are picked up as outliers for further evaluation.

The value of the threshold should not be too low or too high. If the threshold is too low, noises are likely to be regarded as signal (false positive). If the threshold is too high, signals are likely to be missed (false negative). An appropriate threshold can either be determined by noise statistics or by Monte-Carlo (MC) simulations. A common approach is to find out a threshold with which most of the signals are captured, while moderate amount of noises are also captured. In other words, this threshold corresponds to a certain confidence level (C.L.), such as 95 %. For example, if noise and signal are both Gaussian random variables with consistent standard deviation σ and different means (mean of noise being zero), a threshold of $3.355\,\sigma$ is found to ensure that, in the presence of a 5σ signal, the chance of capturing the signal is 95 %, and the chance of false positive (false alarm rate) is only 4×10^{-4}. When noise distribution is not Gaussian, the threshold can be determined by MC simulations, in which the noise and signal are simulated with the same statistics as in the experiment.

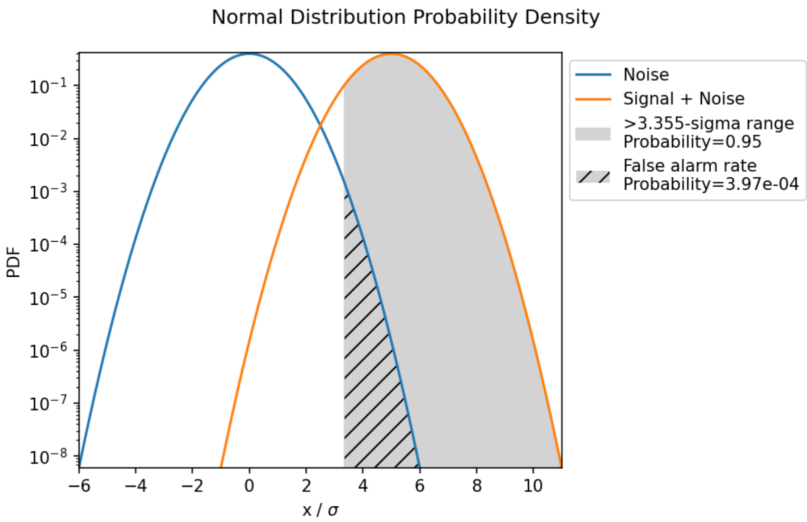

Figure 4.3: Noise and signal amplitude distributions assuming Gaussian noise and linear signal dependence.Yuzhe: to be updated. be careful with digits. check old casper g repo

NMR characterization

The sample and the bias magnetic field are characterized by the following parameters and measurements:

1. Larmor frequency, T_2^*, NMR linewidth and lineshape, polarization, and signal power measured by the free decay signal after a 90° pulse.

2. T_2 measured by CPMG pulse sequence.

These parameters are used to determine the sensitivity to ALP signals in the frequency band of interest. Since these parameters may vary due to changes in the sample temperature, bias magnetic field, and other environmental factors, these measurements are performed at each scan step.

Based on the NMR configuration obtained, and the noise of NPE spectra, the sensitivity to ALP signals can be determined. Yuzhe: this sentence can be adjusted to a different postion

Validation check and results

Outliers that pass the threshold are not necessarily ALP signals. Several methods can be used to validate the outliers. Those that can pass the validation tests are considered ALP signal candidates.

The key principle of the validation check is that ALP signals are expected to be consistent in terms of Compton frequency and coupling strength across measurements. Following this principle, the outliers are examined by the following steps:

1. **Merging outliers within individual scan steps by frequency.** The frequency of the outlier should be consistent within a certain tolerance. This tolerance is determined by the spectral RBW and the ALP linewidth. Outliers that appear at adjacent frequencies are likely to be the same ALP signal. Therefore, the first step in the validation check is to merge the outliers by their frequencies. The merged outlier is represented by the outlier with the largest amplitude.

2. **Obtaining coupling strength.** The coupling strength of the ALP signal and its uncertainty can be obtained from the amplitude of the outlier, the NMR configuration, ALP periodic modulation, and the noise of in the spectra.

3. **Reproducibility.** The outlier should be reproducible across scan steps that cover the same frequency range. Though the coupling strength should be consistent, the amplitude of the outlier may vary due to the periodic modulation of the ALP signal and changes in the NMR configuration, especially the Larmor frequency. The outlier should be observed when and only when the estimated signal amplitude is significant compared to the noise level. This rule can be used to exclude outliers that are not reproducible across scans.

In the end, the analysis sets constraints on the coupling strength at frequencies where there is no ALP candiate as minimum coupling strength that can be observed with a given confidence level. Otherwise, the constraints are set by the coupling strength of the ALP candidate, below which a candidate was observed.

4.2 Milky Way ALP measurement

One proof-of-principle measurement was performed on December 23, 2022. This measurement is ideal for Milky Way ALP search because it scanned a frequency range for three times and the duration of each scan step are long enough to cover multiple ALP coherence times. Relevant calculations can be found at scripts-and-data/Result-MW-ALP.ipynbYuzhe: link to be modified.

4.2.1 Overview

The search covered a frequency band of

$$1.348\,570\,\mathrm{MHz} \pm 120\,\mathrm{Hz}\,,$$

corresponding to proton Larmor frequencies driven by a bias field $|\mathbf{B}_0| \approx 31.67\,\mathrm{mT}$. This maps to ALP masses in a range of

$$5.577\,237\,\mathrm{neV/c^2} \pm 496\,\mathrm{feV/c^2}\,.$$

Three sequential scans over the same range were performed. In each scan, there were over 40 scan steps, and the frequency step size was 6 Hz, approximately half the NMR linewidth, so that the NMR resonant windows of adjacent steps overlap.

At each frequency step the following actions were performed in order:

1. Free decay measurement (one $\pi/2$ excitation pulse followed by free decay signal) to determine the Larmor frequency, linewidth, and NMR signal power. A Lorentzian fit to the resulting PSD peak extracts these parameters.

2. T_2 measurement by CPMG pulse sequence.

3. 100 s measurement without any pulses.

4. Frequency step: the Larmor frequency is increased by $\approx$ 6 Hz by ramping the Helmholtz coils.

Delays to allow ring-down noise to decay or the magnetization to return close to its equilibrium are applied between steps. The measurement procedure is illustrated in Figure 4.4. An example of the PSD spectra and NMR characterization from one scan step is shown in Figure 4.5. The experimental parameters and the relevant NMR and ALP characteristics are summarized in Table 4.1.

For a virialized Milky Way ALP field, the expected linewidth near 1.35 MHz is approximately 1 Hz, corresponding to a coherence time of approximately 0.3 s. Thus, each 100 s acquisition spans roughly 300 coherence times and measures the time-averaged stochastic signal power.

Over the six-hour measurement, the ALP signal power varied between scan steps mainly due to Earth's rotation changes the component of the ALP-field gradient perpendicular to the bias field. This predictable modulation, together with the measured NMR response at each step, is included when converting a spectral excess into an estimated coupling strength. A plot of the the effect of the Earth's motion on the expected ALP signal is shown in Figure 4.6. The annual modulation

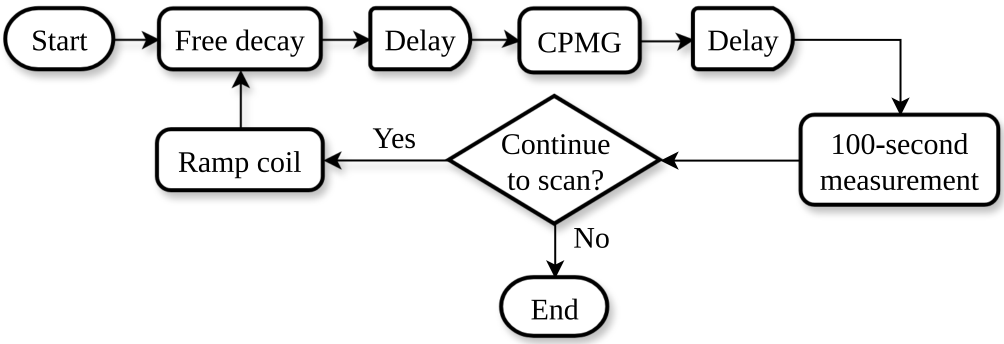

Figure 4.4: Schematic of the MW ALP measurement procedure.

due to the Earth's motion in the solar system is considered in calculation but it did not have a significant impact over the course of the measurement.

4.2.2 Data analysis

The technical details of the data analysis, following the procedure described in Section 4.1.2, are presented.

After the time-series sanity checks, each 100 s record with masks with windowing was Fourier transformed and corrected for the lock-in low-pass response. The spectral baseline was estimated with a SG filter and removed to form the NPE spectrum. For this search, RBW = $1/100\,\mathrm{s} = 0.01\,\mathrm{Hz}$, whereas the expected ALP linewidth is approximately 1 Hz; the signal is therefore resolved over many frequency bins. The optimal filter was applied to the NPE spectrum, using the expected ALP lineshape as the filter kernel. One example of the spectra after different filtering steps is shown in Figure 4.7.

Monte Carlo simulations were performed to determine the threshold for recovering a $5\,\sigma$ ALP signal with 95 % probability. Fake $5\,\sigma$ ALP signals were injected into simulated noise spectra before the optimal filtering, and the amplitudes of the filtered NPE was recorded. Probability distributions of the filtered NPE were obtained from many realizations of the simulated noise and signal, and the outlier threshold Θ_{RE} was determined from the distribution of the filtered NPE in the presence of a $5\,\sigma$ ALP signal. The simulations yielded $\Theta_{\mathrm{RE}} = 3.43$ with a noise-only tail probability of approximately 0.001, as shown in Figure 4.8.

The total number of outliers in each scan exceeds the expectation for white noise, owing to non-white-noise structures that are not removed by the SG filters. Adjacent outliers within one ALP linewidth were first merged. By comparing the

Table 4.1: Technical parameters of the MW ALP measurement on December 23, 2022 (Central European Time).

Parameter	Name or value
Sample	thermally polarized methanol
Number of scans	3
Number of steps per scan	45
Sweep coil current increment	0.8 mA
Frequency increment per step	6 Hz
Frequency scan range	1.348 570 MHz ± 120 Hz
NMR linewidth in one step	≈ 10 Hz
T_2	0.72(5) s
ALP mass scan range	5.577 237 neV/c^2 ± 496 feV/c^2
ALP linewidth $\Delta\nu_a$	≈ 1 Hz
ALP coherence time τ_a	≈ 0.3 s
Duration per step	100 s ≈ 300 τ_a
Total duration	6 h
spectral factor u	≈ 10 %

resonant window (within ±5 NMR linewidths of the Larmor frequency) with off-resonance background, and by cross-checking across scans, these off-resonance outliers can be excluded. The on-resonance outliers were then compared across the three scans at consistent Compton frequencies. No candidate appeared in all three scans at consistent frequency within four ALP linewidths. The search therefore finds no statistically significant evidence for Milky Way ALP dark matter at the achieved sensitivity. Table 4.2 summarizes the outlier counts and ALP signal candidates from all three scans.

Table 4.2: Results of the three ALP scans. N_{full}: total NPE bins. X_{full}: outliers above rescan threshold. $\langle X_{\text{full}} \rangle$: expected value assuming white noise. N_{res}: bins in resonant windows. X_{res}: ALP signal candidates. $\langle X_{\text{res}} \rangle$: expected value. X'_{res}: candidates after merging those within one ALP linewidth. X''_{res}: candidates in this scan also appearing in the primary scan within four ALP linewidths. Parenthetical fractions give the fraction of N_{full}.

	Primary scan	Scan A	Scan B
N_{full}	481 680	481 680	481 680
X_{full}	6393 (1.32 %)	5760 (1.20 %)	6202 (1.29 %)
$\langle X_{\text{full}} \rangle$	145 (0.03 %)	145 (0.03 %)	145 (0.03 %)
N_{res}	8430	8430	8430
X_{res}	5	7	3
$\langle X_{\text{res}} \rangle$	3	3	3
X'_{res}	5	5	3
X''_{res}	—	0	0
$\langle X''_{\text{res}} \rangle$	—	0	0

4.2.3 Results

In the absence of a detection, an upper limit on the ALP-proton gradient coupling is obtained from

$$|g_{\text{ap, lim}}| = \sqrt{\frac{e_5}{\eta_{\text{rec}}\, P_{\perp,1}}}\,, \tag{4.7}$$

where e_5 is the signal power corresponding to the $5\,\sigma$ NPE threshold, $P_{\perp,1}$ is the reference signal power for $|g_{\text{ap}}| = 1\,\text{GeV}^{-1}$, and η_{rec} is the analysis efficiency defined above. The resulting exclusion limits are plotted in Figure 4.9.

The measurement sets the limit $|g_{\text{ap}}| \lesssim 3 \times 10^{-2}\,\text{GeV}^{-1}$ over the scanned ALP mass range $5.577\,237\,\text{neV/c}^2 \pm 496\,\text{feV/c}^2$. The sensitivity is limited by the polarization of the methanol sample; hyperpolarized samples are expected to improve it by multiple orders of magnitude.

4.3 Earth-bound ALP measurement

4.3.1 Experiment

One proof-of-principle measurement for the Earth-bound ALP halo model was performed on December 14, 2022 (Central European Time). This measurement is suited to the Earth-bound search because each ALP-search acquisition per step lasted 1 h, giving a narrow RBW for long-coherence-time signals. Relevant calculations can be found at scripts-and-data/Result-EarthHalo.ipynbYuzhe: link to be modified.

The measurement scanned

$$1.348\,585\,\mathrm{MHz} \pm 24\,\mathrm{Hz}$$

by sweeping the magnitude of the bias field $|\mathbf{B}_0| \approx 31.674\,03\,\mathrm{mT}$. This maps to ALP masses in a range of

$$5.577\,299\,\mathrm{neV/c^2} \pm 50\,\mathrm{feV/c^2}\,.$$

One scan over this range was performed. There were 8 scan steps, and the frequency step size was 6 Hz, approximately half the NMR linewidth, so that the NMR resonant windows of adjacent steps overlap.

At each frequency step the following actions were performed in order:

1. Set sweep coil current.
2. Free decay measurement (one $\pi/2$ pulse followed by free decay signal) to determine the Larmor frequency, linewidth $\Delta\nu$ (and hence $T_2^* = 1/(\pi\Delta\nu)$), and NMR signal power. A Lorentzian fit to the resulting PSD peak extracts these parameters and calibrates the sensitivity at each step.
3. T_2 measurement by CPMG pulse sequence.
4. A 1-hour ALP measurement without any pulses.
5. Free decay measurement like step 2.
6. T_2 measurement like step 3.

Delays ($\gg T_1$) were inserted between pulsed-NMR measurements to allow the magnetization to return close to its equilibrium value along $\mathbf{B}_0$. The measurement procedure is illustrated in Figure 4.10. Note that the 1-hour ALP measurement duration is much longer than the 100-second duration used in the data taking described in Section 4.2. The long measurement duration results in a much narrower RBW, which is beneficial for the sensitivity to ALP signals with long coherence time, such as the Earth-bound ALP halo. The experimental parameters and the relevant NMR and ALP characteristics are summarized in Table 4.3.

Table 4.3: Technical parameters of the Earth-bound ALP measurement (December 14, 2022).

Parameter	Name or value
Sample	thermally polarized methanol
Number of scans	1
Number of scan steps	8
Sweep coil current increment	0.8 mA
Frequency increment per step	6 Hz
Frequency scan range	1.348 585 MHz ± 24 Hz
NMR linewidth in one step	$\approx$ 10 Hz
ALP mass scan range	5.577 299 neV/c^2 ± 50 feV/c^2
ALP linewidth $\Delta\nu_a$	$\lesssim$ 1 mHz
ALP coherence time τ_a	$\gtrsim$ 1000 s
T_2	$\approx$ 2 s
Measurement duration per step	1 h $\lesssim \tau_a$
spectral factor u	$\lesssim 10^{-4}$

4.3.2 ALP signal

Since the state population of the Earth-bound ALP halo is unknown and left as a free parameter, a few scenarios where measurements are sensitive to ALPs are considered here. Specifically, the six lowest-energy states of the Earth-bound ALP halo, where the ALP density is significant at Earth radius, are considered. Scenarios of ALP distributions are: ALPs distributed in individual eigenstates, or uniformly over the six lowest-energy eigenstates, as listed in Table 4.4. The empirical results of these scenarios are presented in this section.

Table 4.4: Frequency offsets and expected signal strengths for selected Earth-bound ALP eigenstates. Scenarios of ALP distributed in individual eigenstates or uniformly over the six lowest-energy eigenstates are considered. The frequency offset is calculated from the eigenenergy relative to the ALP rest energy, $\Delta\nu_a = E_{\mathrm{eigen}}/h - \nu_a$, where ALP Compton frequency ν_a is chosen to be 1.348 585 MHz, the central frequency of the scan range. The rms Rabi frequencies are calculated over the course of the measurement. The Rabi-frequency values are left open to be filled after the signal normalization is finalized.

Eigenstate population	$\Delta\nu_a$ (mHz)	rms $\Omega_a/(2\pi)$ (mHz)
1s	0.198	Yuzhe: TBD
2s	0.539	Yuzhe: TBD
3s	0.757	Yuzhe: TBD
2p	0.788	Yuzhe: TBD
4s	0.895	Yuzhe: TBD
5s	1.005	Yuzhe: TBD
uniform over 1s–5s	-	Yuzhe: TBD

For the scenarios of ALPs distributed in one single eigenstate, the expected ALP signal is a single frequency bin in the frequency spectra, in which case the optimal filter is not applicable. For the scenario of ALPs distributed uniformly over the six lowest-energy eigenstates, the expected ALP signal is expected to appear at the six discrete frequencies. The differences between the eigen-frequencies of the Earth-bound ALP halo are on the order of 1 mHz, while the difference between 3s to 5s is 0.25 mHz For the 1 h recorded data used here, RBW = 1/1 h = 0.28 mHz, which is roughly the same order as the ALP eigen-frequency differences. Therefore, the signal is not well resolved in the frequency spectra, and the optimal filter is not applicable for this scenario either.

In the reference frame where quantization axis is along the Sun-Earth direction, due to the daily rotation and the motion of the Earth, the component of the ALP-field gradient perpendicular to $\mathbf{B}_0$ varies over the course of the measurement. Using the ALP signal model established in Chapter 2.2, the expected Rabi frequencies of the pseudomagnetic field is calculated. In Figure 4.11, the evolution of the expected Rabi frequency of the pseudomagnetic field is plotted. The rms Rabi frequencies are summarized in Table 4.4. Since the Rabi frequency $\Omega_a \approx 2\pi \times 2\,\mathrm{mHz}$, the tipping angle of the magnetization can be estimated as $\theta \approx \Omega_a T_2 \approx 0.004\,\pi$.

4.3.3 Data analysis

The technical details of the data analysis, following the procedure described in Section 4.1.2, are presented as follows.

No problem was found in the time-series sanity check except for jumps. One example of the jumps is shown in Figure 4.12. The jumps are unlikely to be statistical fluctuations, since the jumps appear in clusters rather than randomly distributed in time. One possible explanation is that the jumps are magnetic noise caused by transient mechanical vibrations. Another possible explanation is that the jumps are exotic signals, such as ALP domain walls. The analysis of these transient signals is included in the Chapter 5.

After the jumps were removed by applying windowing, the Fourier transform were performed to generate the frequency spectra. The NPE spectrum was later formed after lock-in low-pass correction and the baseline subtraction. Yuzhe: Add one example of the spectra after different filtering steps.

The time-series signal from two channels of LIA are independent and Gaussian, so the NPE follows a χ^2 distribution with 2 degrees of freedom, also denoted as χ^2_2. As discussed in Section 4.1.2, a one-sided $5\,\sigma$ outlier in Gaussian statistics corresponds to a p-value of 2.87×10^{-7}. This p-value corresponds to $15.06\,\sigma$ in the χ^2_2 distribution. In the presence of a $15.06\,\sigma$ ALP signal, the NPE distribution is a non-central χ^2_2 distribution. If the outlier threshold is set at $7.809\,\sigma$Yuzhe: check digits, the probability of capturing a $15.06\,\sigma$ ALP signal is 95 %, while the probability of false positive is 4×10^{-4}. Therefore, the outlier threshold is set at $\Theta = 7.809\,\sigma$ for the Earth-bound ALP search. The threshold and corresponding probabilities are illustrated in Figure 4.13.

Yuzhe: Add analysis efficiency.

? outliers were found in the NPE spectra of 8 scan steps. Since the expected Earth-bound ALP linewidth is narrower than the RBW, the ALP signal would not appear in more than two adjacent frequency bins. The outliers were merged if they are adjacent. The corresponding coupling strengths of the merged outliers were obtained by the NMR characterization and Equation Yuzhe: add equation ref here. Outliers appearing at consistent frequencies with consistent coupling strengths were considered as candidates. The results of the validation check are summarized in Table 4.5.

? candidates were found after the validation check. One candidate is shown in

Figure 4.14 as an example. Note that there was only one scan performed over the scan range, so the candidates cannot be cross-checked by multiple scans.

Yuzhe: check if the frequency spectra is reversed Yuzhe: Add Earth-bound candidate counts and consistency checks. Yuzhe: Add a table summarizing the outlier counts and ALP signal candidates from the Earth-bound ALP scan.

Table 4.5: Results of the Earth-bound ALP scan. N_{full}: total NPE bins. X_{full}: outliers above threshold. $\langle X_{\mathrm{full}} \rangle$: expected value assuming white noise. N_{res}: bins in resonant windows. X_{res}: ALP signal candidates. $\langle X_{\mathrm{res}} \rangle$: expected value. Yuzhe: Adjust columns after the Earth-bound candidate definition is finalized.

	Earth-bound scan
N_{full}	Yuzhe: TBD
X_{full}	Yuzhe: TBD
$\langle X_{\mathrm{full}} \rangle$	Yuzhe: TBD
N_{res}	Yuzhe: TBD
X_{res}	Yuzhe: TBD
$\langle X_{\mathrm{res}} \rangle$	Yuzhe: TBD

4.3.4 Empirical results

Yuzhe: add description of the scenarios

In the absence of detections, constraints on the ALP-proton coupling can be derived from the measured NPE noise spectra and the expected signal power for the Earth-bound ALP halo model.Yuzhe: add equation ref here. At frequencies where candidates were found, the lower limit of constraints are set to the corresponding coupling strengths of the candidates. The resulting exclusion limits are plotted in Figure 4.15.

Yuzhe: Add the Earth-bound limit equation, including the reference signal power and analysis efficiency for the selected state-population model.

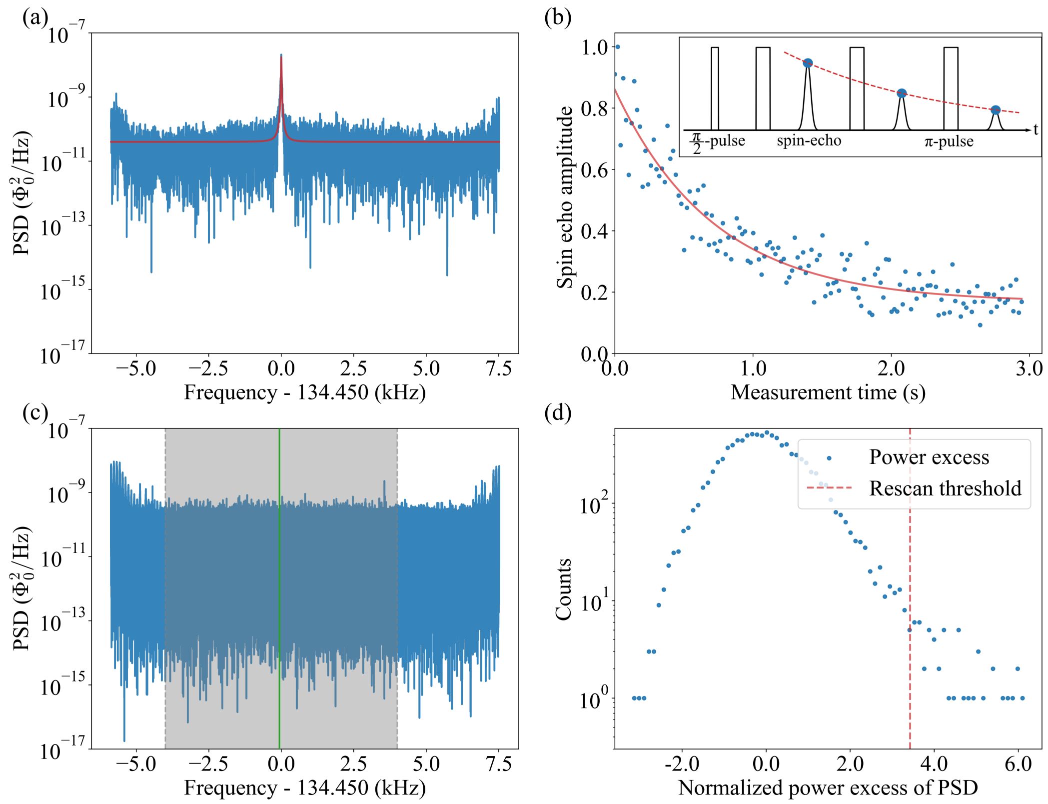

Figure 4.5: (a) PSD after a $\pi/2$ pulse. The NMR peak is fitted with a Lorentzian (red line); the center is at 1.348 449 21(5) MHz with a FWHM of 14.79(1) Hz. **(b)** CPMG spin-echo amplitudes (blue dots) and exponential fit (red), giving $T_2 = 0.72(5)$ s. **(c)** PSD of a 100 s ALP-search step. The gray analysis window (8 kHz wide) excludes filter artifacts near the band edges; the green resonant window (±5 NMR linewidths = ±70 Hz) is where ALP signal candidates are sought. **(d)** Histogram of normalized power excess (NPE) derived from (c), with the rescan threshold Θ_{RE} (red line) at the 95 % confidence level for a 5 σ ALP detection. Reproduced from Ref. [41].Yuzhe: a narrow frequency range for a and c? this figure may have to go to a different section

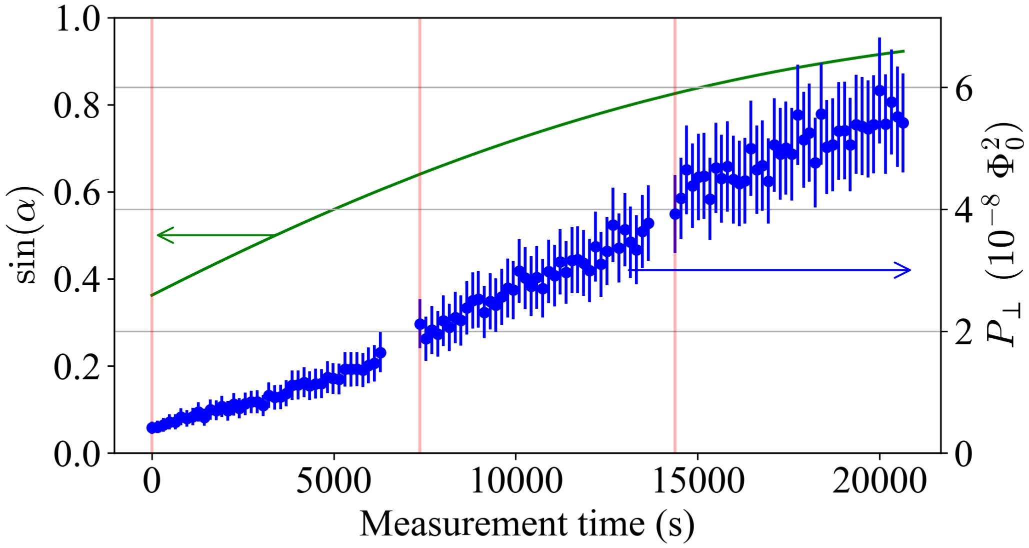

Figure 4.6: ALP parameters modulated by Earth's motion. Plotted for the start times of each measurement step for each of the three scans in seconds. Zero time corresponds to start of the data taking on December 23, 2022, 15:47:30 (Central European Time, UTC+1). Vertical dashed red lines mark the beginning time of a scan. Green line: $\sin\alpha$ oscillates with a period of one day. Blue dot with error bar: expected ALP signal powers in the PSD, according to Eq.Yuzhe: put an equation here, for the reference case of an ALP gradient coupling to proton of $g_{ap} = 10^{-9}\,\mathrm{GeV}^{-1}$. Uncertainties for these values were propagated from errors of the spin relaxation times T_2 and T_2^*, sample Larmor frequency ν_L and NMR signal power P_{NMR}. Yuzhe: use sin alpha without brackets. adjust ylabel paddings

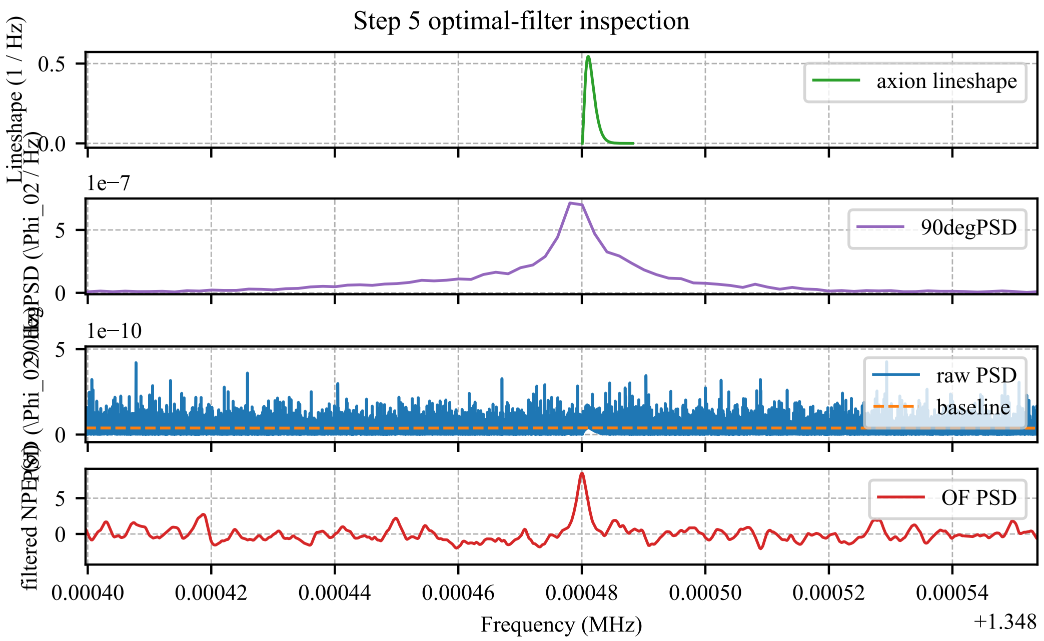

Figure 4.7: Example of data processing. Top to bottom: spectra of normalized ALP field, free decay signal, no-pulse signal, and NPE. The baseline of the no-pulse signal spectrum is indicated by a dashed line. Yuzhe: modify the figure. do we need fake ALP signal?

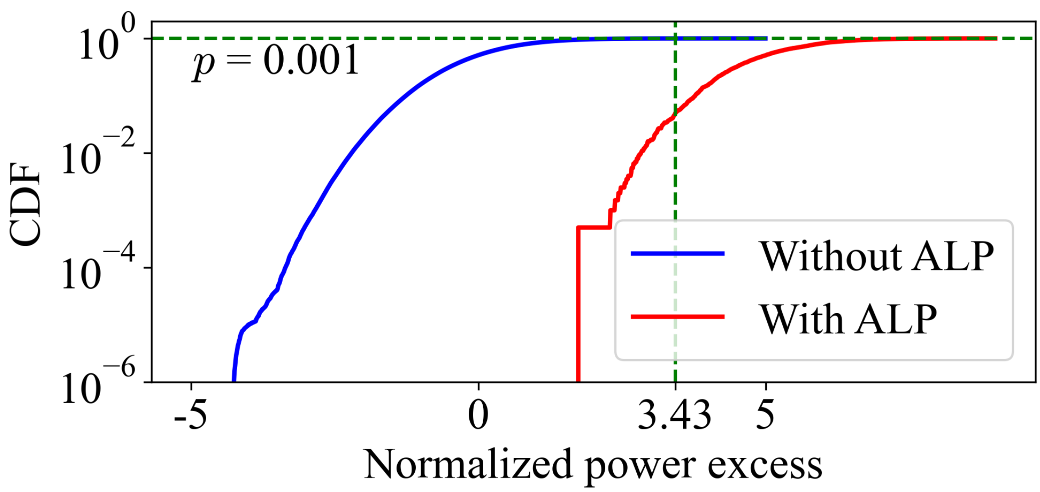

Figure 4.8: Empirical CDF obtained from the NPE distribution of spectra. The vertical dashed line represents the threshold for rescan, Θ_{RE}, corresponding to the null hypothesis of a $5\,\sigma$ ALP detection at the $95\,\%$ confidence level. The p-value corresponding to the threshold (horizontal dashed line) is approximately 0.001.

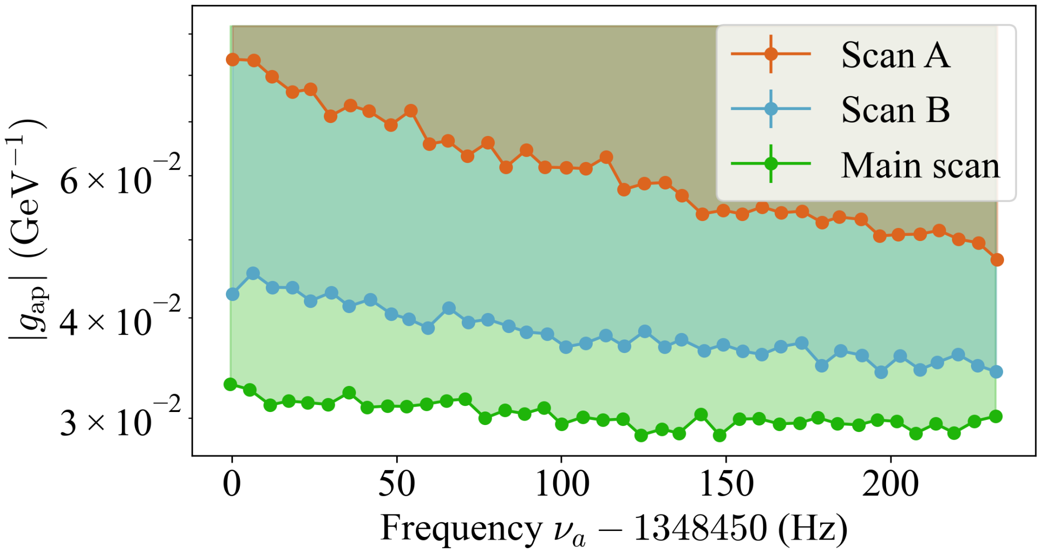

Figure 4.9: ALP gradient-proton coupling excluded at 95 % confidence level by the three sequential scans. Each data point represents the sensitivity for one scan step. The variation in sensitivity between scans reflects the daily modulation of the ALP wind direction relative to the bias field. Reproduced from Ref. [41].

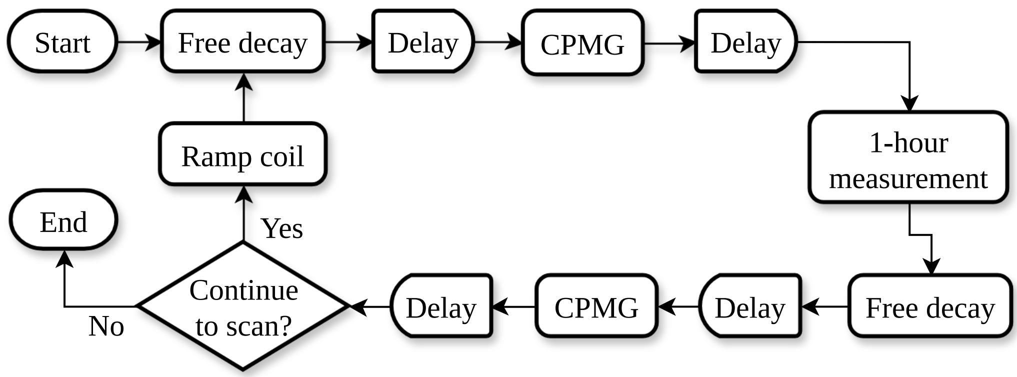

Figure 4.10: Schematic of the Earth-bound ALP measurement procedure.

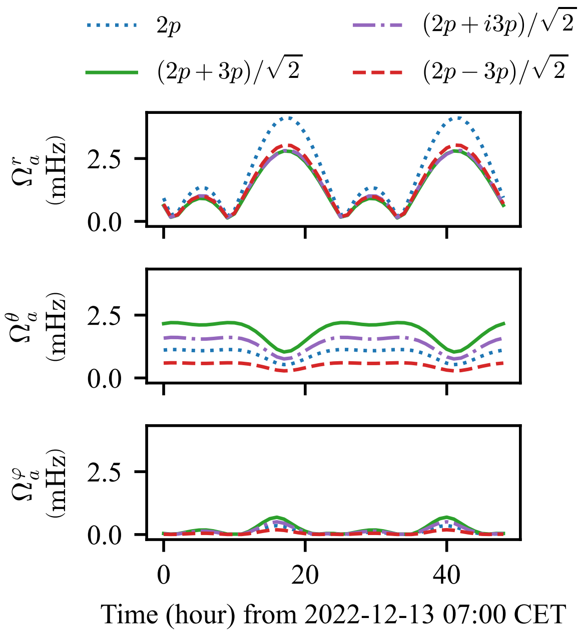

Figure 4.11: Expected evolution of the ALP-induced Rabi frequencies for the Earth-bound ALP halo model. Top to bottom: Rabi frequencies induced by gradients and Lorentz boost in radial, polar, and azimuthal directions. Scenarios of different state populations are considered as shown in the legend. The amplitude modulation is driven by the rotation of the laboratory frame relative to the ALP-field gradient.Yuzhe: highlight the course of the measurement in the figure.

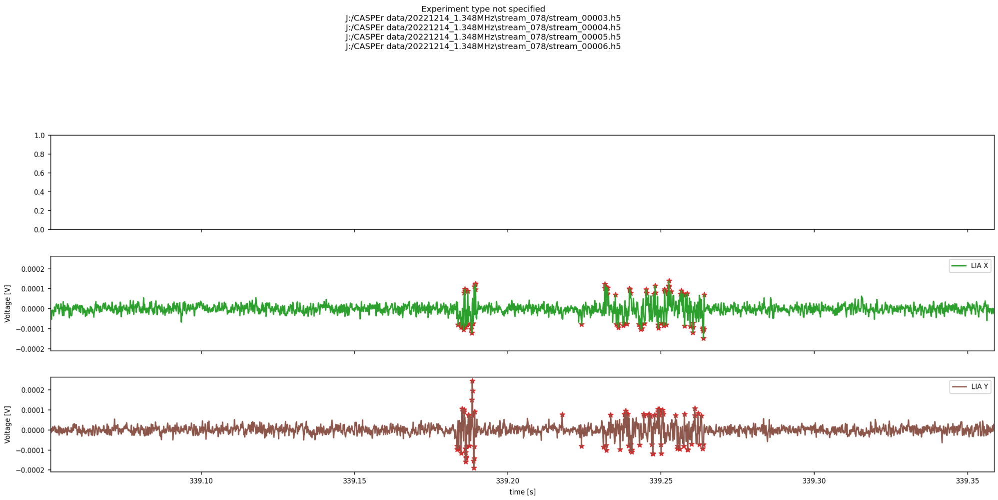

Figure 4.12: Example of jumps found in the time-series sanity check.Yuzhe: update this figure later.

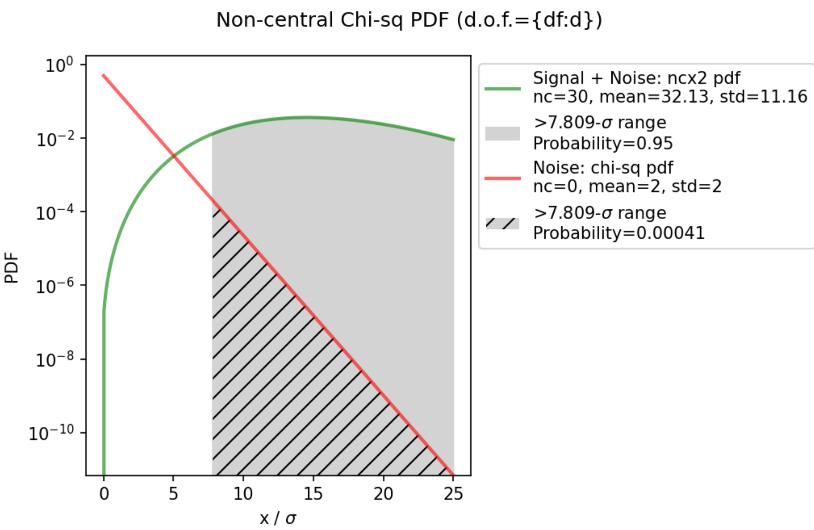

Figure 4.13: Distribution of the NPE under the null hypothesis (white noise) and alternative hypothesis (ALP signal).

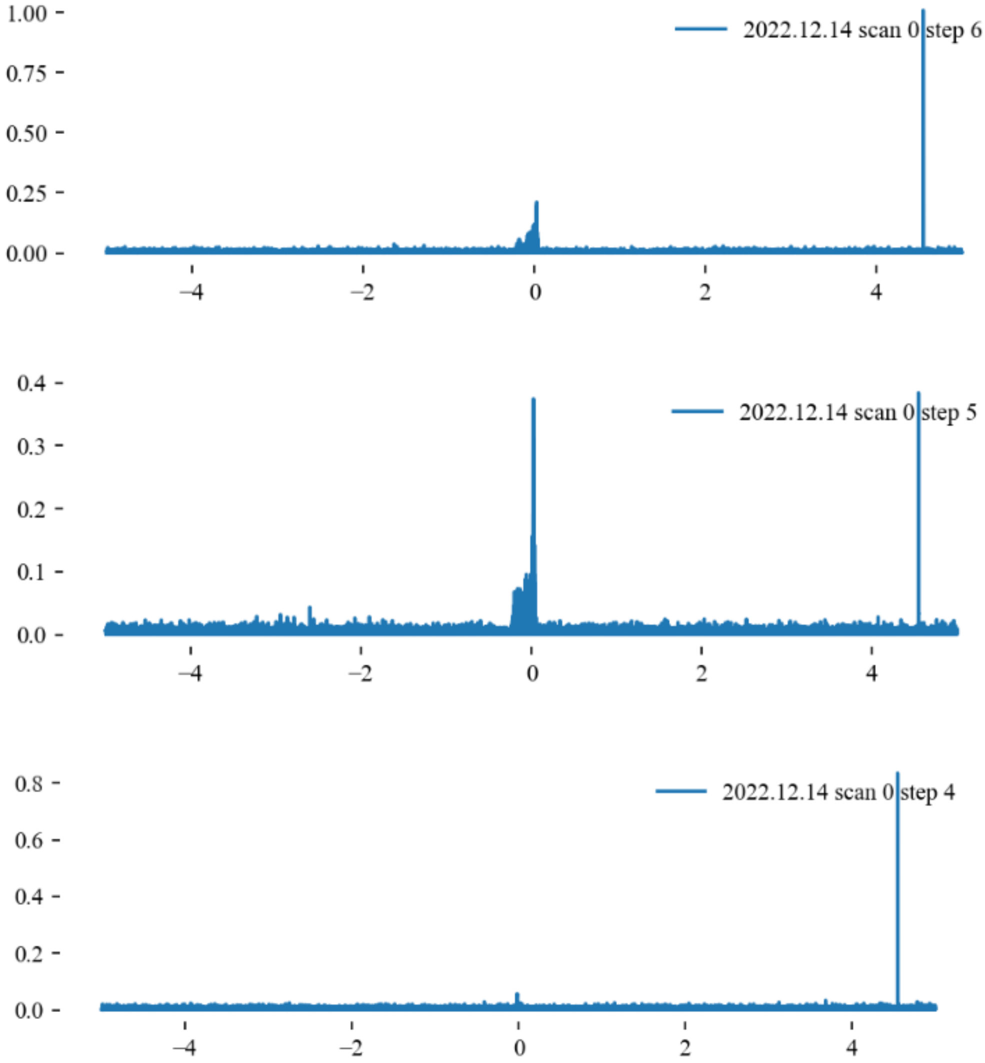

Figure 4.14: Example of an Earth-bound ALP signal candidate. Yuzhe: modify the figure

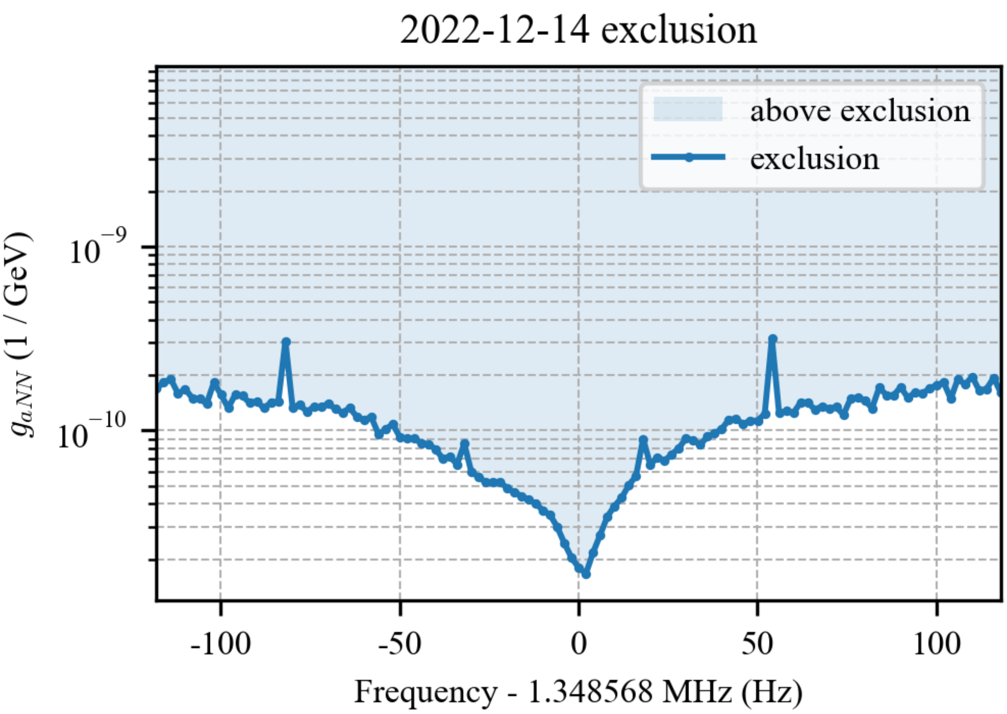

Figure 4.15: ALP gradient-proton coupling excluded at 95 % confidence level by the Earth-bound ALP scan. Yuzhe: Specify the assumed Earth-bound state population and total bound mass normalization.

5 Conclusions & Outlook

5.1 Summary of results

This thesis investigated axionlike-particle dark matter through the nuclear-gradient coupling, using nuclear magnetic resonance as the resonant detector. In this interaction, the spatial gradient of the ALP field acts as an oscillating pseudomagnetic field on nuclear spins. When the Larmor frequency of the sample is tuned to the ALP Compton frequency, this weak drive can generate a transverse magnetization that is detected by the CASPEr-Gradient-Low-Field apparatus.

The theoretical work developed the signal model and the scanning strategy needed for such a search. Two dark-matter scenarios were considered. For virialized Milky Way halo ALPs, the field is a stochastic superposition of many velocity modes, giving a finite coherence time and a linewidth set by the standard-halo velocity distribution. For Earth-bound ALPs, the field is instead described by gravitational bound states in the Earth's potential, leading to discrete eigenstates, longer coherence times, and characteristic daily and annual modulations of the local gradient.

The frequency-scanning analysis identified four signal regimes determined by the ordering of T_2^*, T_2, and the ALP coherence time τ_a. In all long-dwell-time regimes considered here, the coupling sensitivity scales only as $T_{\mathrm{tot}}^{-1/4}$ with total measurement time. This slow scaling makes the main experimental lesson clear: improving spin polarization, field homogeneity, T_2^*, T_2, and detector noise is more powerful than simply accumulating longer data sets. The optimal frequency step is the sensitive bandwidth of one measurement, and the optimal dwell time is uniform across scan steps except in the long-coherence regime with $T_2^* = T_2 \ll \tau_a$, where $T \propto \nu^{-1}$.

Experimentally, the thesis applied this framework to proof-of-principle measurements with thermally polarized methanol. The December 23, 2022 Milky Way ALP search scanned a 240 Hz band centered at 1.348 570 MHz, corresponding to ALP masses $5.577\,237\,\mathrm{neV/c^2} \pm 496\,\mathrm{feV/c^2}$. Three sequential scans were performed, with 100 s ALP-search acquisitions at each frequency step. The data were processed through sanity checks, Fourier transformation, lock-in response correction, Savitzky–Golay baseline removal, matched filtering with the expected ALP line-shape, and normalized-power-excess thresholding.

No statistically significant Milky Way ALP candidate survived the cross-scan consistency checks. The resulting 95 % confidence-level constraint is

$$|g_{\mathrm{ap}}| \lesssim 3 \times 10^{-2}\,\mathrm{GeV}^{-1} \tag{5.1}$$

over the scanned mass range. This narrow-band search serves as a commissioning measurement for CASPEr-Gradient-LF and, to the best of our knowledge, is the first dedicated laboratory search in this mass range.

The December 14, 2022 Earth-bound ALP measurement used the same apparatus and sample but emphasized longer dwell time. It scanned 1.348 585 MHz ± 24 Hz with 1 h no-pulse acquisitions at each of eight scan steps, targeting the narrower linewidth and longer coherence time expected for Earth-bound ALP states. The analysis framework for this data set was structured analogously to the Milky Way search, but with the important distinction that an unresolved one-bin statistic may replace the matched-filter convolution when the Earth-bound linewidth is smaller than the Fourier resolution. The final Earth-bound candidate counts and coupling limits remain to be completed once the candidate validation, signal normalization, and analysis efficiency are finalized.

5.2 Outlook

The most direct path to improved sensitivity is higher nuclear-spin polarization. The thermally polarized methanol sample used here has a proton polarization of only approximately $1.8{\times}10^{-7}$, which dominates the sensitivity budget. Hyperpolarized methanol, produced for example by brute-force prepolarization, parahydrogen-induced polarization, or dynamic nuclear polarization, could improve the projected coupling sensitivity to the 10^{-6}–$10^{-5}\,\mathrm{GeV}^{-1}$ range. Liquid ^{129}Xe polarized by spin-exchange optical pumping, combined with long T_2, offers a still more powerful route, with projected sensitivity at the 10^{-12}–$10^{-11}\,\mathrm{GeV}^{-1}$ level. Recent progress on ^{129}Xe hyperpolarization and on the temperature-control sample probe for the HF setup suggests that hyperpolarized samples should also improve the LF sensitivity.

Further gains should come from longer coherence in the sample and a quieter, more homogeneous apparatus. Improving the bias-field homogeneity from the present 10 ppm scale toward the projected 2 ppm level would lengthen T_2^* and increase the fraction of spins resonant with a narrow ALP signal. The LF setup is also expected to benefit from an upgraded variable-temperature insert (VTI), which should improve temperature stability and operating flexibility. Lower electronic noise and larger effective sample volume would improve the apparatus prefactor in the sensitivity equations. Larger sample volume and improved pickup-coil performance would further enhance the sensitivity. These improvements are complementary: polarization increases the available magnetization, while long relaxation times increase the accumulated ALP-induced tipping angle.

The Earth-bound ALP analysis opens a second direction for future work. Because bound states can have long coherence times and geometry-dependent gradients, the optimal statistic, signal normalization, and validation tests differ from the Milky Way halo case. Completing this analysis requires finalizing the state-population assumptions, the reference signal power, the recovery efficiency, and the candidate consistency criteria. The transient jumps seen in the one-hour records should also be catalogued systematically. They are likely instrumental in origin, but the same tools may become useful for searches for transient exotic signals such as ALP domain walls.

Finally, the numerical and statistical framework developed here can be extended beyond the specific methanol measurements. The `axionbloch` simulations provide a way to test Bloch dynamics for realistic relaxation, field inhomogeneity, hyperpolarized T_1 decay, and stochastic ALP fields. The Neyman–Pearson and optimal-filter treatment provides a general basis for future search statistics. Together with the broader frequency reach of CASPEr-Gradient-LF and the high-field apparatus, these tools prepare CASPEr for wider scans of ALP parameter space and for comparisons between different dark-matter models, spin species, and coupling structures. A particularly interesting direction is the dual-NMR-sample concept, which could expand the experimental reach to correlated searches for axions and other new-physics signals such as dark photons [**dror2025axionproductiondetectionusing**].

Data and source code availability

The LaTeX source of this dissertation, including all analysis scripts and figures, is publicly available in the GitHub repository of the CASPEr Collaboration:

https://github.com/CASPEr-Collaboration/ZhangYuzhe-Dissertation

The numerical simulation package `axionbloch` used in Appendix A is available as open-source software [45]:

https://github.com/Yuzhe98/AxionBloch

The raw experimental data underlying the methanol ALP search presented in Chapter 4 are associated with Ref. [41]. Requests for access to the raw data should be directed to the corresponding authors of that publication.

The scanning-optimization analysis underlying Sections 3.7–3.9 is associated with Ref. [47].

A Numerical simulation with axionbloch

This appendix is adapted from Ref. [46].

Analytical estimates of the ALP-induced spin-precession signal (Chapter 2) are valid in the long-dwell-time limit $T \gg \tau_a, T_2^*, T_2$. At lower Compton frequencies, the ALP coherence time becomes comparable to or longer than experimentally accessible dwell times, and the scaling laws no longer apply. Numerical simulation is then the tool of choice. This appendix describes `axionbloch`, an open-source Python package that simulates spin dynamics under both magnetic and axion-induced pseudomagnetic fields.

A.1 Software architecture

The package uses an object-oriented design separating three concerns: experimental configuration (`Sample`, `Magnet`), field generation (`MagField`), and simulation management (`Simulation`, `Simulations`). Axion-field models are encapsulated in dedicated classes (e.g., `MilkyWayAxionHalo`), allowing new models to be added without modifying the simulation framework. The architecture is illustrated in Figure A.1.

A.2 Calibration

Three standard NMR experiments are used to benchmark the simulation against known analytical results.

A.2.1 Pulsed NMR: free decay and spin echo

After a 90° pulse, the magnetization tips into the transverse plane and decays with the dephasing time T_2^* (free-induction decay). A subsequent 180° pulse refocuses the inhomogeneous dephasing, producing a spin echo whose amplitude decays with the true T_2. The simulated signals reproduce both behaviors, calibrating the pulse-induced rotation, Larmor precession, and transverse relaxations. The simulated magnetization is shown in Figure A.2 (a).

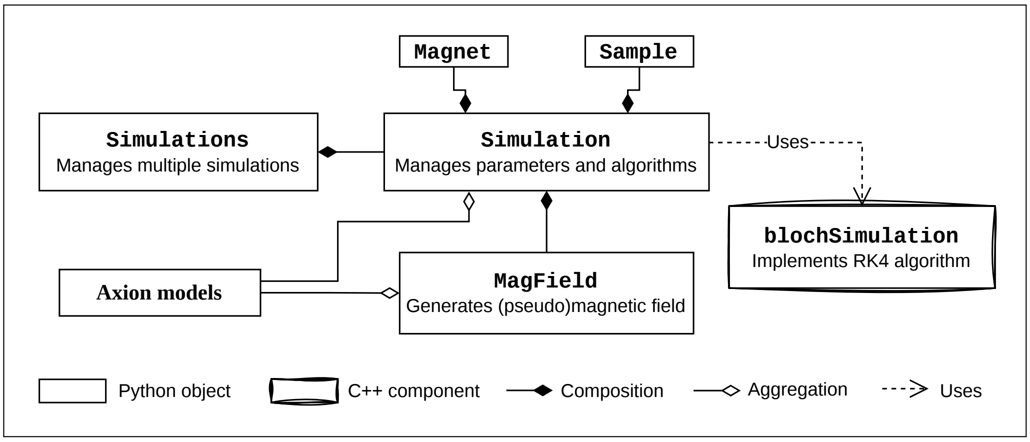

Figure A.1: Architecture of `axionbloch`. Each `Simulation` instance receives parameters from `Magnet`, `Sample`, and `MagField`. For axion simulations, an axion-model object is required both to configure the simulation and to generate the pseudomagnetic field. Numerical integration is performed via the `blochSimulation` API, which provides Python access to the C++ backend. Reproduced from Ref. [46].

A.2.2 Continuous-wave NMR

For a weak oscillating field on resonance ($\gamma B T_2^* \ll 1$), the transverse magnetization grows as $\gamma B T_2^* (1 - e^{-t/T_2^*})$ [1]. The simulation reproduces this analytical form, as shown in Figure A.2 (b). This benchmark calibrates both the Larmor precession and the T_2^* relaxation under continuous driving—the same physical regime as the ALP-induced signal.

A.2.3 Hyperpolarized sample and T_1 relaxation

Hyperpolarization boosts the initial spin polarization above its thermal value. In the simulation, the initial polarization is set as a parameter of the `Sample` class. A sample polarized significantly above thermal equilibrium relaxes toward $M_{\rm eqb}$ with the longitudinal time constant T_1. The simulated $M_z(t) = M_{\rm eqb}[1 - (M_{\rm eqb}^{\rm init}/M_{\rm eqb} - 1)e^{-t/T_1}]$ reproduces the expected exponential approach to equilibrium, as illustrated in Figure A.3.

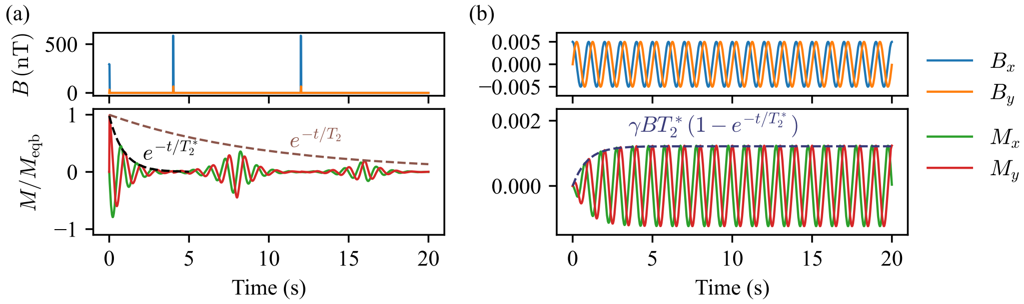

Figure A.2: Calibration of `axionbloch` against analytical NMR results. Upper rows: applied transverse magnetic fields; lower rows: normalized simulated magnetization. **(a)** After 90° and 180° pulses, the simulation produces free-induction decay (timescale T_2^*) and a spin echo (timescale T_2). **(b)** Under a weak continuous-wave drive on resonance, the transverse magnetization grows as $\gamma B T_2^* (1 - e^{-t/T_2^*})$ (dashed line). Reproduced from Ref. [46].

A.3 Axion simulation example

As a demonstration, the `MilkyWayAxionHalo` model simulates the pseudomagnetic field from virialized galactic dark matter following the standard halo model [20, 24]. The axion field oscillates at the Compton frequency with a Maxwell-Boltzmann velocity distribution setting the spectral linewidth. The stochasticity of the field amplitude is accounted for by generating many independent realizations of the pseudomagnetic field.

Figure A.4 shows results for a ^{129}Xe sample polarized to 50 %. A single realisation (left) illustrates the stochastic fluctuations in both the pseudomagnetic field and the induced transverse magnetization. An ensemble of 1000 realizations (right) yields the mean signal and its standard deviation band. The signal initially grows as the ALP field drives the magnetization away from equilibrium, and then decays on the T_1 timescale as the hyperpolarization is lost.

These simulations validate the three-regime picture Yuzhe: add reference to the analytical three-regime expression: at low frequencies where $\tau_a \gg T$, the signal grows linearly in dwell time; in the intermediate regime $\tau_a \sim T$, the signal departs from the analytical long-dwell-time scaling; and at high frequencies where $T \gg \tau_a$, the analytical results apply directly.

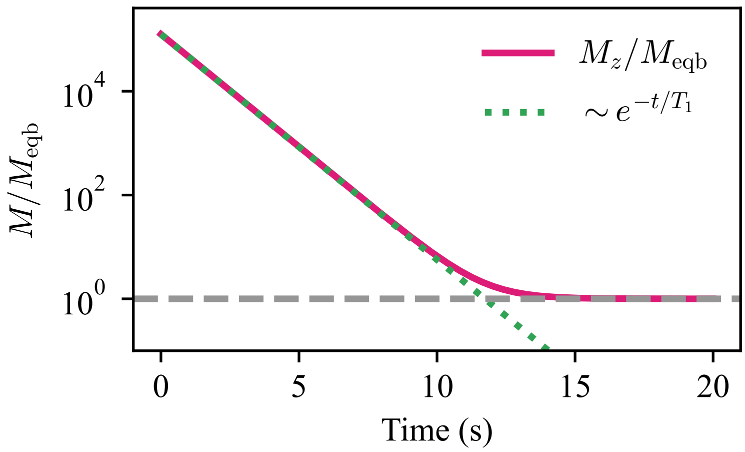

Figure A.3: Simulated T_1 relaxation of a hyperpolarized sample in a static bias field. The axial magnetization M_z is normalized by the equilibrium magnetization M_{eqb}. The exponential approach to equilibrium calibrates the longitudinal relaxation in the simulation. Reproduced from Ref. [46].

A.4 Statistical hypothesis test for ALP detection

A.5 The Neyman–Pearson Lemma

The *Neyman–Pearson lemma* is a foundational result in statistical hypothesis testing. It establishes the form of the most powerful test for discriminating between two simple hypotheses at a fixed significance level.

A.5.1 Problem setup

Suppose we observe data x drawn from a probability distribution. We wish to test

$$H_0 : x \sim f_0(x)$$

against

$$H_1 : x \sim f_1(x),$$

where both $f_0(x)$ and $f_1(x)$ are fully specified probability density functions. The goal is to construct a test that maintains a fixed false-alarm probability (Type I error) while maximizing the probability of correctly detecting the alternative hypothesis (power).

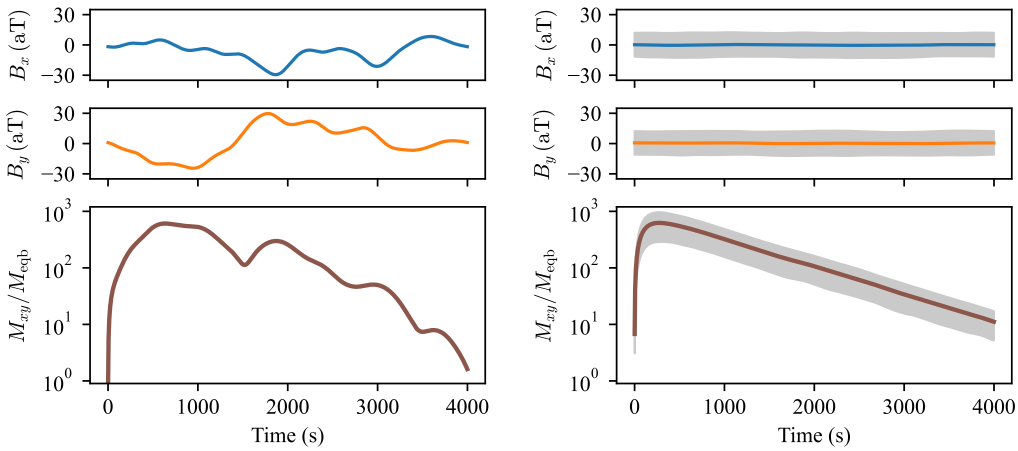

Figure A.4: Simulated axion-induced NMR signals for a virialized Milky Way axion dark matter halo. Upper panels: pseudomagnetic fields B_x and B_y in the transverse plane. Lower panels: amplitude of the induced transverse magnetization M_{xy}, normalized by M_{eqb}. Left: single simulation realisation. Right: ensemble of 1000 realizations; solid lines show the mean and shaded bands show $\pm\sigma$. The sample is ^{129}Xe initially polarized to 50 %. Reproduced from Ref. [46].

A.5.2 Likelihood ratio and the lemma

The key quantity is the likelihood ratio

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)}.$$

For a fixed significance level α, the Neyman–Pearson lemma states that the most powerful test rejects H_0 when $\Lambda(x) > k$, where k satisfies $P(\Lambda(x) > k \mid H_0) = \alpha$. The likelihood-ratio test is therefore optimal among all tests with the same false-positive rate.

A.5.3 Type I error and power

The significance level $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$ is the false-alarm probability. The power $1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ true})$ is the probability of correctly detecting an ALP signal. The Neyman–Pearson lemma guarantees that the likelihood-ratio test maximizes the power for fixed α.

A.5.4 Application to the ALP search

In the ALP data analysis described in Section 4.1, the normalized power excess (NPE) plays the role of the test statistic. The null hypothesis is that a $5\,\sigma$ ALP signal is present; the alternative is that no signal is present. Monte Carlo simulations of the NPE distribution under the null hypothesis determine the threshold $\Theta_{\mathrm{RE}} = 3.43$ corresponding to the 95 % confidence level (see Section 4.1).

A.5.5 Gaussian example

For $X \sim \mathcal{N}(\mu, \sigma^2)$ with $H_0 : \mu = 0$ versus $H_1 : \mu = 1$, the log-likelihood ratio simplifies to $\log \Lambda(x) = x/\sigma^2 - 1/(2\sigma^2)$, so rejecting for large $\Lambda(x)$ is equivalent to rejecting for large x. This confirms that the sample mean is the optimal test statistic for Gaussian location testing.

A.6 Optimal filter as a weighted average

The optimal filter used in the ALP search can be understood as the linear combination of frequency bins that maximizes the signal-to-noise ratio. Consider a baseline-subtracted spectrum with bin values x_i near a trial ALP frequency. If the expected ALP lineshape in these bins is s_i, the signal model can be written as

$$x_i = A s_i + n_i \,, \tag{A.1}$$

where A is the signal amplitude and n_i is noise. The noise in a power spectrum follows a χ^2 distribution with two degrees of freedom before averaging. After subtracting the local mean, the noise fluctuations have zero mean but are not Gaussian. The following result therefore states optimality among linear filters, rather than optimality among all possible statistical tests.

For a general linear statistic $T = \sum_i w_i x_i$, the signal-to-noise ratio is

$$\mathrm{SNR}(T) = \frac{A \sum_i w_i s_i}{\left(\sum_{ij} w_i C_{ij} w_j\right)^{1/2}} \,, \tag{A.2}$$

where $C_{ij} = \langle n_i n_j \rangle$ is the noise covariance matrix. Maximizing this expression gives

$$\mathbf{w} \propto \mathbf{C}^{-1}\mathbf{s} \,. \tag{A.3}$$

If the frequency bins are independent, $C_{ij} = \sigma_i^2 \delta_{ij}$, and the optimal linear estimator combines the bins as

$$\hat{A} = \frac{\sum_i s_i x_i/\sigma_i^2}{\sum_i s_i^2/\sigma_i^2}\,. \tag{A.4}$$

Equivalently, the filtered spectrum is a weighted average of the neighboring bins, with weights proportional to s_i/σ_i^2. If the noise variance is approximately constant across the linewidth, the weights are proportional to the expected ALP lineshape. Thus the optimal linear filter is matched to the expected ALP lineshape and reduces to a lineshape-weighted average when the local noise level is uniform.

A.7 Optimal dwell time in frequency scanning

This appendix derives the optimal dwell-time allocation scheme for a frequency scan, following the treatment of Ref. [47]. The goal is to maximize the sensitive area in the $\log \nu$–$\log |g_{\mathrm{aNN}}|$ parameter space over a total measurement time T_{tot}.

To simplify the notation, the frequency is normalized: $\hat{\nu} = \nu/\nu_{\mathrm{start}}$, with $\hat{\nu} \in [1, \hat{\nu}_{\mathrm{end}}]$ where $\hat{\nu}_{\mathrm{end}} = \nu_{\mathrm{end}}/\nu_{\mathrm{start}} > 1$. The dwell time is parameterized as

$$T = \kappa\,\hat{\nu}^{\,n}\,, \tag{A.5}$$

where κ is determined by T_{tot} and n, and the optimal value of n is determined below.

A.7.1 Regimes (1)–(3): step size proportional to ν

In regimes (1), (2), and (3), the frequency-step size is proportional to $\hat{\nu}$ (either one ALP linewidth $\hat{\nu}/Q_a$ or one NMR linewidth $\delta_\nu \hat{\nu}$). The scan rate is

$$\frac{d\hat{\nu}}{dt} = \frac{\Gamma_{\mathrm{scan}}}{T} \propto \hat{\nu}^{1-n}\,, \tag{A.6}$$

so that

$$dt = \kappa\, C\, \hat{\nu}^{n-1}\, d\hat{\nu}\,, \tag{A.7}$$

where $C = Q_a$ in regime (1) and $C = \delta_\nu^{-1}$ in regimes (2) and (3). Integrating over $[0, T_{\mathrm{tot}}]$ and $[1, \hat{\nu}_{\mathrm{end}}]$ gives

$$\kappa = \frac{n\,T_{\mathrm{tot}}}{C(\hat{\nu}_{\mathrm{end}}^{n} - 1)}\,, \qquad n \neq 0\,. \tag{A.8}$$

At $n = 0$: $\kappa|_{n=0} = T_{\mathrm{tot}}/[C\ln(\hat{\nu}_{\mathrm{end}})]$.

A.7.2 Regime (4): step size independent of ν

In regime (4) ($T_2^* = T_2 \ll \tau_a$), the step size is one NMR linewidth $2/T_2$, independent of frequency. The scan rate is proportional to $\hat{\nu}^{-n}$, giving

$$\kappa = \frac{2(n+1)\,T_{\mathrm{tot}}}{(\hat{\nu}_{\mathrm{end}}^{n+1} - 1)\,T_2}\,, \qquad n \neq -1\,. \tag{A.9}$$

At $n = -1$: $\kappa|_{n=-1} = 2T_{\mathrm{tot}}/[\ln(\hat{\nu}_{\mathrm{end}})\,T_2]$.

A.7.3 Optimal exponent n

The sensitive area in log-log scale is

$$A(n) = -\int_1^{\hat{\nu}_{\mathrm{end}}} \log f(\hat{\nu},\, n)\, d\log\hat{\nu}\,, \tag{A.10}$$

where $f(\hat{\nu},\, n)$ is the exclusion threshold from Equation (3.10) with the dwell time from Equation (A.5). Taking the derivative $A'(n)$ and setting it to zero shows:

- For regimes (1), (2), and (3): $A'(n) > 0$ for $n < 0$ and $A'(n) < 0$ for $n > 0$, so the maximum is at $n = 0$ (uniform dwell time).
- For regime (4): the analysis shifts the effective exponent from n to $n + 1$, giving an optimum at $n = -1$ (dwell time $T \propto \hat{\nu}^{-1}$).

An intuitive restatement: when the sensitive bandwidth per step grows with frequency (regimes 1–3), the number of steps per unit log ν is constant, and uniform dwell time is optimal. When the bandwidth is independent of frequency (regime 4), spending less time at higher frequencies—where the ALP coherence time is shorter and the signal integrates faster—maximizes the covered area.

A.8 ALP candidate lists

The outliers appeared in at least two independent scan steps with consistent frequencies and coupling strengths are regarded as ALP candidates. This appendix lists the ALP candidates identified in the Milky Way and Earth-halo searches.

Table A.1: Milky Way ALP candidates.

Index	Step index	Frequency (Hz)	Mass (neV/c^2)
1	5, 27	1348668.58	5.57764506
2	66, 86, 102	1348485.70	5.57688873
3	111, 117, 128	1348792.89	5.57815918
4	126, 140	1348872.72	5.57848936
5	7, 10	1348686.37	5.57771866
6	29, 95	1348769.02	5.57806045

Table A.2: Earth-halo ALP candidates.

Index	Step index	Frequency (Hz)	Mass (neV/c^2)
1	7	1348672.0739	5.5776595297
2	2	1348646.2182	5.5775525993
3	7	1348646.2294	5.5775526455
4	7	1348646.2375	5.5775526792
5	7	1348646.2472	5.5775527189
6	3	1348514.7177	5.5770087568
7	4	1348505.8949	5.5769722685
8	5	1348505.3398	5.5769699730
9	5	1348505.3520	5.5769700234
10	5	1348505.8094	5.5769719150
11	5	1348505.8120	5.5769719257
12	5	1348505.8628	5.5769721358
13	5	1348505.9028	5.5769723014
14	5	1348505.9126	5.5769723419

Index	Step index	Frequency (Hz)	Mass (neV/c^2)
15	5	1348505.9161	5.5769723564
16	5	1348595.9240	5.5773445990
17	5	1348595.9522	5.5773447157
18	5	1348595.9537	5.5773447217
19	5	1348595.9770	5.5773448182
20	5	1348595.9908	5.5773448753
21	5	1348596.0179	5.5773449876
22	5	1348596.0282	5.5773450299
23	5	1348646.2350	5.5775526685
24	5	1348646.2589	5.5775527675
25	6	1348505.8143	5.5769719353
26	6	1348505.8432	5.5769720548
27	6	1348505.8699	5.5769721653
28	6	1348505.8731	5.5769721784
29	6	1348505.8740	5.5769721821
30	6	1348505.8809	5.5769722106
31	6	1348505.8885	5.5769722423
32	6	1348505.8896	5.5769722469
33	6	1348505.8963	5.5769722745
34	6	1348595.9210	5.5773445867
35	6	1348595.9736	5.5773448042
36	6	1348596.0053	5.5773449353
37	6	1348596.0214	5.5773450017
38	6	1348596.0231	5.5773450090

Bibliography

[1] Anatole Abragam and Herman Y. Carr. **The Principles of Nuclear Magnetism**. *Physics Today* 14 (1961), 56–58 (see pages 2, 86).

[2] PAR Ade, Nabila Aghanim, Yashar Akrami, PK Aluri, M Arnaud, M Ashdown, J Aumont, C Baccigalupi, AJ Banday, RB Barreiro, et al. **Planck 2015 results-XVI. Isotropy and statistics of the CMB**. *Astronomy & Astrophysics* 594 (2016), A16 (see page 5).

[3] Deniz Aybas, Janos Adam, Emmy Blumenthal, Alexander V Gramolin, Dorian Johnson, Annalies Kleyheeg, Samer Afach, John W Blanchard, Gary P Centers, Antoine Garcon, et al. **Search for axionlike dark matter using solid-state nuclear magnetic resonance**. *Physical Review Letters* 126:14 (2021), 141802 (see pages 45, 51, 57).

[4] Deniz Aybas, Hendrik Bekker, John W Blanchard, Dmitry Budker, Gary P Centers, Nataniel L Figueroa, Alexander V Gramolin, Derek F Jackson Kimball, Arne Wickenbrock, and Alexander O Sushkov. **Quantum sensitivity limits of nuclear magnetic resonance experiments searching for new fundamental physics**. *Quantum Science and Technology* 6:3 (June 2021), 034007. DOI: 10.1088/2058-9565/abfbbc. URL: https://dx.doi.org/10.1088/2058-9565/abfbbc (see pages 2, 47).

[5] Deniz Aybas, Hendrik Bekker, Dmitry Budker, Wei Ji, On Kim, Younggeun Kim, Derek F. Jackson Kimball, Jia Liu, Xiaolin Ma, Chiara P. Salemi, Yannis K. Semertzidis, Alexander O. Sushkov, Kai Wei, Arne Wickenbrock, and Yuzhe Zhang. **Cavity, Lumped Circuit, and Spin-Based Detection of Axion Dark Matter: Differences and Similarities**. *Universe* 12:4 (2026). ISSN: 2218-1997. DOI: 10.3390/universe12040106. URL: https://www.mdpi.com/2218-1997/12/4/106 (see page 18).

[6] Abhishek Banerjee, Dmitry Budker, Joshua Eby, Victor V. Flambaum, Hyungjin Kim, Oleksii Matsedonskyi, and Gilad Perez. **Searching for Earth/Solar Axion Halos**. *Journal of High Energy Physics* 2020:9 (Sept. 2020). arXiv:1912.04295 [astro-ph, physics:hep-ph, physics:physics], 4. ISSN: 1029-8479. DOI: 10.1007/JHEP09(2020)004. URL: http://arxiv.org/abs/1912.04295 (visited on 07/24/2023) (see page 21).

[7] Abhishek Banerjee, Dmitry Budker, Joshua Eby, Hyungjin Kim, and Gilad Perez. **Relaxion Stars and their detection via Atomic Physics**. *Communications Physics* 3:1 (Jan. 2020). arXiv:1902.08212 [astro-ph, physics:hep-ph, physics:physics], 1. ISSN: 2399-3650. DOI: 10.1038/s42005-019-0260-3. URL: http://arxiv.org/abs/1902.08212 (visited on 07/24/2023) (see page 21).

[8] Lars Bergström, Piero Ullio, and James H. Buckley. **Observability of γ rays from dark matter neutralino annihilations in the Milky Way halo**. *Astroparticle Physics* 9:2 (1998), 137–162. ISSN: 0927-6505. DOI: https://doi.org/10.1016/S0927-6505(98)00015-2. URL: https://www.sciencedirect.com/science/article/pii/S0927650598000152 (see page 10).

[9] F. Bloch. **Nuclear Induction**. *Phys. Rev.* 70 (7-8 Oct. 1946), 460–474. DOI: 10.1103/PhysRev.70.460. URL: https://link.aps.org/doi/10.1103/PhysRev.70.460 (see page 2).

[10] B. M. Brubaker, L. Zhong, S. K. Lamoreaux, K. W. Lehnert, and K. A. van Bibber. **HAYSTAC axion search analysis procedure**. *Physical Review D* 96:12 (Dec. 2017). ISSN: 2470-0029. DOI: 10.1103/physrevd.96.123008. URL: http://dx.doi.org/10.1103/PhysRevD.96.123008 (see pages 51, 57).

[11] Dmitry Budker, Peter W. Graham, Micah Ledbetter, Surjeet Rajendran, and Alexander O. Sushkov. **Proposal for a Cosmic Axion Spin Precession Experiment (CASPEr)**. *Phys. Rev. X* 4 (2 May 2014), 021030. DOI: 10.1103/PhysRevX.4.021030. URL: https://link.aps.org/doi/10.1103/PhysRevX.4.021030 (see pages 11, 47).

[12] Bernard Carr and Florian Kühnel. **Primordial Black Holes as Dark Matter: Recent Developments**. *Annual Review of Nuclear and Particle Science* 70 (2020), 355–394. DOI: 10.1146/annurev-nucl-050520-125911. URL: https://doi.org/10.1146/annurev-nucl-050520-125911 (see page 7).

[13] Gary P. Centers, John W. Blanchard, Jan Conrad, Nataniel L. Figueroa, Antoine Garcon, Alexander V. Gramolin, Derek F. Jackson Kimball, Matthew Lawson, Bart Pelssers, Joseph A. Smiga, Alexander O. Sushkov, Arne Wickenbrock, Dmitry Budker, and Andrei Derevianko. **Stochastic fluctuations of bosonic dark matter**. *Nature Communications* 12:1 (Dec. 2021), 7321. ISSN: 2041-1723. DOI: 10.1038/s41467-021-27632-7. URL: https://doi.org/10.1038/s41467-021-27632-7 (see pages 11, 16, 17).

[14] Douglas Clowe, Maruša Bradač, Anthony H. Gonzalez, Maxim Markevitch, Scott W. Randall, Christine Jones, and Dennis Zaritsky. **A Direct Empirical Proof of the Existence of Dark Matter**. *The Astrophysical Journal* 648:2 (Aug. 2006), L109–L113. ISSN: 1538-4357. DOI: 10.1086/508162. URL: http://dx.doi.org/10.1086/508162 (see page 5).

[15] Michael Dine, Willy Fischler, and Mark Srednicki. **A Simple Solution to the Strong *CP* Problem with a Harmless Axion**. *Physics Letters B* 104:3 (Aug. 1981), 199–202. DOI: 10.1016/0370-2693(81)90590-6. URL: https://doi.org/10.1016/0370-2693(81)90590-6 (see page 7).

[16] Pieter van Dokkum, Shany Danieli, Yotam Cohen, Allison Merritt, Aaron J. Romanowsky, Roberto Abraham, Jean Brodie, Charlie Conroy, Deborah Lokhorst, Lamiya Mowla, Ewan O'Sullivan, and Jielai Zhang. **A galaxy lacking dark matter**. *Nature* 555:7698 (Mar. 2018), 629–632. DOI: 10.1038/nature25767. URL: https://doi.org/10.1038/nature25767 (see page 6).

[17] Pieter van Dokkum, Zili Shen, Michael A. Keim, Sebastian Trujillo-Gomez, Shany Danieli, Dhruba Dutta Chowdhury, Roberto Abraham, Charlie Conroy, J. M. Diederik Kruijssen, Daisuke Nagai, and Aaron J. Romanowsky. **A trail of dark-matter-free galaxies from a bullet-dwarf collision**. *Nature* 605:7910 (May 2022), 435–439. DOI: 10.1038/s41586-022-04665-6. URL: https://doi.org/10.1038/s41586-022-04665-6 (see page 6).

[18] Adam M. Dziewonski and Don L. Anderson. **Preliminary reference Earth model**. *Physics of the Earth and Planetary Interiors* 25:4 (1981), 297–356. ISSN: 0031-9201. DOI: https://doi.org/10.1016/0031-9201(81)90046-7. URL: https://www.sciencedirect.com/science/article/pii/0031920181900467 (see page 21).

[19] James Eills, Dmitry Budker, Silvia Cavagnero, Eduard Y. Chekmenev, Stuart J. Elliott, Sami Jannin, Anne Lesage, Jörg Matysik, Thomas Meersmann, Thomas Prisner, Jeffrey A. Reimer, Hanming Yang, and Igor V. Koptyug. **Spin Hyperpolarization in Modern Magnetic Resonance**. *Chemical Reviews* 123:4 (2023). PMID: 36701528, 1417–1551. DOI: 10.1021/acs.chemrev.2c00534. eprint: https://doi.org/10.1021/acs.chemrev.2c00534. URL: https://doi.org/10.1021/acs.chemrev.2c00534 (see page 2).

[20] N. Wyn Evans, Ciaran A. J. O'Hare, and Christopher McCabe. **Refinement of the standard halo model for dark matter searches in light of the Gaia Sausage**. *Phys. Rev. D* 99 (2 Jan. 2019), 023012. DOI: 10.1103/PhysRevD.99.023012. URL: https://link.aps.org/doi/10.1103/PhysRevD.99.023012 (see pages 8, 9, 16, 87).

[21] Antoine Garcon, John W. Blanchard, Gary P. Centers, Nataniel L. Figueroa, Peter W. Graham, Derek F. Jackson Kimball, Surjeet Rajendran, Alexander O. Sushkov, Yevgeny V. Stadnik, Arne Wickenbrock, Teng Wu, and Dmitry Budker. **Constraints on bosonic dark matter from ultralow-field nuclear magnetic resonance**. *Science Advances* 5:10 (2019), eaax4539 (see page 12).

[22] Peter W. Graham and Surjeet Rajendran. **New observables for direct detection of axion dark matter**. *Phys. Rev. D* 88 (3 Aug. 2013), 035023. DOI: 10.1103/PhysRevD.88.035023. URL: https://link.aps.org/doi/10.1103/PhysRevD.88.035023 (see page 10).

[23] Alexander V. Gramolin, Deniz Aybas, Dorian Johnson, Janos Adam, and Alexander O. Sushkov. **Search for axion-like dark matter with ferromagnets**. *Nature Physics* 17:1 (Jan. 2021), 79–84. ISSN: 1745-2481. DOI: 10.1038/s41567-020-1006-6. URL: https://doi.org/10.1038/s41567-020-1006-6 (see page 51).

[24] Alexander V. Gramolin, Arne Wickenbrock, Deniz Aybas, Hendrik Bekker, Dmitry Budker, Gary P. Centers, Nataniel L. Figueroa, Derek F. Jackson Kimball, and Alexander O. Sushkov. **Spectral signatures of axionlike dark matter**. *Phys. Rev. D* 105 (3 Feb. 2022), 035029. URL: https://link.aps.org/doi/10.1103/PhysRevD.105.035029 (see pages 17–19, 46, 87).

[25] Giovanni Grilli di Cortona, Edward Hardy, Javier Pardo Vega, and Giovanni Villadoro. **The QCD Axion, Precisely**. *Journal of High Energy Physics* 2016:1 (Jan. 2016), 34. ISSN: 1029-8479. DOI: 10.1007/JHEP01(2016)034. URL: https://doi.org/10.1007/JHEP01(2016)034 (see page 7).

[26] G. Heinzel, A. Rüdiger, and R. Schilling. **Spectrum and Spectral Density Estimation by the Discrete Fourier Transform (DFT), Including a Comprehensive List of Window Functions and Some New Flat-Top Windows**. Tech. rep. Max-Planck-Institut für Gravitationsphysik, Feb. 2002. URL: https://holometer.fnal.gov/GH_FFT.pdf (see page 55).

[27] Lam Hui, Jeremiah P. Ostriker, Scott Tremaine, and Edward Witten. **Ultralight scalars as cosmological dark matter**. *Phys. Rev. D* 95 (4 Feb. 2017), 043541. DOI: 10.1103/PhysRevD.95.043541. URL: https://link.aps.org/doi/10.1103/PhysRevD.95.043541 (see page 7).

[28] Derek F Jackson Kimball and Karl Van Bibber. **The search for ultralight bosonic dark matter**. Springer Nature, 2023 (see pages 7, 11).

[29] Gerard Jungman, Marc Kamionkowski, and Kim Griest. **Supersymmetric Dark Matter**. *Physics Reports* 267 (1992), 195–373 (see page 10).

[30] Jihn E. Kim. **Weak-Interaction Singlet and Strong *CP* Invariance**. *Physical Review Letters* 43:2 (July 1979), 103–107. DOI: 10.1103/PhysRevLett.43.103. URL: https://doi.org/10.1103/PhysRevLett.43.103 (see page 7).

[31] David J. E. Marsh. **Axion Cosmology**. *Physics Reports* 643 (2016), 1–79. DOI: 10.1016/j.physrep.2016.06.005 (see page 7).

[32] Morgan W. Mitchell and Silvana Palacios Alvarez. **Colloquium: Quantum limits to the energy resolution of magnetic field sensors**. *Rev. Mod. Phys.* 92 (2 Apr. 2020), 021001. DOI: 10.1103/RevModPhys.92.021001. URL: https://link.aps.org/doi/10.1103/RevModPhys.92.021001 (see page 2).

[33] National Institute of Standards and Technology. *Fundamental Physical Constants from NIST.* Accessed 2026-07-31. 2024. URL: https://physics.nist.gov/cuu/Constants/ (see page 2).

[34] R. D. Peccei and Helen R. Quinn. **CP Conservation in the Presence of Pseudoparticles**. *Phys. Rev. Lett.* 38 (25 June 1977), 1440–1443. URL: https://link.aps.org/doi/10.1103/PhysRevLett.38.1440 (see page 7).

[35] R. D. Peccei and Helen R. Quinn. **Constraints imposed by CP conservation in the presence of pseudoparticles**. *Phys. Rev. D* 16 (6 Sept. 1977), 1791–1797. URL: https://link.aps.org/doi/10.1103/PhysRevD.16.1791 (see page 7).

[36] Vera C Rubin and W Kent Ford Jr. **Rotation of the Andromeda nebula from a spectroscopic survey of emission regions**. *Astrophysical Journal, vol. 159, p. 379* 159 (1970), 379 (see page 5).

[37] Mikhail A. Shifman, Arkady I. Vainshtein, and Valentin I. Zakharov. **Can Confinement Ensure Natural *CP* Invariance of Strong Interactions?** *Nuclear Physics B* 166:3 (Apr. 1980), 493–506. DOI: 10.1016/0550-3213(80)90209-6. URL: https://doi.org/10.1016/0550-3213(80)90209-6 (see page 7).

[38] Yoshiaki Sofue and Vera Rubin. **Rotation Curves of Spiral Galaxies**. *Annual Review of Astronomy and Astrophysics* 39:1 (2001), 137–174 (see pages 5, 10).

[39] Simona Vegetti, Simon D. M. White, John P. McKean, Devon M. Powell, Cristiana Spingola, Davide Massari, Giulia Despali, and Christopher D. Fassnacht. **A possible challenge for cold and warm dark matter**. *Nature Astronomy* 10 (Mar. 2026), 440–447. DOI: 10.1038/s41550-025-02746-w. URL: https://doi.org/10.1038/s41550-025-02746-w (see page 6).

[40] Julian Walter, Hendrik Bekker, John Blanchard, Dmitry Budker, Nataniel L. Figueroa, Arne Wickenbrock, Yuzhe Zhang, and Pengyu Zhou. **Fast Shimming Algorithm Based on Bayesian Optimization for Magnetic Resonance Based Dark Matter Search**. *Annalen der Physik* 536:1 (2024), 2300258. DOI: https://doi.org/10.1002/andp.202300258. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/andp.202300258 (see page 37).

[41] Julian Walter, Olympia Maliaka, Yuzhe Zhang, John W. Blanchard, Gary Centers, Arian Dogan, Martin Engler, Nataniel L. Figueroa, Younggeun Kim, Derek F. Jackson Kimball, Matthew Lawson, Declan W. Smith, Alexander O. Sushkov, Dmitry Budker, Hendrik Bekker, and Arne Wickenbrock. **Search for axionlike dark matter using liquid-state nuclear magnetic resonance**. *Phys. Rev. D* (2025). CASPEr-Gradient-LF methanol proof-of-concept measurement (see pages 37, 41, 51, 57, 70, 73, 83).

[42] Steven Weinberg. **A New Light Boson?** *Physical Review Letters* 40:4 (Jan. 1978), 223–226. DOI: 10.1103/PhysRevLett.40.223. URL: https://doi.org/10.1103/PhysRevLett.40.223 (see page 7).

[43] Frank Wilczek. **Problem of Strong *P* and *T* Invariance in the Presence of Instantons**. *Physical Review Letters* 40:5 (Jan. 1978), 279–282. DOI: 10.1103/PhysRevLett.40.279. URL: https://doi.org/10.1103/PhysRevLett.40.279 (see page 7).

[44] Teng Wu, John W. Blanchard, Gary P. Centers, Nataniel L. Figueroa, Antoine Garcon, Peter W. Graham, Derek F. Jackson Kimball, Surjeet Rajendran, Yevgeny V. Stadnik, Alexander O. Sushkov, Arne Wickenbrock, and Dmitry Budker. **Search for Axionlike Dark Matter with a Liquid-State Nuclear Spin Comagnetometer**. *Phys. Rev. Lett.* 122 (19 May 2019), 191302 (see page 12).

[45] Yuzhe Zhang. *AxionBloch*. Version v0.1. 2026. URL: https://github.com/Yuzhe98/AxionBloch (see page 83).

[46] Yuzhe Zhang et al. *axionbloch: an Open-Source Python Package for Simulating Axion-Induced Spin Dynamics*. https://github.com/Yuzhe98/AxionBloch. 2024 (see pages 85–89).

[47] Yuzhe Zhang, Deniz Aybas Tumturk, Hendrik Bekker, Dmitry Budker, Derek F. Jackson Kimball, Alexander O. Sushkov, and Arne Wickenbrock. **Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments**. *Annalen der Physik* 536:1 (2024), 2300223. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/andp.202300223 (see pages 11, 13, 37, 47, 48, 83, 91).

[48] A. R. Zhitnitsky. **The Weinberg Model of *CP* Violation and *T*-Odd Correlations in Weak Decays**. *Soviet Journal of Nuclear Physics* 31 (1980). In Russian; original publication: Yadernaya Fizika 31, 1024–1035 (1980), 529–534 (see pages 7, 8).

[49] Zurich Instruments AG. *MFIA User Manual: 500 kHz / 5 MHz Impedance Analyzer*. Release 26.04; see Section 6.4, Discrete-Time Filters. Zurich Instruments AG. 2026. URL: https://docs.zhinst.com/pdf/ziMFIA_UserManual.pdf (see page 55).